

Supporting Information:

Developing Pharmaceutically Relevant Pd-Catalyzed C–N Coupling Reactivity Models Leveraging High-Throughput Experimentation

Seung Kyun Ha,^{†1} Dipannita Kalyani,^{*†3} Michael S. West,^{†3} Jessica Xu,^{†1,2} Yu-hong Lam,³ Thomas Struble,³ Spencer Dreher,³ Shane W. Krska,³ Stephen L. Buchwald,² Klavs F. Jensen^{*1}

¹ Department of Chemical Engineering, Massachusetts Institute of Technology, Cambridge, MA 02139, USA

² Department of Chemistry, Massachusetts Institute of Technology, Cambridge, MA 02139, USA

³ Discovery Chemistry, Merck & Co., Inc., Rahway, NJ 07065, USA

[†] Equal authorship

^{*} D. Kalyani: dipannita.kalyani@merck.com. Klavs F. Jensen: kfjensen@mit.edu

Table of Contents

I. General Information	p. S3–S5
II. HTE Optimization Screen Procedures	p. S6–S20
III. HTE Procedures for Aryl Halide and Amine Scopes	p. S21–S28
IV. Procedures for Amine and Aryl Halide Quality Assessment	p. S29–S34
V. Substrate Selection and Nanoscale Procedures for Validation Libraries	p. S35–S54
VI. Product Scale-up and Characterization	p. S55–S68
VII. Selection of Secondary Amines and Aryl Halides for Scope Experiments	p. S69–S70
VIII. Details of Featurization, Pistachio data curation, t-SNE, and UMAP	p. S71–S73
IX. Additional Analyses	p. S74–S81
X. Data Splitting	p. S82–S88
XI. Modeling Details	p. S89–S91
XII. Modeling Results and Additional Discussions	p. S92–S104
XIII. References	p. S105–S106
XIV. NMR Spectra	p. S107–S117

I. General Information

Materials. All reagents were used as received from vendors (Merck & Co., Inc., Rahway, NJ, USA Building Block Collection (MBBC), Aldrich, Fisher, Enamine, Combi-Blocks). The bases were stored in a glovebox. Precatalysts were either synthesized using literature procedures or purchased from Aldrich or Strem Chemicals. Solvents were purchased from Aldrich, anhydrous, sure-seal quality, and used without further purification.

Robotics. All robots were housed inside an MBraun glovebox under an atmosphere of N₂. A TECAN Freedom Evo robot was used to create reaction source plates and a TTP Labtech Mosquito HTS® robot was used to dose nanomole-scale chemistry reactions from these source plates into a reaction plate. A Thermo Matrix 3x2 Platemate liquid handling robot was then used to quench and dilute reaction mixtures after the reactions were complete.

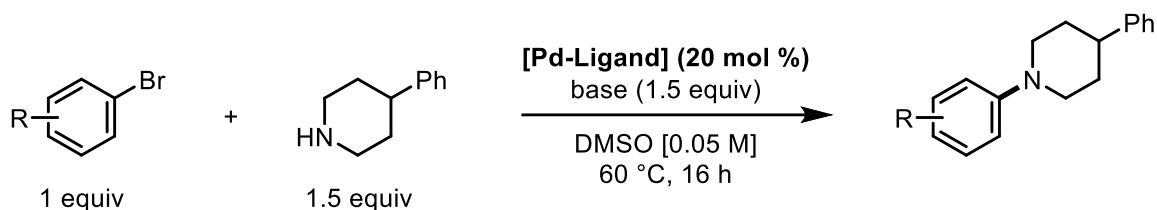
General Experimental. All reactions were set up inside an MBraun glovebox operating with a constant N₂ purge. Nanoscale reactions (50 nmol) were run using Corning 384-well plates (Product No. 3657, clear, polypropylene, round bottom) as source plates, Beckman Coulter 1536-well plates (Product No. 001-7913, Echo qualified cyclic olefin-copolymer (COC), clear, low dead volume, flat bottom) as reaction plates, and Analytical Sales 384-well plates (Product No. 38120, clear, polypropylene, clear, square wells, flat bottom) as analytical plates for UPLC-MS analysis. Dosing of reaction components into the 1536-well plates was accomplished in the glovebox using a Mosquito HTS liquid handling robot (TTP Labtech, 4.5 mm pitch tip spool) with no special modifications, and using the TTP Labtech native software. After dosing was complete, the 1536-well reaction plates were then heat sealed inside the glovebox with peelable aluminum seal (Agilent, Product No. 24210-001) using an Agilent Velocity11 PlateLoc Automated Microplate Sealer. The resulting sealed plate was centrifuged for 30 seconds to ensure that reaction components were mixed at the bottom of the wells. The sealed plate was covered by a perfluoroalkoxy alkane (PFA) mat (Analytical Sales, Cat. No. 96981), followed by a silicon rubber mat (Analytical Sales, Cat. No. 96982) and then secured with a clamp (Arctic White, Cat. No. AWS-051001).

UPLC-UV-MS Analysis. Reaction data was collected using a Waters™ Acquity UPLC I-Class system (Waters Corp., Milford, MA, USA) equipped with a binary pump, a sample manager, column manager, a photodiode array detector, SQD 2 single quadrupole mass spectrometer with an electrospray ionization (ESI) source, and Masslynx® software. A Waters™ BEH C18 column (50 x 1.0 mm I.D., 1.7 µm particle size) set at 55 °C was used for chromatographic separations. The mobile phase used for the analysis was 0.1% v/v trifluoroacetic acid in water (Fisher Optima LC/MS Optima grade) for MPA + 0.1% v/v trifluoroacetic acid in acetonitrile (Fisher Optima LC/MS Optima grade) for MPB. The flow rate was set 0.35 mL/min. The analyses were conducted using a mobile phase gradient with the following program: an initial hold for 0.25 min at 2% MPB, 2%–100% MPB from 0.25 min to 2.1 min using a linear gradient, hold at 100% MPB until 2.30 min and return to 2% MPB at 2.31 min. The UV was scanned between 200–500 nm. An autosampler program using 1 µL injections was made for all samples, standards, and diluent injections. The total time per run for each injection was 2.5 minutes. Virscidian Analytical Studio™ (Virscidian Inc., Cary, NC, USA) was used for data processing of high-throughput data LC/MS analysis. LCAP (Liquid Chromatography Area Percent) was used to gauge reaction performance and can be calculated as: (area of product peak / total area of all integrated peaks) × 100. Data visualizations were created using Spotfire™ or with Python packages.

Compound Characterization. ^1H NMR spectra were recorded on a Bruker 500 MHz spectrometer at ambient probe temperature and are internally referenced to residual protio solvent signals as noted. Data for ^1H NMR are reported as follows: chemical shift (δ ppm), multiplicity (s = singlet, d = doublet, t = triplet, q = quartet, m = multiplet, dd = doublet of doublets, dt = doublet of triplets, br = broad), coupling constant (Hz), and integration. ^{13}C NMR spectra were recorded on a Bruker 500 MHz (126 MHz) spectrometer at ambient probe temperature and are internally referenced to residual protio solvent signals as noted. ^{19}F NMR spectra were recorded on a Bruker Avance III HD 400 (376 MHz). High-resolution mass spectrometry was performed on a Waters Xevo G2-XS Q-Time-of-flight (Tof) using electrospray ionization (ESI+).

II. HTE Optimization Screen Procedures

Nanoscale procedure for catalyst screen (Manuscript Figure 3A, Step 1)



In a nitrogen-filled glovebox, to a 1536 plastic COC plate (Corning EchoTM qualified, Cat. No. 3730, Cyclic Olefin-Copolymer COC, 12.5 μ L-wells, flat bottom, clear) using the TTP Labtech Mosquito robot, was added aryl bromide (0.2 μ L, 0.25 M solution in DMSO, 0.05 μ mol, 1.0 equiv), 4-phenylpiperidine (0.19 μ L, 0.4 M solution in DMSO, 0.075 μ mol, 1.5 equiv), precatalyst (0.4 μ L, 0.025 M mixture in DMSO, 0.01 μ mol, 0.2 equiv), and base (0.273 μ L, 0.275 M solution in DMSO, 0.075 μ mol, 1.5 equiv). Once the dosing was completed, the reaction plate was heat sealed, centrifuged, enclosed in an aluminum block, and taken out of the glovebox. The aluminum block containing the reaction plate was placed between two preheated thermocouples (bottom thermocouple at 50 °C and top thermocouple at 70 °C). This setup was agitated at 10 G using the Resodyn acoustic mixing LabRam (PharmaRAM II Mixer) for 16 h. The aluminum block containing the reaction plate was cooled to room temperature and taken into the glovebox. The reaction plate was taken out of the aluminum block and centrifuged. The seal was removed, and the reaction mixture was transferred using 100 μ L of 1% AcOH/DMSO into a 384 flat-bottom COC (Cyclic-Olefin Copolymer) analytical plate using the Thermo Matrix Platemate robot. The analytical plate was heat sealed in the glovebox and the reaction mixtures were analyzed using UPLC-UV/MS to obtain the product LCAPs.

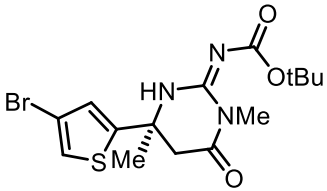
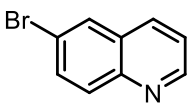
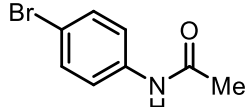
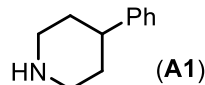
<u>Aryl Halides</u>  Informer X13  6-bromoquinoline (Br-1)  4-bromoacetanilide (Br-2)	<u>Pd Catalysts</u> <table> <tr> <th>Pd G3 catalysts</th><th>Pd G4 catalysts</th></tr> <tr> <td>XPhos Pd G3 (P1)</td><td>XPhos Pd G4 (P8)</td></tr> <tr> <td><i>t</i>BuXPhos Pd G3 (P2)</td><td>BrettPhos Pd G4 (P9)</td></tr> <tr> <td><i>t</i>BuBrettPhos Pd G3 (P3)</td><td>DavePhos Pd G4 (P10)</td></tr> <tr> <td>AdBrettPhos Pd G3 (P4)</td><td>RuPhos Pd G4 (P11)</td></tr> <tr> <td>RuPhos Pd G3 (P5)</td><td>CPhos Pd G4 (P12)</td></tr> <tr> <td>DavePhos Pd G3 (P6)</td><td><i>t</i>BuPhCPhos Pd G4 (P13)</td></tr> <tr> <td>JackiePhos Pd G3 (P7)</td><td></td></tr> </table> <u>Pd G6 catalysts</u> <table> <tr> <td>XPhos Pd G6 (P14)</td><td>DavePhos Pd G6 (P19)</td></tr> <tr> <td><i>t</i>BuXPhos Pd G6 (P15)</td><td>JackiePhos Pd G6 (P20)</td></tr> <tr> <td><i>t</i>BuBrettPhos Pd G6 (P16)</td><td>AlPhos Pd G6 (P21)</td></tr> <tr> <td>AdBrettPhos Pd G6 (P17)</td><td>GPhos Pd G6 (P22)</td></tr> <tr> <td>RuPhos Pd G6 (P18)</td><td></td></tr> </table>	Pd G3 catalysts	Pd G4 catalysts	XPhos Pd G3 (P1)	XPhos Pd G4 (P8)	<i>t</i> BuXPhos Pd G3 (P2)	BrettPhos Pd G4 (P9)	<i>t</i> BuBrettPhos Pd G3 (P3)	DavePhos Pd G4 (P10)	AdBrettPhos Pd G3 (P4)	RuPhos Pd G4 (P11)	RuPhos Pd G3 (P5)	CPhos Pd G4 (P12)	DavePhos Pd G3 (P6)	<i>t</i> BuPhCPhos Pd G4 (P13)	JackiePhos Pd G3 (P7)		XPhos Pd G6 (P14)	DavePhos Pd G6 (P19)	<i>t</i> BuXPhos Pd G6 (P15)	JackiePhos Pd G6 (P20)	<i>t</i> BuBrettPhos Pd G6 (P16)	AlPhos Pd G6 (P21)	AdBrettPhos Pd G6 (P17)	GPhos Pd G6 (P22)	RuPhos Pd G6 (P18)	
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RuPhos Pd G6 (P18)																											
<u>Amine</u>  (A1)	<u>Bases</u> <table> <tr> <td>NaO<i>t</i>Bu (B11)</td><td>BTMG (B17)</td></tr> <tr> <td>DBU (B15)</td><td>P2-Et (B35)</td></tr> </table>	NaO <i>t</i> Bu (B11)	BTMG (B17)	DBU (B15)	P2-Et (B35)																						
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Figure S1. Reagents used for catalyst screen (Manuscript Figure 3A, Step 1).

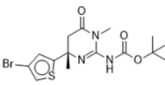
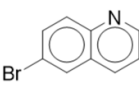
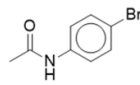
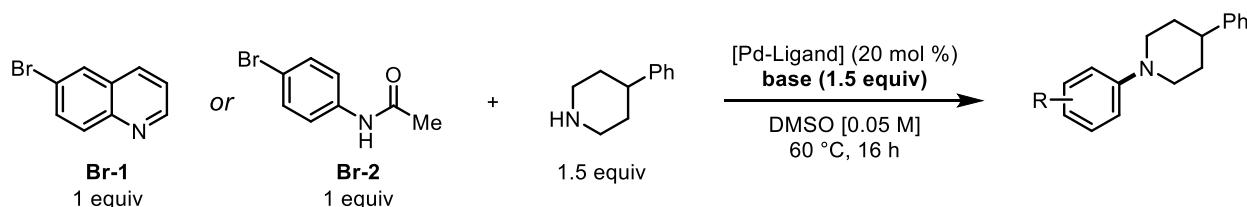
	 X13				 Br-1				 Br-2			
	NaOtBu (B11)	DBU (B15)	BTMG (B17)	P2-Et (B35)	NaOtBu (B11)	DBU (B15)	BTMG (B17)	P2-Et (B35)	NaOtBu (B11)	DBU (B15)	BTMG (B17)	P2-Et (B35)
XPPhos Pd G3 (P1)	1.20	1.13	2.65	1.55	36.30	6.19	14.71	14.84	5.67	0.00	0.84	4.76
tBuXPPhos Pd G3 (P2)	4.60	0.46	1.09	4.02	50.90	30.84	57.57	56.53	5.54	3.16	14.10	0.02
tBuBrettPhos Pd G3 (P3)	2.11	3.84	1.17	1.84	5.02	6.28	1.09	2.52	5.38	0.04	0.11	4.07
AdBrettPhos Pd G3 (P4)	0.65	0.57	0.40	0.42	4.27	6.36	0.99	6.17	2.87	0.69	0.65	1.99
RuPhos Pd G3 (P5)	1.37	0.41	2.77	1.29	36.81	3.27	40.77	77.83	5.84	0.00	0.82	5.59
DavePhos Pd G3 (P6)	0.92	1.58	0.87	0.75	23.31	6.67	28.96	51.42	2.84	0.00	0.71	1.96
JackiePhos Pd G3 (P7)	0.12	0.24	0.08	0.12	5.69	8.84	2.00	4.36	0.23	0.17	0.09	0.16
XPPhos Pd G4 (P8)	2.04	0.98	3.20	2.16	69.38	1.69	41.37	84.60	17.46	0.00	2.79	15.17
BrettPhos Pd G4 (P9)	0.99	0.14	0.89	0.39	33.66	5.07	25.92	70.11	8.38	0.00	1.16	3.42
DavePhos Pd G4 (P10)	1.27	1.56	0.90	0.75	28.51	0.70	56.29	75.61	5.70	0.00	2.46	6.80
RuPhos Pd G4 (P11)	2.80	0.21	3.80	2.49	54.86	1.08	29.44	83.36	15.79	0.00	3.62	12.51
CPhos Pd G4 (P12)	4.90	0.55	3.61	4.37	77.77	1.53	61.26	84.29	45.90	0.05	0.31	50.61
tBuPhCPhos Pd G4 (P13)	6.67	1.15	2.74	6.70	50.10	8.34	52.24	70.41	61.12	0.70	6.90	57.83
XPPhos Pd G6 (P14)	2.72	0.62	1.47	1.84	76.72	0.94	52.02	87.80	14.80	0.00	3.01	14.40
tBuXPPhos Pd G6 (P15)	5.62	0.33	1.07	6.88	56.70	22.96	68.62	58.17	3.81	1.69	15.95	3.15
tBuBrettPhos Pd G6 (P16)	3.73	0.45	0.83	3.26	3.64	0.51	0.93	1.08	3.67	0.03	0.05	2.93
AdBrettPhos Pd G6 (P17)	4.31	0.55	1.39	5.80	2.10	0.26	1.02	0.45	1.95	0.25	0.25	1.33
RuPhos Pd G6 (P18)	3.62	0.21	3.04	2.79	50.80	1.09	41.86	87.17	11.44	0.00	5.30	11.06
DavePhos Pd G6 (P19)	2.91	0.74	2.69	2.11	32.01	12.44	49.39	77.06	5.78	0.24	4.89	6.10
JackiePhos Pd G6 (P20)	0.31	0.44	0.49	0.27	27.91	1.21	12.05	28.99	0.00	0.00	0.00	0.00
AlPhos Pd G6 (P21)	1.26	0.71	0.51	1.19	6.99	2.48	2.55	3.12	2.72	0.18	0.27	2.45
GPhos Pd G6 (P22)	4.92	0.74	2.04	3.56	27.87	4.14	9.01	17.05	14.05	0.08	0.27	10.84

Figure S2. Results from catalyst screen (Manuscript Figure 3A, Step 1). Numbers are product LCAPs (UV254).

Nanoscale procedure for base screen using Br-1 or Br-2 (Manuscript Figure 3A, Step 2)



In a nitrogen-filled glovebox, to a 1536 plastic COC plate (Corning Echo™ qualified, Cat. No. 3730, Cyclic Olefin-Copolymer COC, 12.5 μ L-wells, flat bottom, clear) using the TTP Labtech Mosquito robot, was added aryl bromide (0.167 μ L, 0.30 M solution in DMSO, 0.05 μ mol, 1.0 equiv), 4-phenylpiperidine (0.214 μ L, 0.35 M solution in DMSO, 0.075 μ mol, 1.5 equiv), precatalyst (0.4 μ L, 0.025 M mixture in DMSO, 0.01 μ mol, 0.2 equiv), and base (0.273 μ L, 0.275 M solution in DMSO, 0.075 μ mol, 1.5 equiv). Once the dosing was completed, the reaction plate was heat sealed, centrifuged, enclosed in an aluminum block, and taken out of the glovebox. The aluminum block containing the reaction plate was placed between two preheated thermocouples (bottom thermocouple at 50 °C and top thermocouple at 70 °C). This setup was agitated at 10 G using the Resodyn acoustic mixing LabRam (PharmaRAM II Mixer) for 16 h. The aluminum block containing the reaction plate was cooled to room temperature and taken into the glovebox. The reaction plate was taken out of the aluminum block and centrifuged. The seal was removed, and the reaction mixture was transferred using 100 μ L of 1% AcOH/DMSO into a 384 flat-bottom

COC (Cyclic-Olefin Copolymer) analytical plate using the Thermo Matrix Platemate robot. The analytical plate was heat sealed in the glovebox and the reaction mixtures were analyzed using UPLC-UV/MS to obtain the product LCAPs.

Note: the reaction plates for **Br-1** and **Br-2** were prepared separately.

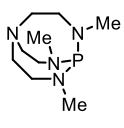
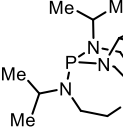
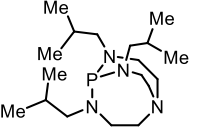
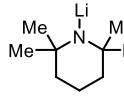
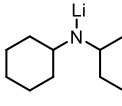
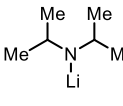
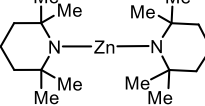
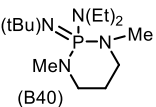
<u>Siloxy bases</u>	<u>Organic bases</u>	<u>Verkade's superbases</u>	
B1. LiOTMS	B15. DBU		
	B16. MTBD		
<u>HMDS bases</u>	B17. BTMG		
B2. LiHMDS	B18. TMG		
B3. NaHMDS	B19. DABCO		
B4. KHMDS	B20. Et ₃ N		
B5. Zn(HMDS) ₂	B21. diisopropylethylamine		
B6. Sn(HMDS) ₂	B22. 4-piperidinopyridine	<u>Amido bases</u>	
B7. Y(HMDS) ₃	B23. 2,6-di-t-butylpyridine		
B8. Sm(HMDS) ₃	B24. 2,6-dimethylpyridine		
B9. Mg(HMDS) ₂	B25. 2,2,6,6-tetramethylpiperidine		
	B26. 1,2,2,6,6-pentamethylpiperidine		
<u>Alkoxy bases</u>	B27. Proton Sponge		
B10. LiOtBu	B28. 1,5,7-triazabicyclo[4.4.0]dec-5-ene		
B11. NaOtBu	B29. 1,5-diazabicyclo[4.3.0]non-5-ene		
B12. KOtBu	B30. 3,3,6,6,9,9-pentamethyl-2,10-diazabicyclo[4.4.0]dec-1-ene		
	B31. DBU/NaTFA		
<u>Phenoxy bases</u>	B32. DBU/NaOTf		
B13. NaBHT	B33. MTBD/NaTFA		
B14. Al(OPh) ₃	B34. MTBD/NaOTf		
	<u>Phosphazene bases</u>		
	B35. P ₂ Et		
	B36. P ₁ -H		
	B37. P ₁ -tBu		
	B38. P ₁ -t-Oct		
	B39. P ₁ -tBu-tris(tetramethylene)		
	B40. BEMP		
		<u>Pd precatalysts</u>	
		tBuXPhos Pd G3 (P2)	AdBrettPhos Pd G6 (P17)
		XPhos Pd G4 (P8)	AlPhos Pd G6 (P21)
		CPhos Pd G4 (P12)	GPhos Pd G6 (P22)
		tBuPhCPhos Pd G4 (P13)	

Figure S3. Reagents used for base screen (Manuscript Figure 3A, Step 2).

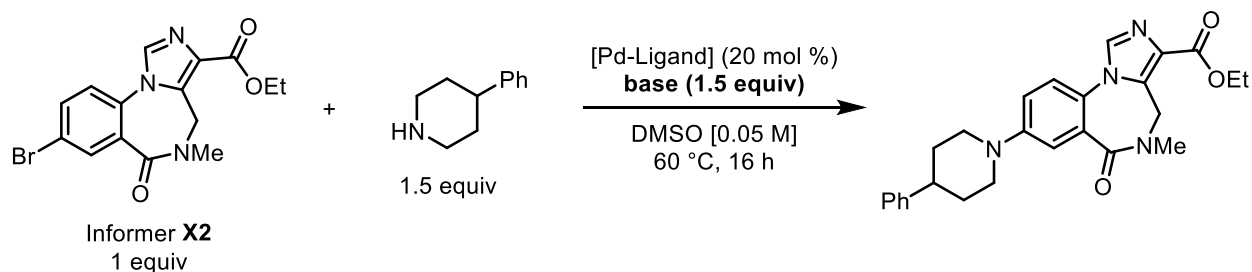
		tBuXPhos Pd G3 (P2)	XPhos Pd G4 (P8)	CPhos Pd G4 (P12)	tBuPhCPhos Pd G4 (P13)	AdBrettPhos Pd G6 (P17)	AlPhos Pd G6 (P21)	GPhos Pd G6 (P22)
Siloxo HMDS	B1	82.11	87.39	85.85	78.06	20.25	7.05	82.51
	B2	52.20	59.97	61.83	43.82	11.66	7.54	23.44
	B3	61.69	39.30	65.40	43.47	15.56	18.18	43.28
	B4	61.31	9.80	44.71	25.46	19.66	20.46	20.75
	B5	6.86	2.14	1.92	3.47	5.69	4.28	1.98
	B6	1.37	0.00	1.30	1.05	0.00	0.00	1.50
	B7	2.36	2.77	3.26	3.02	0.00	0.00	0.00
	B8	14.02	37.00	56.34	20.06	0.00	0.00	25.97
	B9	27.92	29.32	50.91	19.82	0.00	0.00	10.40
Alkoxy	B10	73.74	84.51	83.37	74.76	9.04	6.53	37.71
	B11	55.82	69.76	77.77	56.84	9.43	6.86	39.60
	B12	45.76	10.07	27.43	16.83	10.84	9.94	20.56
Phenoxy	B13	80.87	78.94	81.85	70.67	14.06	13.18	62.84
	B14	7.19	3.89	4.72	4.96	0.93	2.17	2.41
Organic	B15	41.87	3.43	5.76	19.32	0.00	3.62	3.83
	B16	75.90	50.31	71.66	58.19	7.67	7.17	25.38
	B17	72.86	55.40	63.48	56.21	4.08	3.39	14.94
	B18	29.16	6.05	18.57	18.47	4.26	3.70	4.08
	B19	1.78	1.29	1.65	1.32	3.74	0.00	1.48
	B20	4.35	3.22	3.09	3.39	0.00	0.00	2.48
	B21	5.19	3.49	3.51	4.59	3.48	3.30	2.48
	B22	2.09	1.18	0.75	2.74	2.16	0.00	0.00
	B23	4.07	0.92	0.00	1.64	0.00	0.00	1.24
	B24	5.86	1.27	2.43	3.10	5.32	0.00	1.58
	B25	1.97	1.35	1.54	1.57	0.00	3.02	1.50
	B26	2.63	2.39	3.99	2.46	3.27	0.00	1.94
	B27	1.69	0.89	1.15	0.99	2.08	0.00	0.90
	B28	3.46	1.35	3.58	0.00	0.50	0.14	0.98
	B29	26.42	2.13	1.51	6.90	11.98	1.27	1.54
	B30	0.00	7.12	12.65	9.53	3.24	3.17	2.41
	B31	50.39	3.92	6.06	22.12	3.68	3.78	1.81
	B32	40.75	3.22	5.89	21.76	3.39	3.29	1.67
	B33	0.00	55.91	74.58	59.77	5.19	4.98	32.01
	B34	78.07	58.11	73.08	60.63	4.77	5.69	35.18
Phosphazene	B35	67.31	82.89	82.15	76.40	6.91	6.60	42.48
	B36	78.25	46.15	66.33	62.69	4.34	3.20	31.26
	B37	75.63	60.88	70.05	66.66	8.35	6.50	18.07
	B38	10.87	8.87	14.96	12.66	0.00	0.00	6.92
	B39	32.55	34.27	32.76	34.61	3.96	3.60	12.72
	B40	45.50	47.70	49.28	47.33	4.98	4.28	16.47
Verkade's superbases	B41	11.92	5.04	6.37	6.56	5.98	3.50	9.43
	B42	5.91	2.89	6.14	7.53	3.15	1.84	4.82
	B43	2.03	0.00	2.42	2.02	1.97	0.00	0.00
Amido	B44	59.95	55.99	64.78	41.26	22.12	14.78	41.75
	B45	53.72	48.06	60.97	35.75	15.42	12.40	40.99
	B46	7.73	5.96	5.29	6.62	4.32	4.44	3.33
	B47	2.12	1.90	2.50	1.63	0.00	1.87	1.91

Figure S4. Results from base screen using aryl halide **Br-1** (Manuscript Figure 3A, Step 2). Numbers are product LCAPs (UV254).

		tBuXPhos Pd G3 (P2)	XPhos Pd G4 (P8)	CPhos Pd G4 (P12)	tBuPhCPhos Pd G4 (P13)	AdBrettPhos Pd G6 (P17)	AlPhos Pd G6 (P21)	GPhos Pd G6 (P22)
Siloxo HMDS	B1	38.61	23.36	71.64	74.81	7.25	6.13	70.19
	B2	6.91	6.87	46.56	53.83	0.00	2.45	11.28
	B3	9.21	11.01	61.61	64.44	0.00	0.00	32.71
	B4	11.89	14.26	64.16	70.08	9.77	7.05	50.15
	B5	0.00	0.00	0.00	0.09	0.00	0.20	0.00
	B6	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	B7	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	B8	3.37	2.03	10.82	0.00	0.00	1.35	3.83
	B9	2.63	0.00	0.47	0.45	0.30	0.13	0.00
Alkoxy	B10	9.96	26.30	66.25	70.06	2.70	2.65	23.70
	B11	9.45	25.53	71.41	69.45	3.15	2.71	24.03
	B12	9.41	0.00	56.16	62.69	3.15	3.41	27.80
Phenoxy	B13	56.08	34.28	52.04	60.71	3.07	3.17	48.67
Organic	B14	0.09	0.00	0.00	0.06	0.00	0.00	0.00
	B15	5.77	0.06	0.15	0.86	0.27	0.22	0.00
	B16	29.85	3.57	9.36	14.64	0.40	0.41	1.18
	B17	23.09	0.00	7.74	12.59	0.48	0.00	0.59
	B18	0.00	0.09	0.73	1.19	0.00	0.00	0.05
	B19	0.00	0.00	0.00	0.06	0.12	0.00	0.00
	B20	0.05	0.00	0.00	0.07	0.15	0.00	0.00
	B21	0.09	0.00	0.00	0.06	0.13	0.00	0.00
	B22	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	B23	0.00	0.00	0.00	0.00	0.14	0.15	0.00
	B24	0.05	0.00	0.17	0.00	0.00	0.14	0.00
	B25	0.00	0.00	0.00	0.00	0.16	0.00	0.00
	B26	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	B27	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	B28	0.33	0.00	0.20	0.00	0.08	0.09	0.17
	B29	0.00	0.00	0.00	0.19	0.00	0.00	0.00
	B30	1.19	0.17	0.40	0.89	0.08	0.08	0.00
	B31	6.59	0.00	0.00	1.23	0.00	0.00	0.00
	B32	0.00	0.06	0.00	0.00	0.31	0.00	0.07
	B33	31.83	4.22	11.45	14.26	0.50	0.48	0.88
Phosphazene	B34	28.71	4.43	10.91	14.66	0.44	0.49	0.98
	B35	9.70	22.78	53.83	57.05	0.00	0.00	18.97
	B36	39.79	3.29	0.00	23.38	2.06	0.00	2.47
	B37	29.69	6.86	13.68	19.22	0.55	0.56	1.07
	B38	1.34	0.00	2.00	2.28	0.00	0.00	0.00
	B39	7.04	6.30	8.16	8.14	0.00	0.20	0.00
	B40	11.63	5.35	10.57	0.00	0.33	0.31	0.87
Verkade's superbase	B41	1.00	0.00	0.36	0.24	1.70	1.90	0.00
	B42	0.00	0.00	2.31	0.62	0.00	0.07	0.00
	B43	0.00	0.00	0.12	0.00	0.00	0.00	0.00
	B44	7.43	8.91	65.36	65.39	3.23	0.00	11.09
Amido	B45	6.38	4.24	54.63	47.68	1.97	2.15	0.00
	B46	0.00	0.00	0.21	0.00	0.00	0.00	0.83
	B47	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Figure S5. Results from base screen using aryl halide **Br-2** (Manuscript Figure 3A, Step 2). Numbers are product LCAPs (UV254).

Nanoscale procedure for base screen using Informer X2 (Manuscript Figure 3A, Step 2)



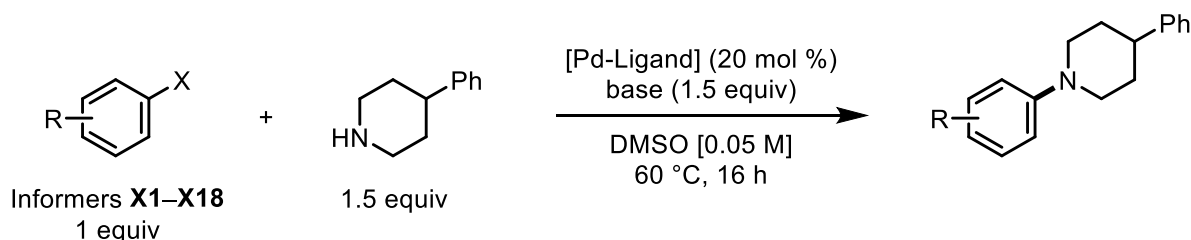
In a nitrogen-filled glovebox, to a 1536 plastic COC plate (Corning Echo™ qualified, Cat. No. 3730, Cyclic Olefin-Copolymer COC, 12.5 µL-wells, flat bottom, clear) using the TTP Labtech Mosquito robot, was added Informer aryl bromide **X2** (1.0 µL, 0.1 M solution in DMSO, 0.1 µmol, 1.0 equiv). The solvent was removed using a Genevac EZ2 (full vacuum, 40 °C, 2 h). To the 1536

plate containing **X2**, using the Mosquito robot, was added 4-phenylpiperidine (0.6 μ L, 0.25 M solution in DMSO, 0.15 μ mol, 1.5 equiv), precatalyst (0.8 μ L, 0.025 M mixture in DMSO, 0.02 μ mol, 0.2 equiv), and base (0.6 μ L, 0.25 M solution in DMSO, 0.15 μ mol, 1.5 equiv). Once the dosing was completed, the reaction plate was heat sealed, centrifuged, enclosed in an aluminum block, and taken out of the glovebox. The aluminum block containing the reaction plate was placed between two preheated thermocouples (bottom thermocouple at 50 °C and top thermocouple at 70 °C). This setup was agitated at 10 G using the Resodyn acoustic mixing LabRam (PharmaRAM II Mixer) for 16 h. The aluminum block containing the reaction plate was cooled to room temperature and taken into the glovebox. The reaction plate was taken out of the aluminum block and centrifuged. The seal was removed, and the reaction mixture was transferred using 200 μ L of 1% AcOH/DMSO into a 384 flat-bottom COC (Cyclic-Olefin Copolymer) analytical plate using the Thermo Matrix Platemate robot. The analytical plate was heat sealed in the glovebox and the reaction mixtures were analyzed using UPLC-UV/MS to obtain the product LCAPs.

		tBuXPhos Pd G3 (P2)	XPhos Pd G4 (P8)	CPhos Pd G4 (P12)	tBuPhCPhos Pd G4 (P13)	AdBrettPhos Pd G6 (P17)	AlPhos Pd G6 (P21)	GPhos Pd G6 (P22)
Siloxo HMDS	B1	44.88	56.83	76.83	41.39	0.51	0.38	49.77
	B2	1.02	3.02	6.96	0.00	1.17	0.86	4.94
	B3	0.16	0.00	0.00	0.00	0.00	0.82	0.30
	B4	0.24	0.00	7.86	1.00	0.75	0.72	1.29
	B5	0.85	0.16	0.09	10.22	0.13	0.43	0.08
	B6	0.51	13.89	0.00	8.63	0.44	1.98	3.20
	B7	0.41	0.00	0.00	13.57	0.00	0.00	0.00
	B8	2.72	8.60	21.84	5.05	2.08	0.20	5.57
	B9	0.09	0.00	0.00	22.22	2.51	0.00	3.41
Alkoxy	B10	9.31	37.41	40.03	0.66	0.26	0.29	1.81
	B11	0.00	0.00	0.00	0.00	1.93	14.91	0.00
	B12	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Phenoxy	B13	7.00	22.34	31.97	16.65	0.84	0.96	6.30
	B14	1.54	13.43	11.73	0.13	0.44	0.25	0.63
Organic	B15	6.07	10.10	0.31	1.47	0.33	0.24	1.57
	B16	37.32	11.47	25.10	13.01	0.29	0.21	3.21
	B17	0.75	0.29	0.15	11.61	0.28	0.09	0.81
	B18	2.52	11.53	1.13	0.89	0.27	0.14	1.00
	B19	0.06	9.78	0.00	8.57	0.10	0.24	1.15
	B20	0.10	0.07	0.10	0.10	0.16	0.07	1.10
	B21	0.16	14.79	0.15	0.10	0.14	0.08	1.09
	B22	0.11	12.26	0.00	8.33	0.11	0.10	0.84
	B23	0.17	0.00	0.00	0.00	0.31	0.11	0.96
	B24	1.80	0.00	0.00	0.00	0.68	0.45	1.13
	B25	0.22	0.00	12.89	5.85	0.16	0.00	0.00
	B26	0.07	0.00	0.00	0.00	0.42	0.15	1.11
	B27	0.13	0.07	13.44	12.01	0.12	0.10	0.84
	B28	0.33	0.06	16.51	0.33	0.25	0.24	0.13
	B29	5.88	9.45	11.76	10.16	0.27	0.12	1.85
	B30	1.97	7.44	0.68	6.22	0.27	0.00	0.16
	B31	8.29	0.27	0.32	1.96	0.32	0.18	1.06
	B32	5.12	0.00	0.00	3.00	0.31	0.18	1.18
	B33	36.21	6.47	35.89	20.49	0.26	0.21	3.87
	B34	48.44	5.67	37.20	14.97	0.29	0.25	5.75
Phosphazene	B35	20.47	64.10	65.99	38.20	0.37	0.63	4.25
	B36	41.97	6.52	22.42	14.10	0.26	0.33	4.11
	B37	17.20	10.64	15.04	10.85	0.08	0.00	1.34
	B38	1.87	0.39	13.41	2.50	0.26	0.09	0.73
	B39	8.38	4.24	3.47	4.51	0.38	0.11	0.92
	B40	11.13	10.38	11.59	8.23	0.07	0.06	1.32
	B41	0.43	0.52	0.55	11.68	0.23	0.30	1.35
Verkade's superbase	B42	0.00	9.05	0.17	7.30	0.16	0.07	0.56
	B43	0.00	0.00	0.00	0.00	0.00	0.12	0.70
Amido	B44	7.47	3.36	17.30	3.34	1.13	1.07	4.58
	B45	0.39	3.12	7.07	0.00	0.98	1.14	4.14
	B46	0.70	1.87	7.93	0.00	1.37	0.00	4.06
	B47	0.07	0.00	0.00	0.00	0.00	0.00	0.00

Figure S6. Results from base screen using Informer aryl bromide **X2** (Manuscript Figure 3A, Step 2). Numbers are product LCAPs (UV254).

Nanoscale procedure for generality screen using Informer halides (Manuscript Figure 3A, Step 3)



In a nitrogen-filled glovebox, to a 1536 plastic COC plate (Corning Echo™ qualified, Cat. No. 3730, Cyclic Olefin-Copolymer COC, 12.5 μL-wells, flat bottom, clear) using the TTP Labtech Mosquito robot, was added Informer halide (0.25 μL, 0.2 M solution in DMSO, 0.05 μmol, 1.0 equiv), 4-phenylpiperidine (0.21 μL, 0.35 M solution in DMSO, 0.075 μmol, 1.5 equiv),

precatalyst (0.33 μ L, 0.03 M mixture in DMSO, 0.01 μ mol, 0.2 equiv), and base (0.27 μ L, 0.28 M solution in DMSO, 0.075 μ mol, 1.5 equiv). Once the dosing was completed, the reaction plate was heat sealed, centrifuged, enclosed in an aluminum block, and taken out of the glovebox. The aluminum block containing the reaction plate was placed between two preheated thermocouples (bottom thermocouple at 50 $^{\circ}$ C and top thermocouple at 70 $^{\circ}$ C). This setup was agitated at 10 G using the Resodyn acoustic mixing LabRam (PharmaRAM II Mixer) for 16 h. The aluminum block containing the reaction plate was cooled to room temperature and taken into the glovebox. The reaction plate was taken out of the aluminum block and centrifuged. The seal was removed, and the reaction mixture was transferred using 100 μ L of 1% AcOH/DMSO into a 384 flat-bottom COC (Cyclic-Olefin Copolymer) analytical plate using the Thermo Matrix Platemate robot. The analytical plate was heat sealed in the glovebox and the reaction mixtures were analyzed using UPLC-UV/MS to obtain the product LCAPs.

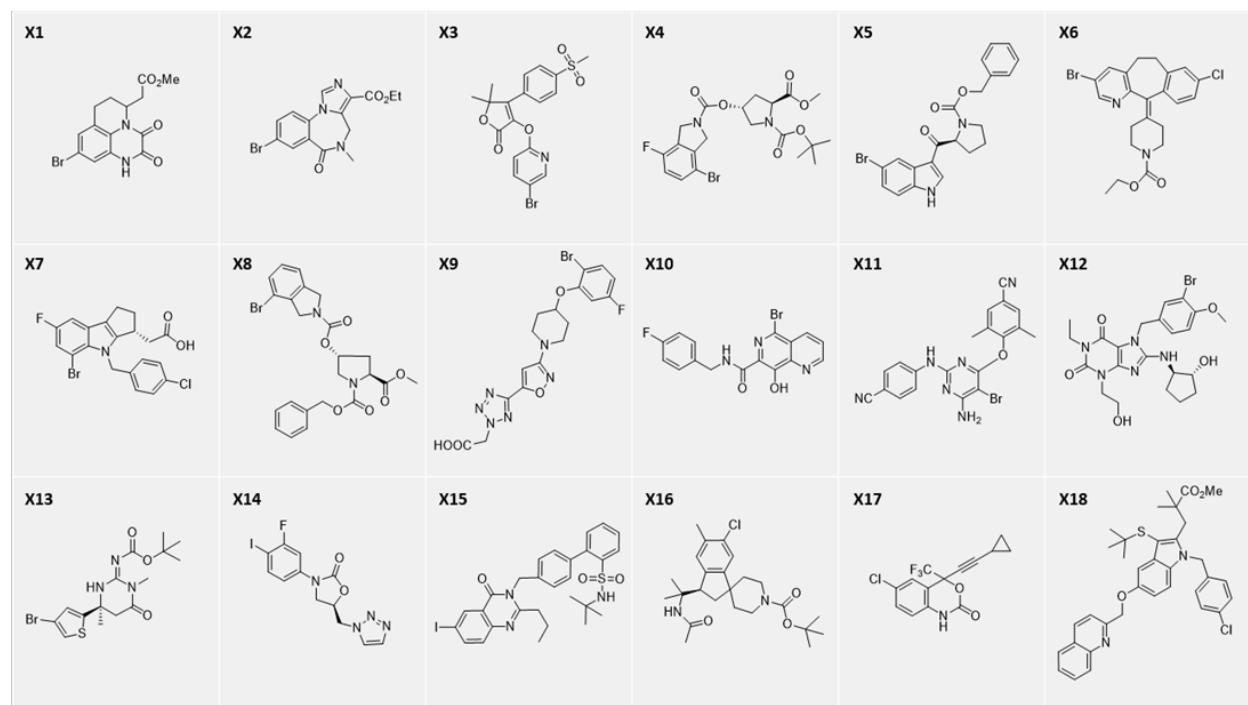


Figure S7. Structures of the Informer halides (Manuscript Figure 3A, Step 3).

<u>Pd precatalysts</u>	<u>Bases</u>
tBuXPhos Pd G3 (P2)	LiOTMS (B1)
XPhos Pd G4 (P8)	LiHMDS (B2)
CPhos Pd G4 (P12)	LiOtBu (B10)
<i>t</i> BuPhCPhos Pd G4 (P13)	NaOtBu (B11)
AdBrettPhos Pd G6 (P17)	NaBHT (B13)
AlPhos Pd G6 (P21)	MTBD (B16)
GPhos Pd G6 (P22)	P ₂ Et (B35)
	P ₁ -H (B36)

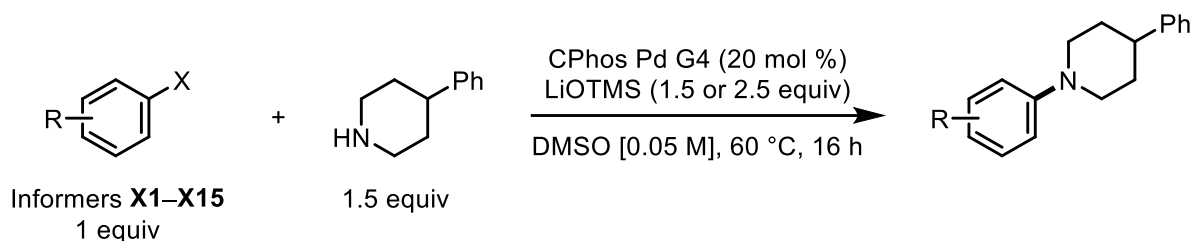
Figure S8. Precatalysts and bases used for generality screen (Manuscript Figure 3A, Step 3).

		tBuXPhos Pd G3 (P2)	XPhos Pd G4 (P8)	CPhos Pd G4 (P12)	tBuPhCPhos Pd G4 (P13)	AdBrettPhos Pd G6 (P17)	AlPhos Pd G6 (P21)	GPhos Pd G6 (P22)
X1	LIOTMS (B1)	25.39	18.14	3.28	35.24	6.02	0.00	11.94
	LIHMDS (B2)	1.01	0.00	0.00	0.00	0.00	0.00	0.00
	LIOTBu (B10)	3.23	0.00	0.00	0.00	0.00	0.00	0.00
	NaOTBu (B11)	2.85	0.00	0.00	0.00	0.00	0.00	0.00
	NaBHT (B13)	42.43	0.00	0.00	32.45	0.00	0.00	17.95
	MTBD (B16)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	P2-Et (B35)	0.02	0.00	0.00	0.00	0.00	0.00	0.00
	P1-H (B36)	0.55	0.00	0.00	0.00	0.00	0.00	0.00
X2	LIOTMS (B1)	10.80	30.78	59.72	4.49	0.34	0.15	38.47
	LIHMDS (B2)	1.27	2.10	9.23	1.98	0.10	0.12	1.98
	LIOTBu (B10)	18.60	57.34	50.23	11.18	0.14	0.00	4.97
	NaOTBu (B11)	3.10	7.92	6.18	2.98	0.41	0.24	0.00
	NaBHT (B13)	44.07	52.07	67.01	33.57	0.00	0.04	11.33
	MTBD (B16)	33.94	9.23	12.30	12.70	0.19	0.31	1.59
	P2-Et (B35)	15.28	60.11	55.60	17.11	0.16	0.30	2.60
	P1-H (B36)	45.70	7.54	10.45	12.57	0.00	0.00	2.23
X3	LIOTMS (B1)	72.87	42.98	54.55	17.40	0.00	0.00	71.63
	LIHMDS (B2)	0.00	1.50	11.86	2.70	0.00	0.00	9.61
	LIOTBu (B10)	13.21	9.24	28.23	6.09	0.00	0.00	33.46
	NaOTBu (B11)	5.52	2.36	6.19	1.81	0.00	0.00	10.08
	NaBHT (B13)	56.65	27.86	55.19	17.94	0.00	0.00	48.95
	MTBD (B16)	54.65	3.23	16.91	12.32	0.00	0.00	0.00
	P2-Et (B35)	58.57	14.79	34.53	11.76	0.00	0.00	35.56
	P1-H (B36)	41.19	2.21	14.20	10.84	0.00	0.00	4.25
X4	LIOTMS (B1)	44.67	6.20	28.16	25.59	0.02	0.00	23.42
	LIHMDS (B2)	2.25	0.00	9.60	3.93	0.00	0.00	5.74
	LIOTBu (B10)	15.98	11.88	28.08	38.58	0.00	0.00	11.59
	NaOTBu (B11)	9.95	0.00	11.77	8.48	0.00	0.00	4.35
	NaBHT (B13)	30.12	17.96	44.45	48.26	0.00	0.00	4.39
	MTBD (B16)	18.03	2.24	11.56	15.26	0.44	0.00	0.00
	P2-Et (B35)	19.14	16.45	44.40	46.19	0.13	0.00	7.23
	P1-H (B36)	18.40	2.64	9.91	12.74	0.00	0.00	1.99
X5	LIOTMS (B1)	3.94	15.92	12.81	21.50	0.00	0.00	0.00
	LIHMDS (B2)	0.92	0.00	10.33	13.50	0.61	0.07	0.00
	LIOTBu (B10)	1.80	15.30	10.25	15.15	0.00	0.00	0.00
	NaOTBu (B11)	2.13	15.58	13.95	15.40	0.00	0.00	3.78
	NaBHT (B13)	3.81	0.00	0.00	15.38	0.00	0.00	0.00
	MTBD (B16)	0.00	0.00	0.00	0.45	0.00	0.00	0.00
	P2-Et (B35)	2.09	14.97	10.66	16.33	0.00	0.00	0.00
	P1-H (B36)	0.28	0.00	0.00	0.45	0.00	0.00	0.00
X6	LIOTMS (B1)	61.43	52.59	73.61	41.83	1.29	0.00	54.23
	LIHMDS (B2)	17.44	26.82	45.96	21.88	0.00	0.00	0.00
	LIOTBu (B10)	31.98	42.29	56.44	32.97	0.00	0.00	11.52
	NaOTBu (B11)	29.00	21.21	44.00	28.11	0.00	0.00	9.39
	NaBHT (B13)	43.30	36.65	47.17	46.19	0.00	1.31	26.59
	MTBD (B16)	14.07	0.00	10.16	10.22	0.94	1.37	1.80
	P2-Et (B35)	19.80	41.27	44.51	29.55	0.00	0.00	0.00
	P1-H (B36)	14.38	3.27	0.00	11.10	0.00	0.87	0.00
X7	LIOTMS (B1)	3.76	0.00	5.03	2.96	0.00	0.00	0.00
	LIHMDS (B2)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	LIOTBu (B10)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	NaOTBu (B11)	7.27	0.00	0.00	0.00	0.00	0.00	0.00
	NaBHT (B13)	0.00	0.00	0.00	2.96	0.00	0.00	0.00
	MTBD (B16)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	P2-Et (B35)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	P1-H (B36)	0.00	0.00	0.00	0.00	0.00	0.07	0.00
X8	LIOTMS (B1)	33.34	30.59	54.56	32.50	0.00	0.00	58.31
	LIHMDS (B2)	0.00	6.19	21.45	8.04	0.00	0.00	17.41
	LIOTBu (B10)	12.86	23.48	50.87	47.31	0.00	0.00	32.03
	NaOTBu (B11)	2.45	2.48	17.45	20.47	0.00	0.00	18.88
	NaBHT (B13)	29.30	38.89	59.89	43.18	0.00	0.00	4.47
	MTBD (B16)	19.93	6.84	19.03	18.97	0.29	0.00	1.63
	P2-Et (B35)	17.09	42.78	65.62	60.56	0.00	0.00	0.02
	P1-H (B36)	17.18	9.05	18.32	20.02	0.00	0.00	3.34
X9	LIOTMS (B1)	53.37	0.00	23.64	37.82	0.49	0.00	2.19
	LIHMDS (B2)	1.32	0.00	0.00	0.00	0.00	0.00	0.00
	LIOTBu (B10)	4.28	0.00	0.80	4.39	0.00	0.00	0.00
	NaOTBu (B11)	6.23	0.00	1.85	4.37	0.00	0.00	0.22
	NaBHT (B13)	8.83	0.00	1.62	11.04	0.00	0.00	0.00
	MTBD (B16)	0.00	0.00	0.00	0.44	0.00	0.00	0.00
	P2-Et (B35)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	P1-H (B36)	0.00	0.00	0.00	0.00	0.00	0.00	0.00

		tBuXPhos Pd G3 (P2)	XPhos Pd G4 (P8)	CPhos Pd G4 (P12)	tBuPhCPhos Pd G4 (P13)	AdBrettPhos Pd G6 (P17)	AlPhos Pd G6 (P21)	GPhos Pd G6 (P22)
X10	LIOTMS (B1)	0.15	0.00	2.63	0.09	1.64	0.06	0.03
	LIHMDS (B2)	0.09	0.05	0.00	0.00	0.00	0.00	0.00
	LIOTBu (B10)	0.00	0.00	0.12	0.00	0.05	0.01	0.05
	NaOTBu (B11)	0.43	0.00	0.63	0.36	0.06	0.07	0.00
	NaBHT (B13)	0.00	0.00	0.00	0.05	0.67	0.05	0.00
	MTBD (B16)	0.09	0.09	0.04	0.08	0.11	0.06	0.04
	P2-Et (B35)	0.00	0.11	0.10	0.00	0.00	0.00	0.06
X11	P1-H (B36)	0.09	0.06	0.10	0.00	0.21	0.00	0.01
	LIOTMS (B1)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	LIHMDS (B2)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	LIOTBu (B10)	0.35	0.00	0.00	0.00	0.00	0.00	0.00
	NaOTBu (B11)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	NaBHT (B13)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	MTBD (B16)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
X12	P2-Et (B35)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	P1-H (B36)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	LIOTMS (B1)	41.47	6.14	67.79	79.21	3.21	0.00	28.05
	LIHMDS (B2)	26.24	3.84	49.16	71.63	3.25	0.00	14.52
	LIOTBu (B10)	37.01	4.28	51.52	76.47	0.00	0.00	16.81
	NaOTBu (B11)	34.54	0.00	59.80	74.58	1.84	0.14	20.79
	NaBHT (B13)	45.82	3.75	35.34	75.25	1.19	0.00	22.14
X13	MTBD (B16)	0.00	0.00	1.89	8.30	1.91	0.04	0.96
	P2-Et (B35)	29.29	4.68	57.37	71.87	1.36	0.00	11.84
	P1-H (B36)	6.40	0.00	3.46	41.56	0.00	0.00	1.08
	LIOTMS (B1)	50.83	2.24	52.96	17.06	6.81	25.96	82.37
	LIHMDS (B2)	6.30	1.14	1.73	2.28	0.00	0.67	0.00
	LIOTBu (B10)	9.22	0.12	3.14	7.18	0.00	1.47	0.00
	NaOTBu (B11)	8.48	1.75	5.05	7.16	0.00	9.03	8.40
X14	NaBHT (B13)	5.41	1.37	1.89	0.68	4.29	26.52	12.80
	MTBD (B16)	0.00	0.00	0.00	2.57	0.00	1.20	0.00
	P2-Et (B35)	5.69	0.00	3.43	8.35	4.99	0.00	5.31
	P1-H (B36)	2.04	0.00	1.46	0.00	0.92	3.69	1.33
	LIOTMS (B1)	31.61	5.51	43.24	34.80	16.54	1.48	45.54
	LIHMDS (B2)	2.73	0.00	3.65	9.71	0.00	0.00	0.00
	LIOTBu (B10)	3.32	0.99	8.75	13.69	0.00	0.00	0.00
X15	NaOTBu (B11)	1.00	0.00	0.43	1.73	0.00	0.00	0.23
	NaBHT (B13)	52.75	6.89	45.89	56.57	14.41	0.00	8.94
	MTBD (B16)	28.60	3.16	14.22	24.73	1.95	0.00	1.55
	P2-Et (B35)	7.26	0.00	3.35	46.02	0.00	0.00	1.39
	P1-H (B36)	19.17	0.00	12.05	26.57	6.34	2.37	1.74
	LIOTMS (B1)	34.78	37.49	41.17	41.08	0.00	0.00	25.06
	LIHMDS (B2)	22.77	27.19	26.61	24.79	0.84	0.00	9.73
X16	LIOTBu (B10)	24.09	29.91	31.72	32.07	0.24	0.00	10.77
	NaOTBu (B11)	24.00	26.61	34.20	35.34	0.46	0.27	8.31
	NaBHT (B13)	28.63	36.78	35.49	29.82	0.12	0.00	14.03
	MTBD (B16)	16.09	12.64	11.38	11.25	0.00	0.00	2.11
	P2-Et (B35)	21.68	25.63	26.75	27.72	0.36	0.20	7.24
	P1-H (B36)	15.91	10.18	10.90	12.30	0.34	0.00	0.01
	LIOTMS (B1)	9.07	0.15	32.60	9.47	0.00	0.00	34.70
X17	LIHMDS (B2)	0.00	6.35	12.72	0.00	0.00	0.00	2.02
	LIOTBu (B10)	4.59	18.51	20.37	0.00	0.00	0.11	3.43
	NaOTBu (B11)	0.00	0.00	4.98	0.07	0.00	0.09	3.32
	NaBHT (B13)	11.45	14.36	26.15	9.27	0.00	0.00	0.00
	MTBD (B16)	0.00	0.00	4.79	0.00	0.00	0.00	0.00
	P2-Et (B35)	10.90	25.51	43.24	12.18	0.00	0.00	7.93
	P1-H (B36)	0.00	0.00	3.63	2.14	0.00	0.00	0.00
X18	LIOTMS (B1)	0.00	3.00	2.13	1.14	0.00	1.39	0.00
	LIHMDS (B2)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	LIOTBu (B10)	0.00	0.00	3.10	0.00	0.00	0.00	0.00
	NaOTBu (B11)	0.00	0.00	3.61	0.00	0.00	0.00	0.00
	NaBHT (B13)	1.74	0.00	0.00	0.00	0.00	0.00	0.00
	MTBD (B16)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	P2-Et (B35)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
X19	P1-H (B36)	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	LIOTMS (B1)	1.18	5.17	5.69	0.00	0.12	0.83	16.09
	LIHMDS (B2)	0.93	1.25	1.95	0.05	0.62	0.42	1.56
	LIOTBu (B10)	3.57	4.80	4.91	0.00	0.07	1.21	10.21
	NaOTBu (B11)	0.00	0.13	0.13	0.00	0.00	0.16	0.00
	NaBHT (B13)	13.14	36.52	19.75	4.49	0.00	0.72	27.06
	MTBD (B16)	3.72	3.13	4.78	1.66	0.29	0.33	1.86
X20	P2-Et (B35)	4.46	14.15	14.76	2.33	0.08	2.09	15.46
	P1-H (B36)	3.80	0.00	4.03	2.22	0.00	0.59	4.86

Figure S9. Results from generality screen using Informer halides (Manuscript Figure 3A, Step 3). Numbers are product LCAPs (UV254).

Effect of equivalents of base (1.5 vs 2.5) on nanoscale coupling of Informer halides



In a nitrogen-filled glovebox, to a 1536 plastic COC plate (Corning EchoTM qualified, Cat. No. 3730, Cyclic Olefin-Copolymer COC, 12.5 μ L-wells, flat bottom, clear) using the TTP Labtech Mosquito robot, was added Informer halide (0.2 μ L, 0.25 M solution in DMSO, 0.05 μ mol, 1.0 equiv), 4-phenylpiperidine (0.21 μ L, 0.35 M solution in DMSO, 0.075 μ mol, 1.5 equiv), CPhos Pd G4 (0.36 μ L, 0.028 M mixture in DMSO, 0.01 μ mol, 0.2 equiv), and lithium trimethylsilanolate (either 0.27 μ L, 0.28 M solution in DMSO, 0.075 μ mol, 1.5 equiv or 0.28 μ L, 0.45 M solution in DMSO, 0.125 μ mol, 2.5 equiv). Once the dosing was completed, the reaction plate was heat sealed, centrifuged, enclosed in an aluminum block, and taken out of the glovebox. The aluminum block containing the reaction plate was placed between two preheated thermocouples (bottom thermocouple at 50 °C and top thermocouple at 70 °C). This setup was agitated at 10 G using the Resodyn acoustic mixing LabRam (PharmaRAM II Mixer) for 16 h. The aluminum block containing the reaction plate was cooled to room temperature and taken into the glovebox. The reaction plate was taken out of the aluminum block and centrifuged. The seal was removed, and the reaction mixture was transferred using 100 μ L of 1% AcOH/DMSO into a 384 flat-bottom COC (Cyclic-Olefin Copolymer) analytical plate using the Thermo Matrix Platemate robot. The analytical plate was heat sealed in the glovebox and the reaction mixtures were analyzed using UPLC-UV/MS to obtain the product LCAPs.

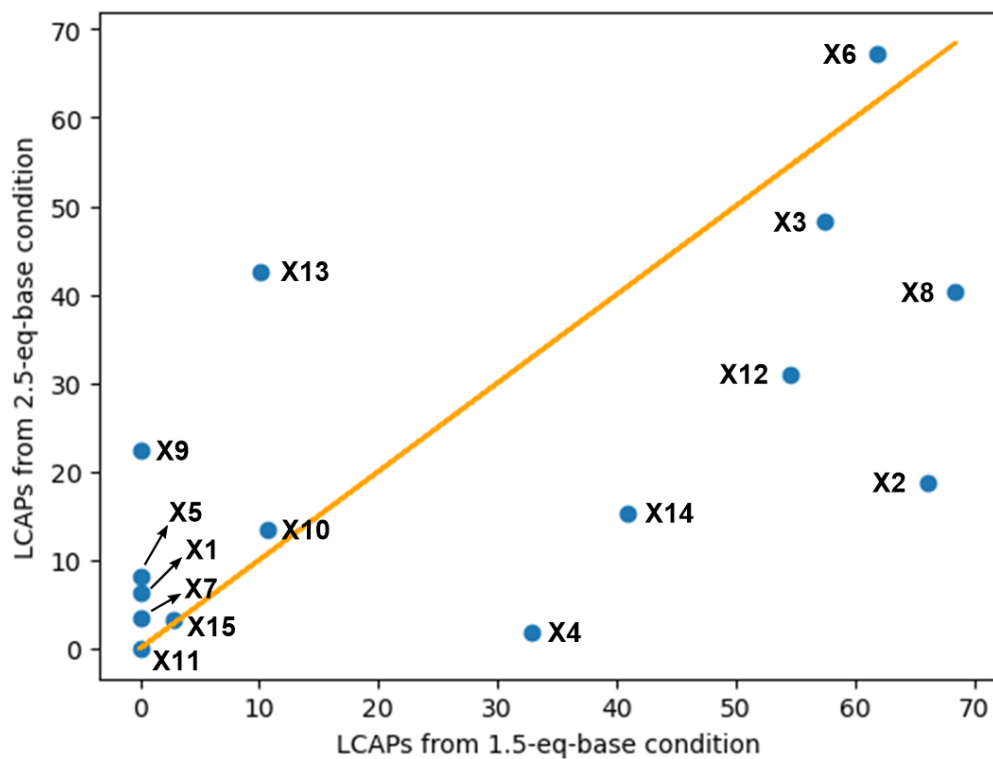


Figure S10. Comparison of LCAPs (average of UV254 and UV210) for Informer halide coupled products using 1.5 or 2.5 equivalents of base.

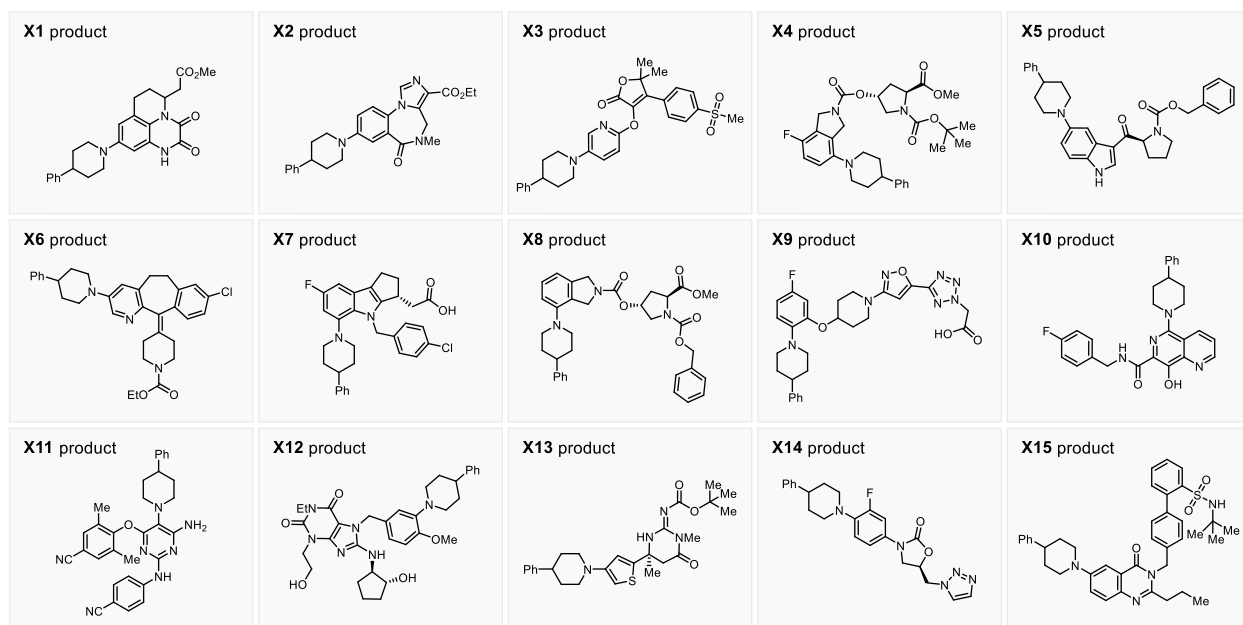


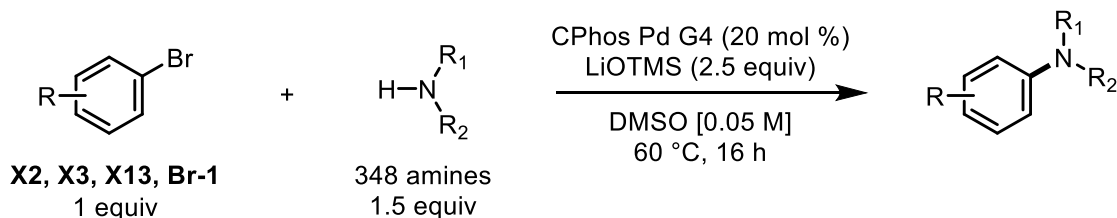
Figure S11. Structures of Informer halide coupled products with 4-phenylpiperidine.

As shown in Figure S10, the equivalents of base used for the coupling has an impact on product LCAPs. For substrates that contain an acidic functional group (e.g., **X9** and **X13**), higher equivalents of base considerably increases the coupling performance leading to higher LCAPs. However, some substrates were found to perform better using a lower equivalents of base (e.g., **X2**, **X4**, and **X14**).

For the amine and aryl halide scope experiments (Manuscript Figure 4B **Part I** and **Part II**), the reaction conditions using 2.5 equiv base was employed because some amines in **Set A** were available only as amine salts and the structural diversity of available amines was prioritized.

III. HTE Procedures for Aryl Halide and Amine Scopes

Nanoscale procedure for aryl halide coupling with diverse amines (Manuscript Figure 4B, Part I)



In a nitrogen-filled glovebox, to a 1536 plastic COC plate (Corning EchoTM qualified, Cat. No. 3730, Cyclic Olefin-Copolymer COC, 12.5 μ L-wells, flat bottom, clear) using the TTP Labtech Mosquito robot, was added aryl halide (0.25 μ L, 0.2 M solution in DMSO, 0.05 μ mol, 1.0 equiv), amine (0.19 μ L, 0.4 M solution in DMSO, 0.075 μ mol, 1.5 equiv), CPhos Pd G4 (0.36 μ L, 0.028 M mixture in DMSO, 0.01 μ mol, 0.2 equiv), and lithium trimethylsilanolate (0.28 μ L, 0.45 M solution in DMSO, 0.125 μ mol, 2.5 equiv). Once the dosing was completed, the reaction plate was heat sealed, centrifuged, enclosed in an aluminum block, and taken out of the glovebox. The aluminum block containing the reaction plate was placed between two preheated thermocouples (bottom thermocouple at 50 °C and top thermocouple at 70 °C). This setup was agitated at 10 G using the Resodyn acoustic mixing LabRam (PharmaRAM II Mixer) for 16 h. The aluminum block containing the reaction plate was cooled to room temperature and taken into the glovebox. The reaction plate was taken out of the aluminum block and centrifuged. The seal was removed, and the reaction mixture was transferred using 100 μ L of 1% AcOH/DMSO into a 384 flat-bottom COC (Cyclic-Olefin Copolymer) analytical plate using the Thermo Matrix Platemate robot. The analytical plate was heat sealed in the glovebox and the reaction mixtures were analyzed using UPLC-UV/MS to obtain the product LCAPs.

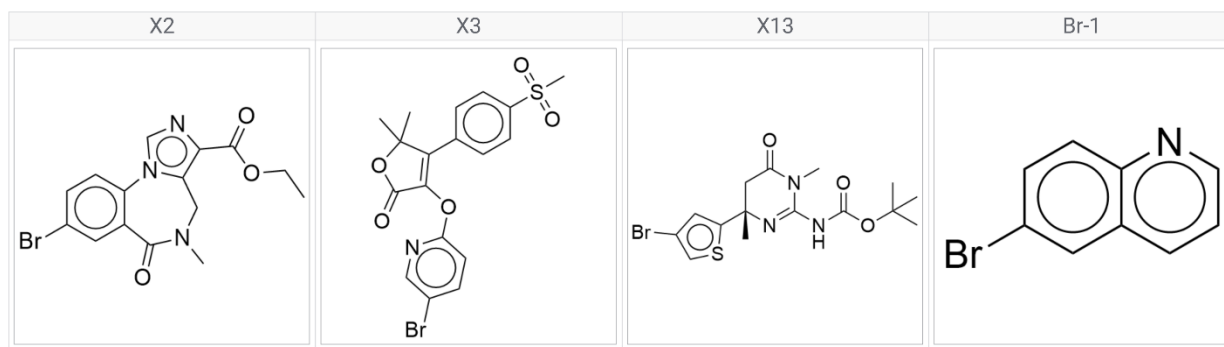


Figure S12. Aryl halide structures used in amine scope (Manuscript Figure 4B, **Part I**).

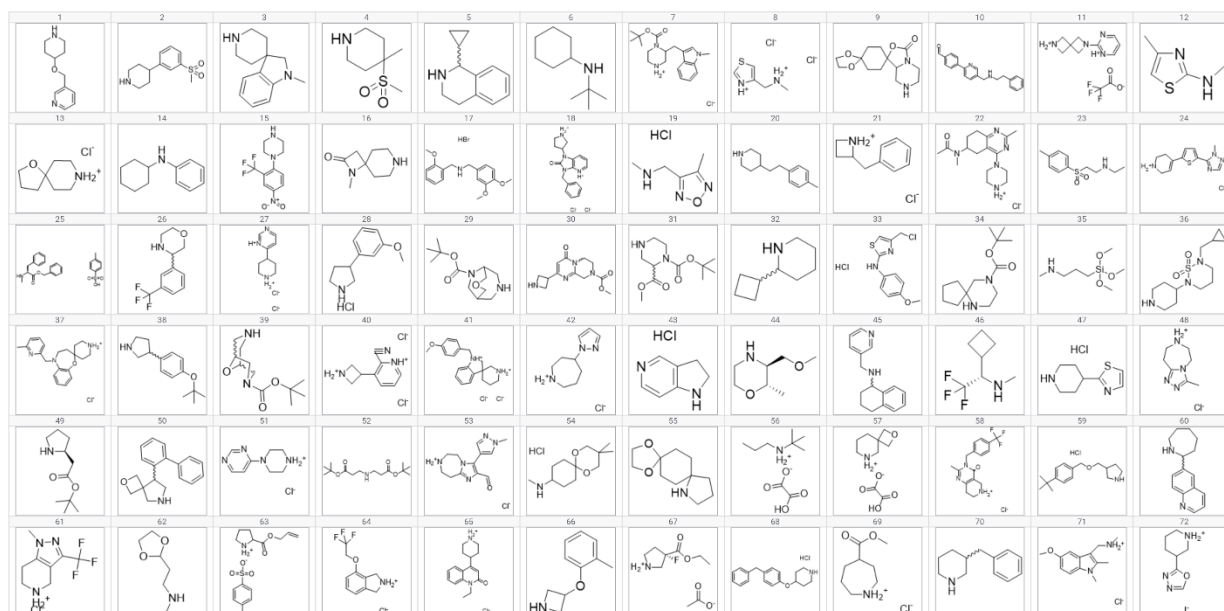
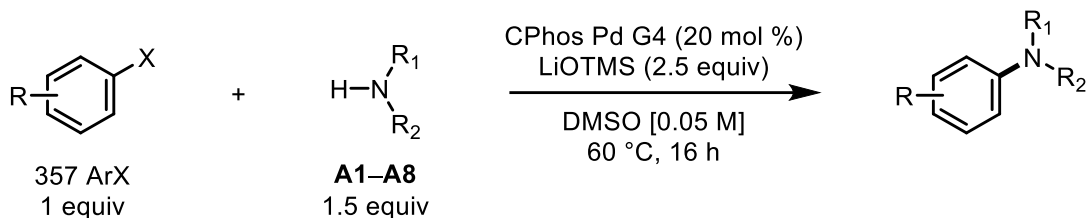






Figure S13. Amine structures used in amine scope (Manuscript Figure 4B, **Part I**).

Nanoscale procedure for amine coupling with diverse aryl halides (Manuscript Figure 4B, Part II)



In a nitrogen-filled glovebox, to a 1536 plastic COC plate (Corning EchoTM qualified, Cat. No. 3730, Cyclic Olefin-Copolymer COC, 12.5 μL -wells, flat bottom, clear) using the TTP Labtech Mosquito robot, was added aryl halide (0.2 μL , 0.25 M solution in DMSO, 0.05 μmol , 1.0 equiv), amine (0.21 μL , 0.35 M solution in DMSO, 0.075 μmol , 1.5 equiv), CPhos Pd G4 (0.36 μL , 0.028 M mixture in DMSO, 0.01 μmol , 0.2 equiv), and lithium trimethylsilanolate (0.28 μL , 0.45 M solution in DMSO, 0.125 μmol , 2.5 equiv). Once the dosing was completed, the reaction plate was heat sealed, centrifuged, enclosed in an aluminum block, and taken out of the glovebox. The aluminum block containing the reaction plate was placed between two preheated thermocouples (bottom thermocouple at 50 $^\circ\text{C}$ and top thermocouple at 70 $^\circ\text{C}$). This setup was agitated at 10 G using the Resodyn acoustic mixing LabRam (PharmaRAM II Mixer) for 16 h. The aluminum block containing the reaction plate was cooled to room temperature and taken into the glovebox. The reaction plate was taken out of the aluminum block and centrifuged. The seal was removed, and the reaction mixture was transferred using 100 μL of 1% AcOH/DMSO into a 384 flat-bottom COC (Cyclic-Olefin Copolymer) analytical plate using the Thermo Matrix Platemate robot. The analytical plate was heat sealed in the glovebox and the reaction mixtures were analyzed using UPLC-UV/MS to obtain the product LCAPs.

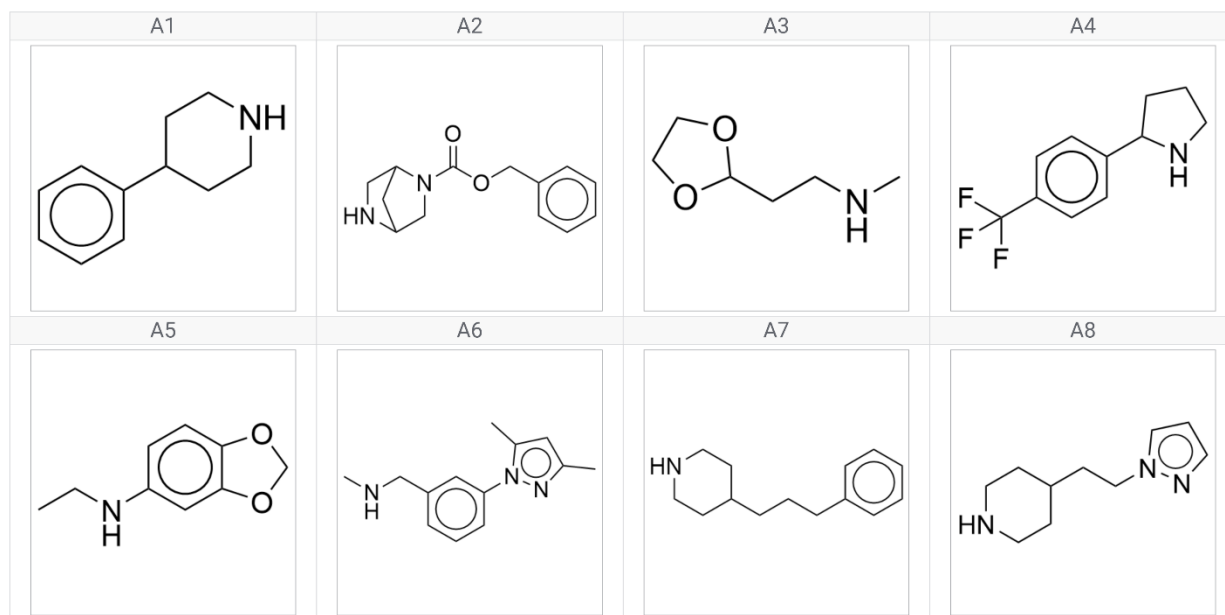
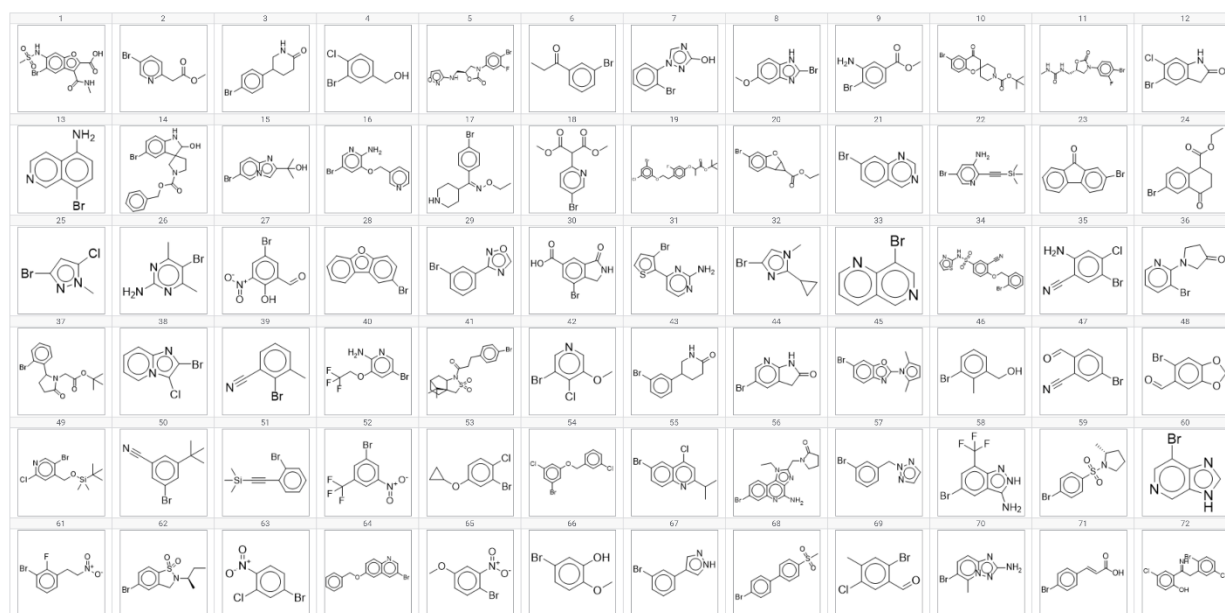


Figure S14. Amine structures used in aryl halide scope (Manuscript Figure 4B, **Part II**).





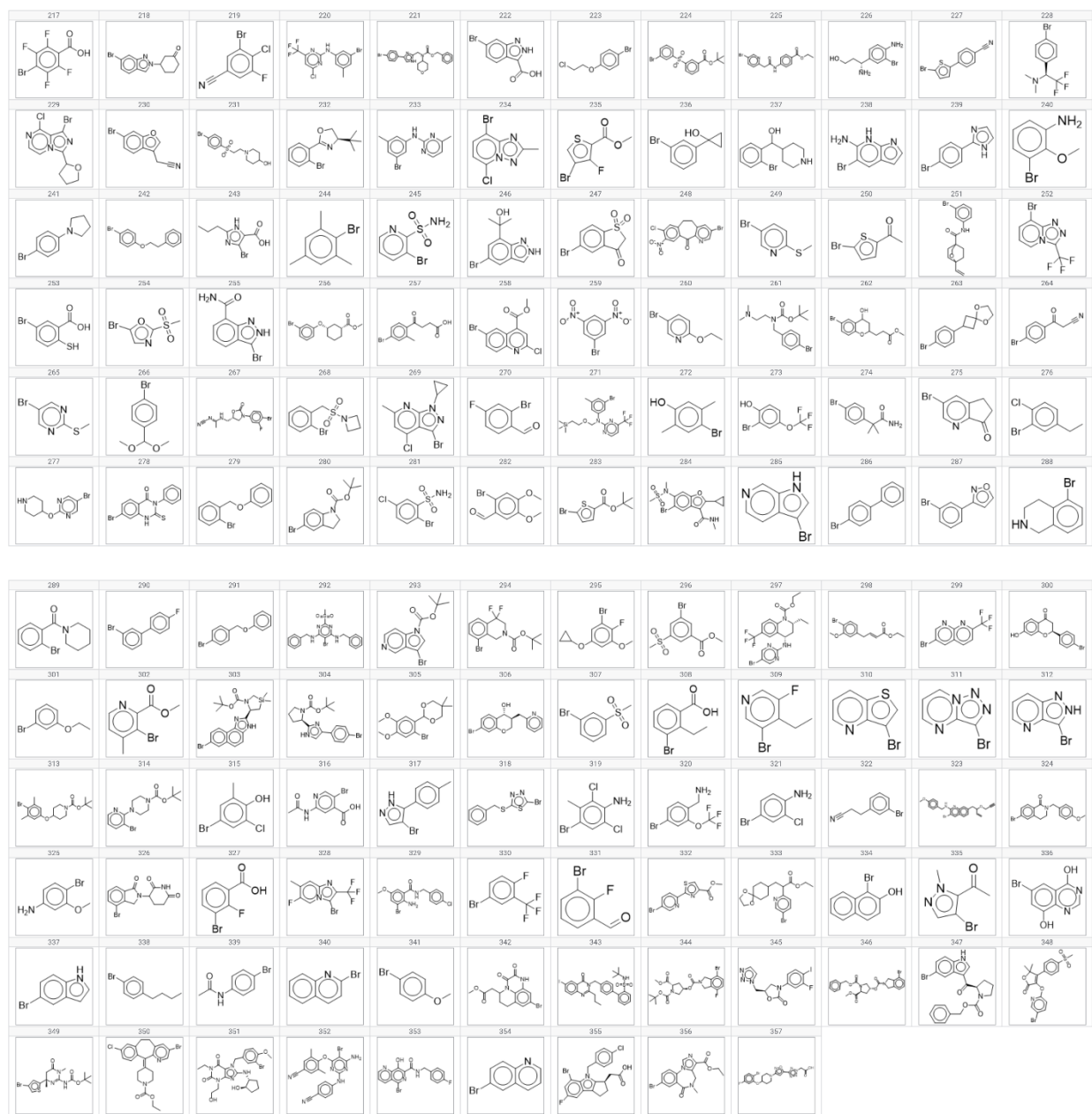
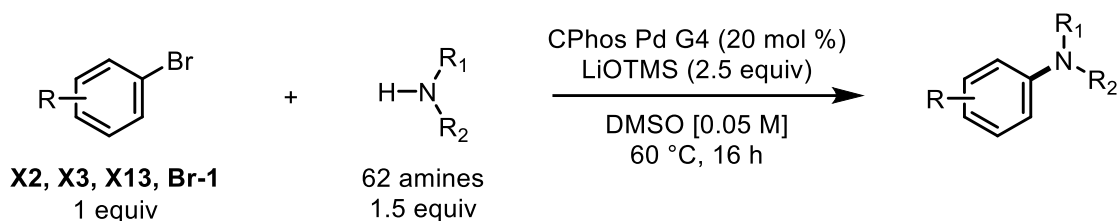


Figure S15. Aryl halide structures used in aryl halide scope (Manuscript Figure 4B, **Part II**).

IV. Procedures for Amine and Aryl Halide Quality Assessment

Nanoscale procedure for amine quality check using C–N coupling



In a nitrogen-filled glovebox, to a 1536 plastic COC plate (Corning EchoTM qualified, Cat. No. 3730, Cyclic Olefin-Copolymer COC, 12.5 μ L-wells, flat bottom, clear) using the TTP Labtech Mosquito robot, was added aryl halide (0.25 μ L, 0.2 M solution in DMSO, 0.05 μ mol, 1.0 equiv), amine (0.19 μ L, 0.4 M solution in DMSO, 0.075 μ mol, 1.5 equiv), CPhos Pd G4 (0.36 μ L, 0.028 M mixture in DMSO, 0.01 μ mol, 0.2 equiv), and lithium trimethylsilanolate (0.28 μ L, 0.45 M solution in DMSO, 0.125 μ mol, 2.5 equiv). Once the dosing was completed, the reaction plate was heat sealed, centrifuged, enclosed in an aluminum block, and taken out of the glovebox. The aluminum block containing the reaction plate was placed between two preheated thermocouples (bottom thermocouple at 50 °C and top thermocouple at 70 °C). This setup was agitated at 10 G using the Resodyn acoustic mixing LabRam (PharmaRAM II Mixer) for 16 h. The aluminum block containing the reaction plate was cooled to room temperature and taken into the glovebox. The reaction plate was taken out of the aluminum block and centrifuged. The seal was removed, and the reaction mixture was transferred using 100 μ L of 1% AcOH/DMSO into a 384 flat-bottom COC (Cyclic-Olefin Copolymer) analytical plate using the Thermo Matrix Platemate robot. The analytical plate was heat sealed in the glovebox and the reaction mixtures were analyzed using UPLC-UV/MS to obtain the product LCAPs.

Note: 62 amines were sourced from Enamine for this experiment – these amines were chosen because they were the only amines from Figure S13 that were commercially available from Enamine.

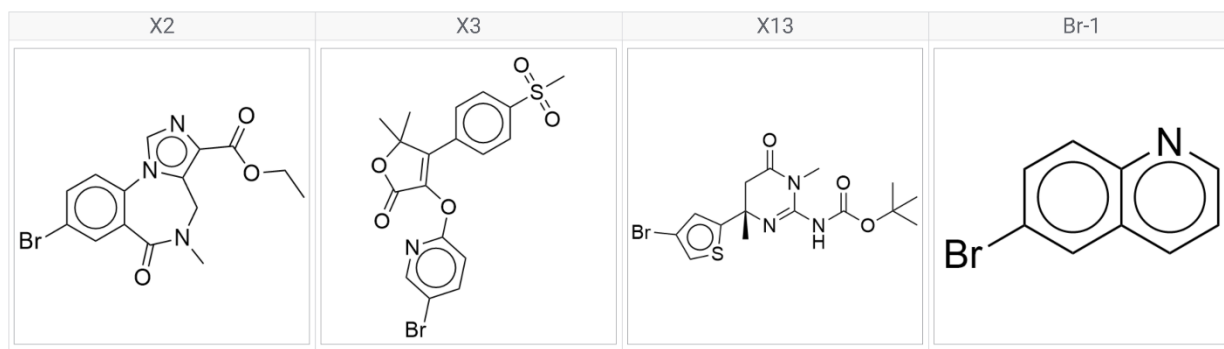


Figure S16. Aryl halide structures used in the amine quality check experiment.

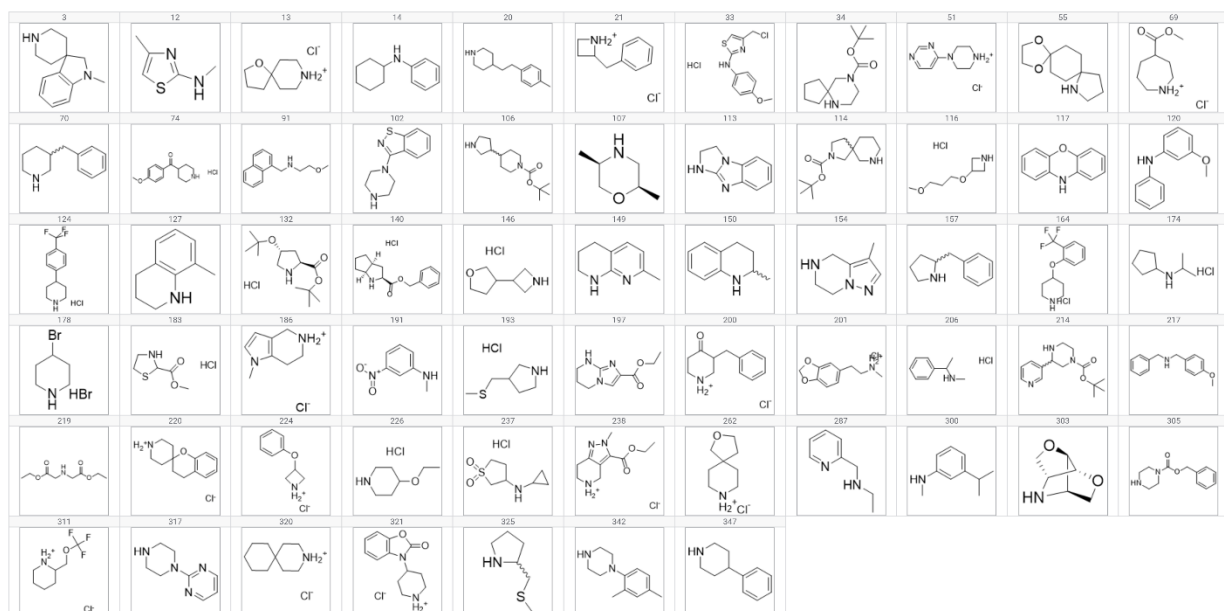


Figure S17. 62 amine structures used in the amine quality check experiment.

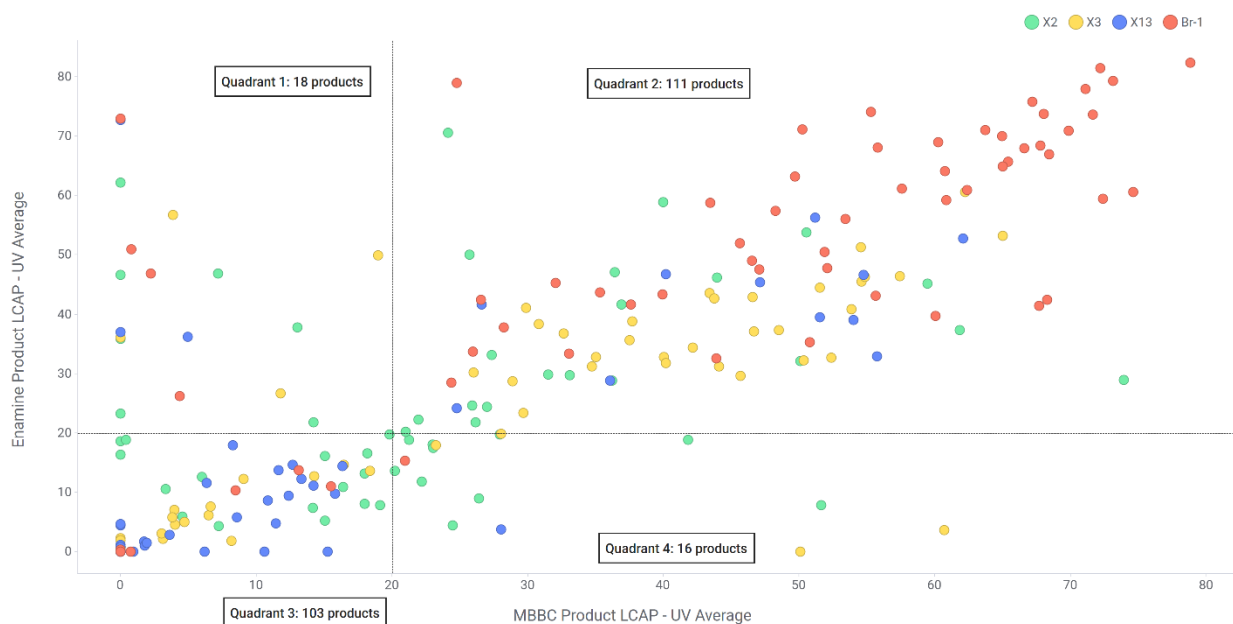


Figure S18. Comparison of product LCAP (average of UV254 and UV210) for C–N couplings using amines sourced from MBBC vs Enamine.

As shown in Figure S18 above, quadrants 1, 2, 3, and 4 contain 18, 111, 103 and 16 products, respectively. Product LCAP is the average of UV254 and UV210 for each individual product. Quadrant 1 consists of products for which the average product LCAP is >20% for the experiment in which the amine was sourced from Enamine but <20% for the experiment in which the same amine was sourced from MBBC. Quadrant 2 consists of products for which the average product LCAP is >20% for both the MBBC and Enamine experiments. Quadrant 3 consists of products for which the average product LCAP is <20% for both the MBBC and Enamine experiments. Quadrant 4 consists of products for which the average product LCAP is >20% for the MBBC experiment but <20% for the Enamine experiment. Figure S19 below shows percentages of matching LCAP classifications (either <20% or >20%) for amines obtained from MBBC vs Enamine. The histogram plot shows the number of amines that fall within each bin for percentage of matching LCAP classifications. In this experiment, each amine was used twice (amine procured from MBBC or amine procured from Enamine) and reacted with 4 different aryl bromide coupling partners. The percentage of reactions (out of 4 reactions) with matching LCAP classifications (MBBC >20% and Enamine >20% or MBBC <20% and Enamine <20%) was calculated. For example, an amine has 75% matching LCAP classifications if 3 out of 4 of its coupling reactions resulted in

classification labels that are both >20% or <20% for MBBC and Enamine experiments (Quadrant 2 or 3). These results show that 38/62 amines have 100% matching classification labels and 18/62 amines have 75% matching classification labels, demonstrating high levels of consistent performance for a majority of the amines tested (i.e., 56/62 (90%) of amines had $\geq 75\%$ matching classification labels).

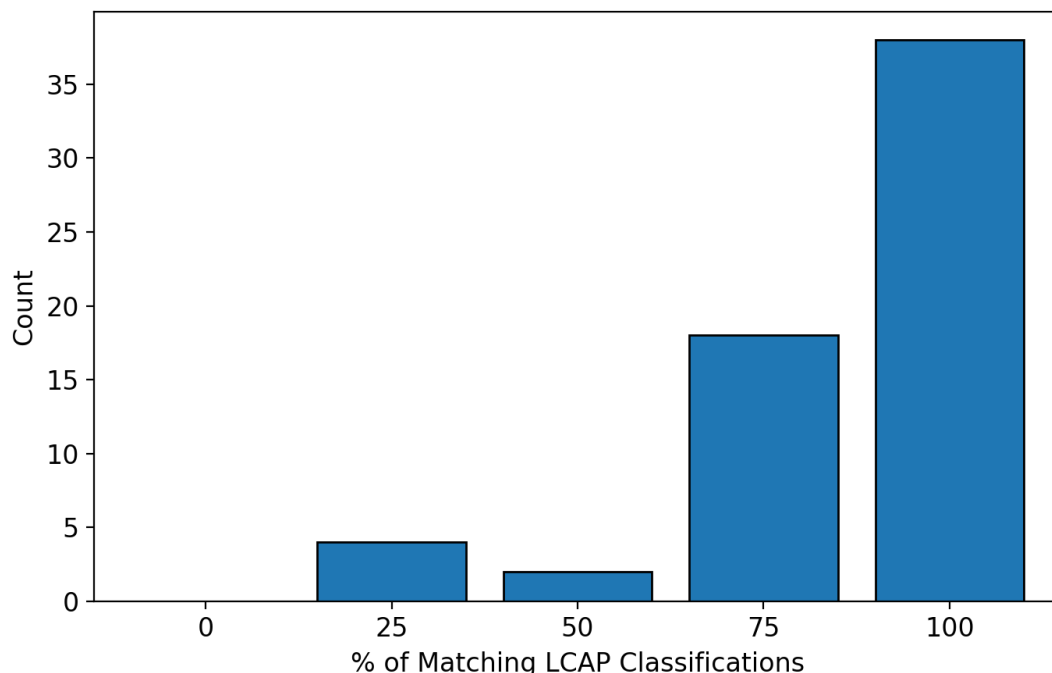
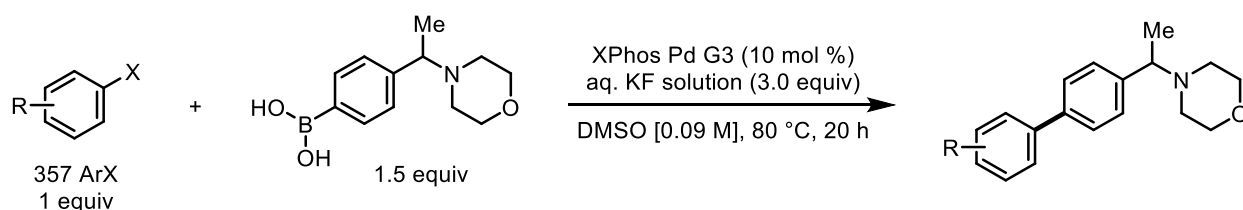


Figure S19. Histogram of percentage of matching LCAP classifications for the MBBC vs Enamine reaction performance with each selected amine.

Nanoscale procedure for aryl halide quality check using Suzuki-Miyaura coupling



In a nitrogen-filled glovebox, to a 1536 plastic COC plate (Corning EchoTM qualified, Cat. No. 3730, Cyclic Olefin-Copolymer COC, 12.5 μ L-wells, flat bottom, clear) using the TTP Labtech Mosquito robot, was added aryl halide (0.40 μ L, 0.25 M solution in DMSO, 0.1 μ mol, 1.0 equiv), (4-(1-Morpholinoethyl)phenyl)boronic acid (0.20 μ L, 0.75 M solution in DMSO, 0.15 μ mol, 1.5

equiv), XPhos Pd G3 (0.3 μ L, 0.033 M mixture in DMSO, 0.01 μ mol, 0.1 equiv), and KF (0.2 μ L, 1.5 M solution in sparged deionized water, 0.3 μ mol, 3.0 equiv). Once the dosing was completed, the reaction plate was heat sealed, centrifuged, enclosed in an aluminum block, and taken out of the glovebox. The aluminum block containing the reaction plate was placed between two preheated thermocouples (bottom thermocouple at 70 $^{\circ}$ C and top thermocouple at 90 $^{\circ}$ C). This setup was agitated at 10 G using the Resodyn acoustic mixing LabRam (PharmaRAM II Mixer) for 20 h. The aluminum block containing the reaction plate was cooled to room temperature and taken into the glovebox. The reaction plate was taken out of the aluminum block and centrifuged. The seal was removed, and the reaction mixture was transferred using 200 μ L of 1% AcOH/DMSO into a 384 flat-bottom COC (Cyclic-Olefin Copolymer) analytical plate using the Thermo Matrix Platemate robot. The analytical plate was heat sealed in the glovebox and the reaction mixtures were analyzed using UPLC-UV/MS to obtain the product LCAPs.

Note: The 357 aryl halide stock solutions used for this experiment were the same stock solutions used for the aryl halide scope experiment described above (Manuscript Figure 3B, **Part II**).

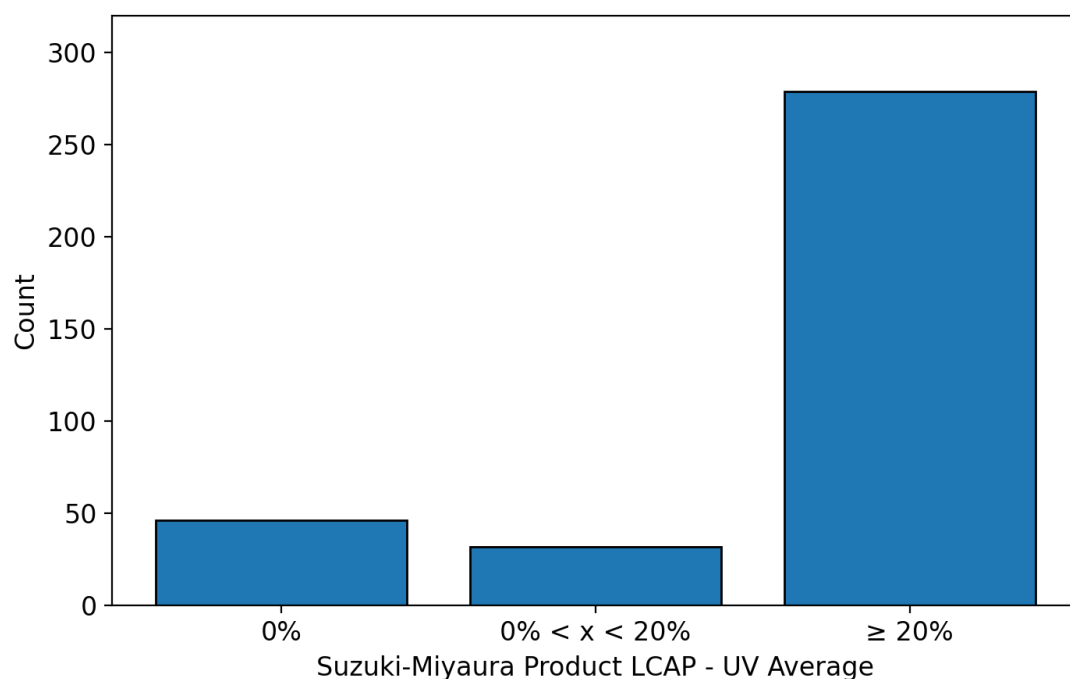


Figure S20. Bar chart showing binned product LCAPs for the Suzuki-Miyaura coupling reactions. Product LCAP is the average of UV254 and UV210.

As shown in Figure S20 above, a high majority of aryl halides (279/357) led to $\geq 20\%$ product LCAP for the Suzuki-Miyaura coupling reactions, providing confidence for the identity of each aryl halide.

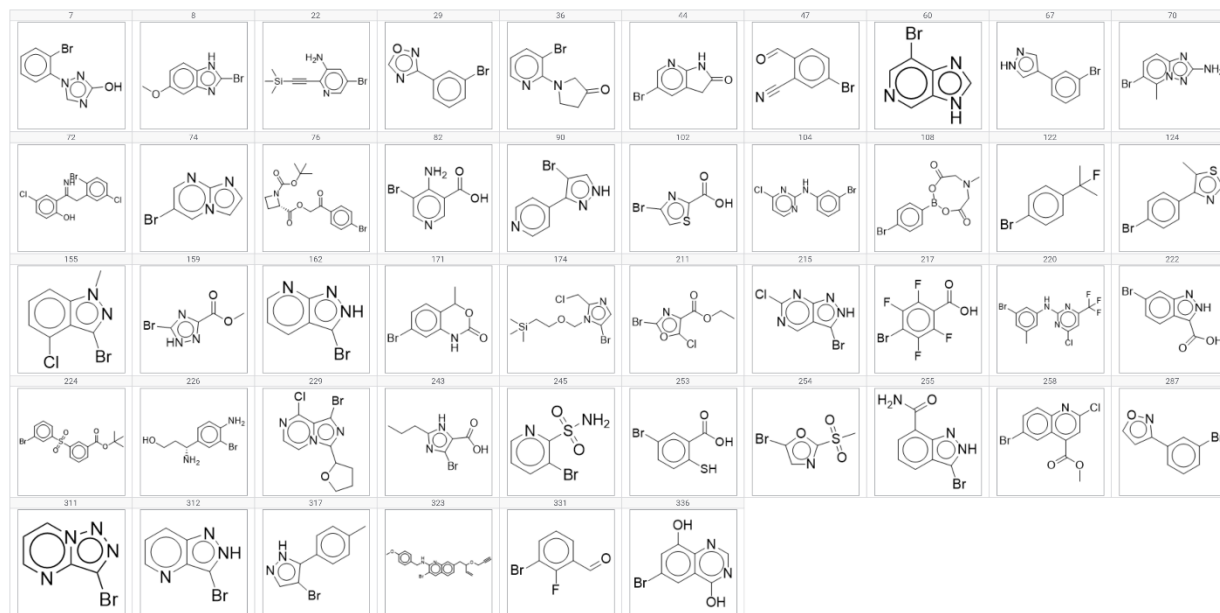


Figure S21. 46 halide structures that resulted in 0% product LCAP in the Suzuki-Miyaura coupling reactions.

V. Substrate Selection and Nanoscale Procedures for Validation Libraries

Substrate selection workflow for validation libraries (Manuscript Figure 8A–C)

To validate the performance of the models, 96-membered libraries were executed for each data split (except random split). These libraries were designed such that 50% of the products were predicted to have <20% LCAP and 50% of the products were predicted to have >20% LCAP (i.e., 48 products of each prediction class). The workflow to select substrates for the validation libraries is outlined below in Figure S22.

Step 1: Randomly select a set of amines and aryl halides that fulfill library criteria and generate product predictions (0 = <20% LCAP prediction, 1 = ≥20% LCAP prediction)

Step 2: Total each column and separate substrates that are always predicted as <20% LCAP (total = 0) or majority are predicted as ≥20% LCAP (high total) into different dataframes.

Step 3: Set a prediction confidence threshold >0.9 (for Amine OOS, ArX OOS, DRS) or >0.8 (for Both OOS) and sort dataframe to identify substrates that have the highest the number of rows that fulfill this criteria (high total). NaN = prediction does not meet confidence threshold. Total = number of predictions with confidence above threshold.

Step 4: Select substrate columns with highest totals, remove any rows with NaN entries, resulting in a matrix of x amines by y ArX with high prediction confidence.

Step 5: Filter out substrates with <0.5 g available in MBBC inventory. For OOS substrates, calculate similarity score against tested scope substrates and filter out substrates with >0.7 similarity.

Step 6: Randomly select appropriate number of substrates for each library from remaining list of filtered substrates

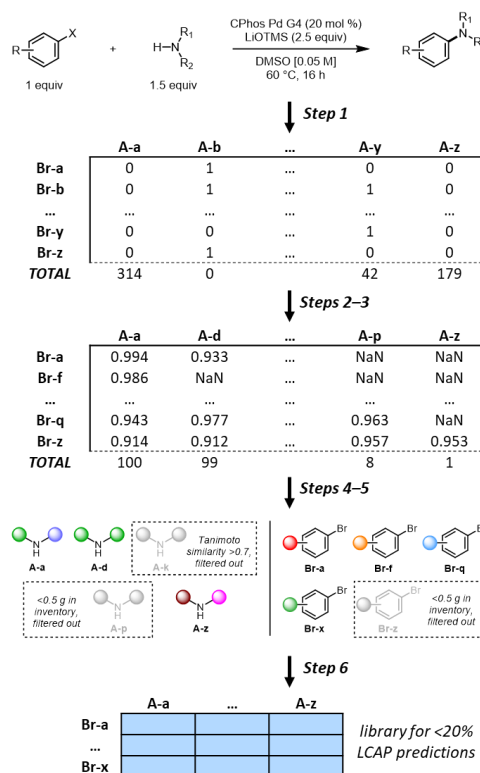


Figure S22. General workflow to select substrates for the validation libraries (Manuscript Figure 8)

For the Amine OOS library, predictions were generated on the products of 500 randomly selected new amines (from 3405 amines in the MBBC chemical space) coupled with the 357 aryl halides used as **Set B** (Manuscript Figure 4B, **Part II**). For the ArX OOS library, predictions were generated on the products of 500 randomly selected new aryl halides (from 5350 aryl halides in the MBBC chemical space) coupled with the 348 amines used as **Set A** (Manuscript Figure 4B, **Part I**). For the Both OOS library, predictions were generated on the products of 1000 randomly

selected new amines with 1000 randomly selected new aryl halides. For the DRS library, predictions were generated on new product combinations of the 348 amines used as **Set A** coupled with the 357 aryl halides used as **Set B**. The substrates were randomly selected from their respective chemical spaces using the Python function `random.sample()`. Predictions for each library was generated from the model trained on the whole experimental data set with the best-performing architecture, input feature, and hyperparameter combinations that were previously identified (see **Modeling Results** section for more details. For DRS validation library, model settings from ‘DRS25’ was used).

A prediction model confidence threshold of >0.9 (for Amine OOS, ArX OOS, DRS) or >0.8 (for Both OOS) was set and the predictions were sorted to identify the substrates that resulted in the highest number of products that cleared the model confidence threshold. For new substrates, Tanimoto similarity was calculated relative to substrates used in the model build (**Set A** or **Set B**) and any substrate with a similarity of >0.7 was filtered out. This was done to ensure that confident model predictions were not solely a result of chemical similarity to seen substrates. Finally, any substrate that generally had less than <0.5 g in the MBBC inventory were filtered out to ensure availability upon ordering. From this list of filtered amine and aryl halide candidates, the appropriate number of substrates required for each library were randomly selected.

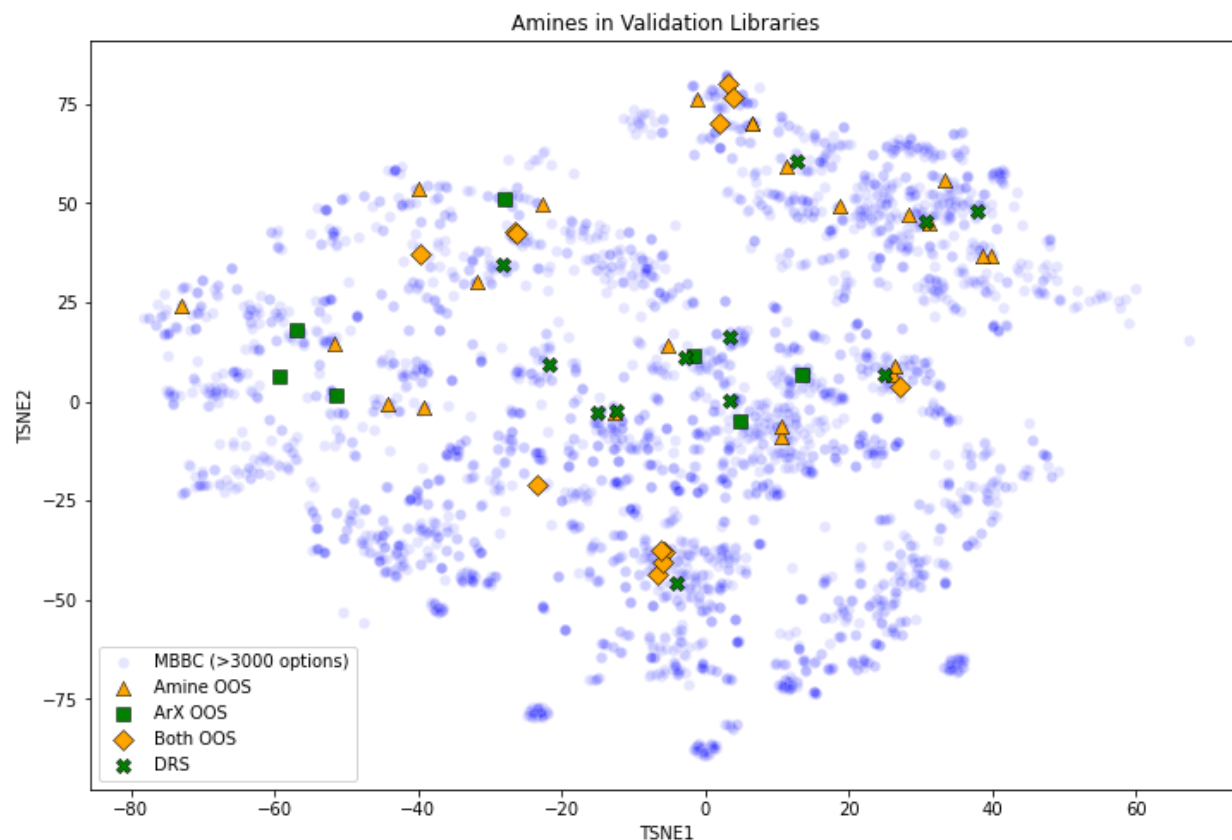


Figure S23. Selected amines for the validation libraries (blue circles = MBBC amine chemical space; triangles = Amine OOS library substrates; squares = ArX OOS library substrates; diamonds = Both OOS library substrates; crosses = DRS library substrates; orange markers = new substrates not used in the data set for the model build; green markers = scope substrates used in the data set for the model build).

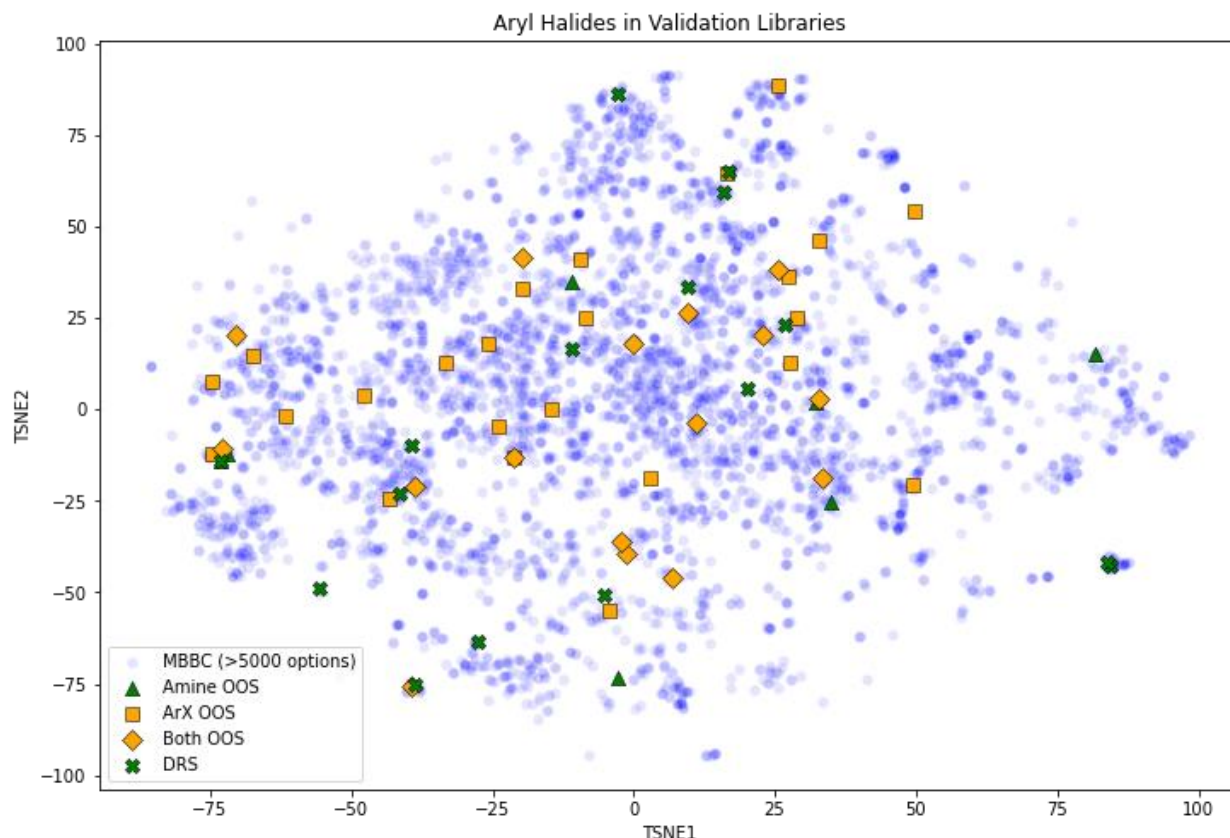
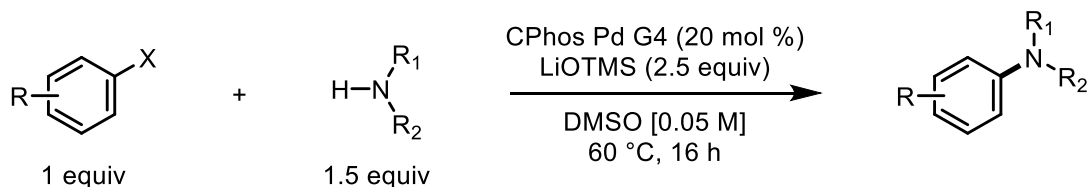


Figure S24. Selected aryl halides for the validation libraries (blue circles = MBBC aryl halide chemical space; triangles = Amine OOS library substrates; squares = ArX OOS library substrates; diamonds = Both OOS library substrates; crosses = DRS library substrates; orange markers = new substrates not used in the data set for the model build; green markers = scope substrates used in the data set for the model build).

General nanoscale procedure for validation libraries (Manuscript Figure 8D)



In a nitrogen-filled glovebox, to a 1536 plastic COC plate (Corning EchoTM qualified, Cat. No. 3730, Cyclic Olefin-Copolymer COC, 12.5 μL -wells, flat bottom, clear) using the TTP Labtech Mosquito robot, was added aryl halide (0.2 μL , 0.25 M solution in DMSO, 0.05 μmol , 1.0 equiv), amine (0.21 μL , 0.35 M solution in DMSO, 0.075 μmol , 1.5 equiv), CPhos Pd G4 (0.36 μL , 0.028 M mixture in DMSO, 0.01 μmol , 0.2 equiv), and lithium trimethylsilanolate (0.28 μL , 0.45 M solution in DMSO, 0.125 μmol , 2.5 equiv). Once the dosing was completed, the reaction plate was

heat sealed, centrifuged, enclosed in an aluminum block, and taken out of the glovebox. The aluminum block containing the reaction plate was placed between two preheated thermocouples (bottom thermocouple at 50 °C and top thermocouple at 70 °C). This setup was agitated at 10 G using the Resodyn acoustic mixing LabRam (PharmaRAM II Mixer) for 16 h. The aluminum block containing the reaction plate was cooled to room temperature and taken into the glovebox. The reaction plate was taken out of the aluminum block and centrifuged. The seal was removed, and the reaction mixture was transferred using 100 µL of 1% AcOH/DMSO into a 384 flat-bottom COC (Cyclic-Olefin Copolymer) analytical plate using the Thermo Matrix Platemate robot. The analytical plate was heat sealed in the glovebox and the reaction mixtures were analyzed using UPLC-UV/MS to obtain the product LCAPs.

Accuracy, precision, and recall are useful metrics for evaluating model performances and were calculated for each validation library using the formulas below (TN = true negative; FN = false negative; FP = false positive; TP = true positive; ‘positive’ meaning >20% LCAP; ‘negative’ meaning <20% LCAP).

$$accuracy = \frac{\text{correct classifications}}{\text{total classifications}} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$precision = \frac{\text{correctly classified actual positives}}{\text{all predicted positives}} = \frac{TP}{TP + FP}$$

$$recall = \frac{\text{correctly classified actual positives}}{\text{all actual positives}} = \frac{TP}{TP + FN}$$

Amine OOS validation library

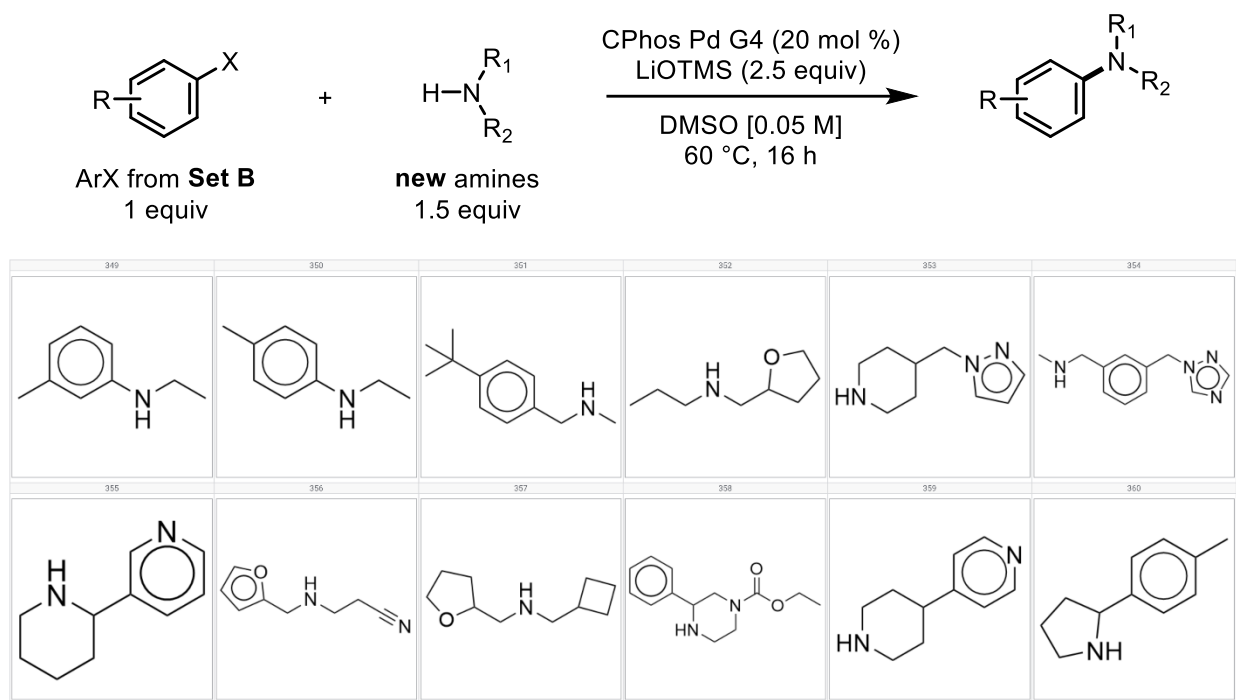


Figure S25. Amine structures for the Amine OOS <20% LCAP library.

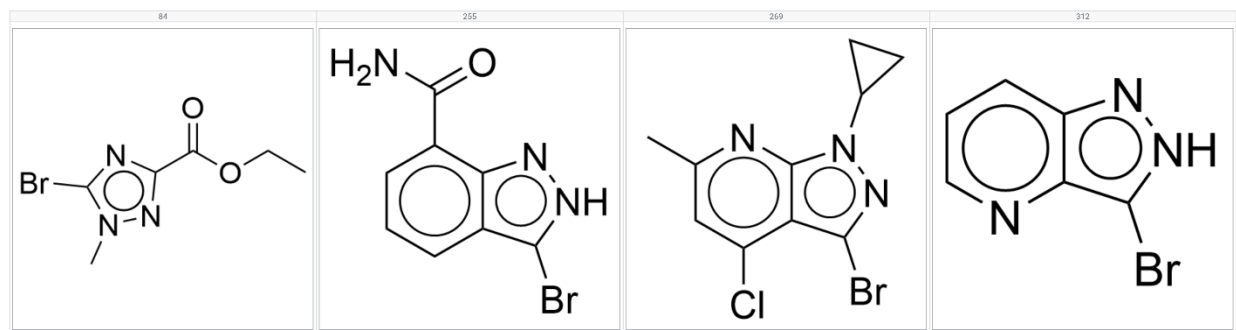


Figure S26. Aryl halide structures for the Amine OOS <20% LCAP library.

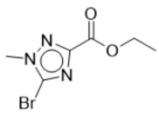
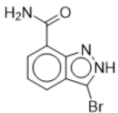
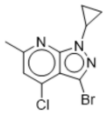
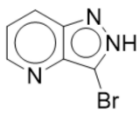
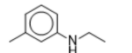
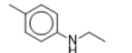
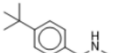
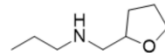
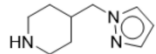
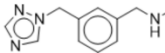
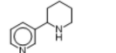
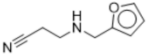
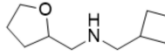
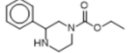
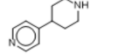
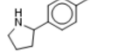
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351		0.00	0.00	0.00	6.01
352		0.00	0.00	0.00	0.00
353		0.00	0.00	0.10	0.00
354		0.00	0.00	0.00	0.00
355		0.00	0.00	0.00	0.00
356		0.00	0.00	0.00	0.00
357		0.00	0.00	4.23	0.00
358		0.00	0.00	0.00	0.00
359		0.00	0.00	0.85	0.00
360		0.00	0.00	0.00	0.00

Figure S27. Results for the Amine OOS <20% LCAP library. Numbers are average product LCAPs (UV254 and U210), blue cells indicating <20% LCAP and green cells indicating >20% LCAP.

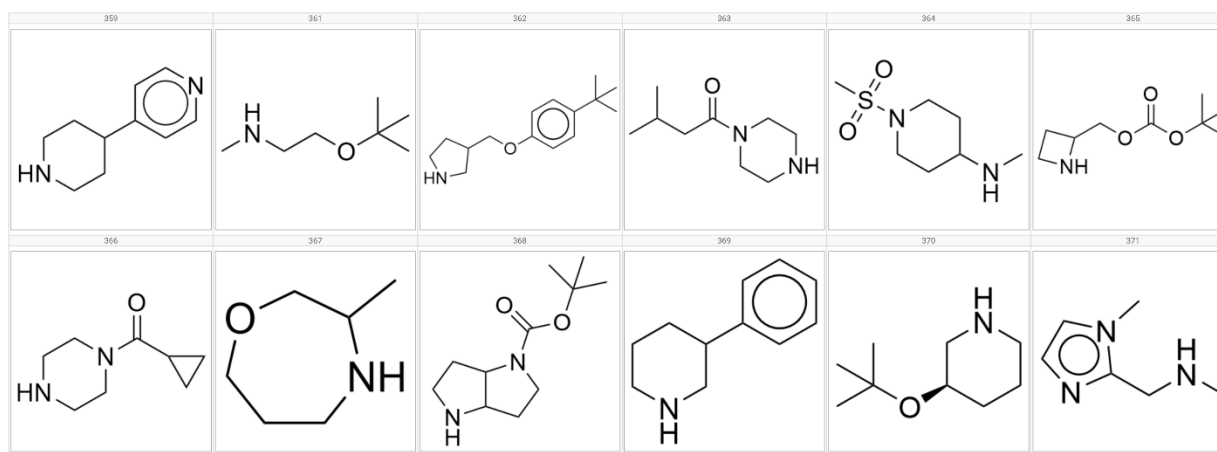


Figure S28. Amine structures for the Amine OOS >20% LCAP library.

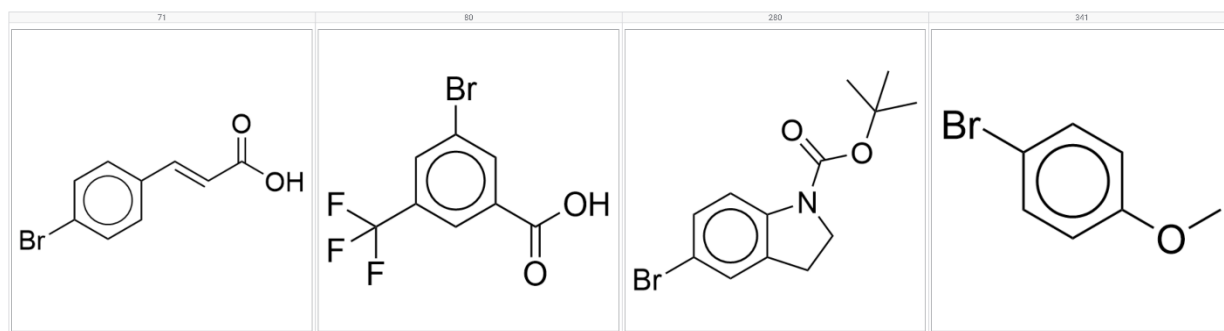


Figure S29. Aryl halide structures for the Amine OOS >20% LCAP library.

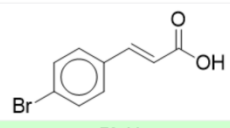
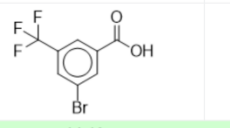
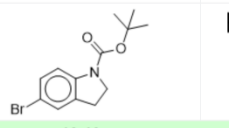
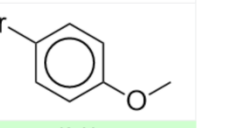
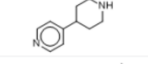
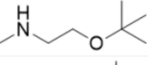
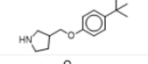
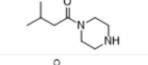
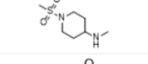
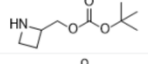
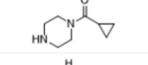
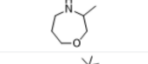
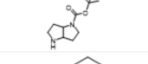
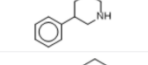
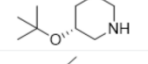
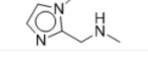
		71	80	280	341
					
359		72.44	64.69	60.69	49.44
361		48.47	39.51	45.91	13.08
362		57.76	50.56	56.48	33.63
363		60.64	66.98	76.22	33.97
364		61.06	44.29	46.25	11.27
365		0.00	0.00	0.00	0.00
366		62.10	63.93	74.72	35.40
367		38.20	38.94	17.69	2.81
368		43.79	78.81	42.83	41.02
369		68.16	47.17	71.36	30.19
370		33.42	54.79	27.71	6.62
371		48.09	28.48	14.97	5.81

Figure S30. Results for the Amine OOS >20% LCAP library. Numbers are average product LCAPs (UV254 and U210), blue cells indicating <20% LCAP and green cells indicating >20% LCAP.

		Actual		
Amine OOS (RF)		<20% LCAP	>20% LCAP	
Predicted	<20% LCAP	48 (TN)	0 (FN)	
	>20% LCAP	11 (FP)	37 (TP)	

accuracy = 0.8854
precision = 0.7708
recall = 1.0000

Figure S31. Confusion matrix with the results of the Amine OOS validation library (TN = true negative; FN = false negative; FP = false positive; TP = true positive).

As summarized in Figure S31 above, the model correctly predicted the performances of all the substrates in the <20% LCAP library (48 TN, 0 FN). For the >20% LCAP library, 37 predictions were correct (TP) and 11 were incorrect (FP). Notably, correct predictions were made for amine **359** which was used in both the <20% LCAP and >20% LCAP libraries (Figure S27 and Figure S30, respectively), which demonstrates that the model is not simply memorizing the reactivity of substrates but is able to predict specific substrate performance in different coupling outcomes.

ArX OOS validation library

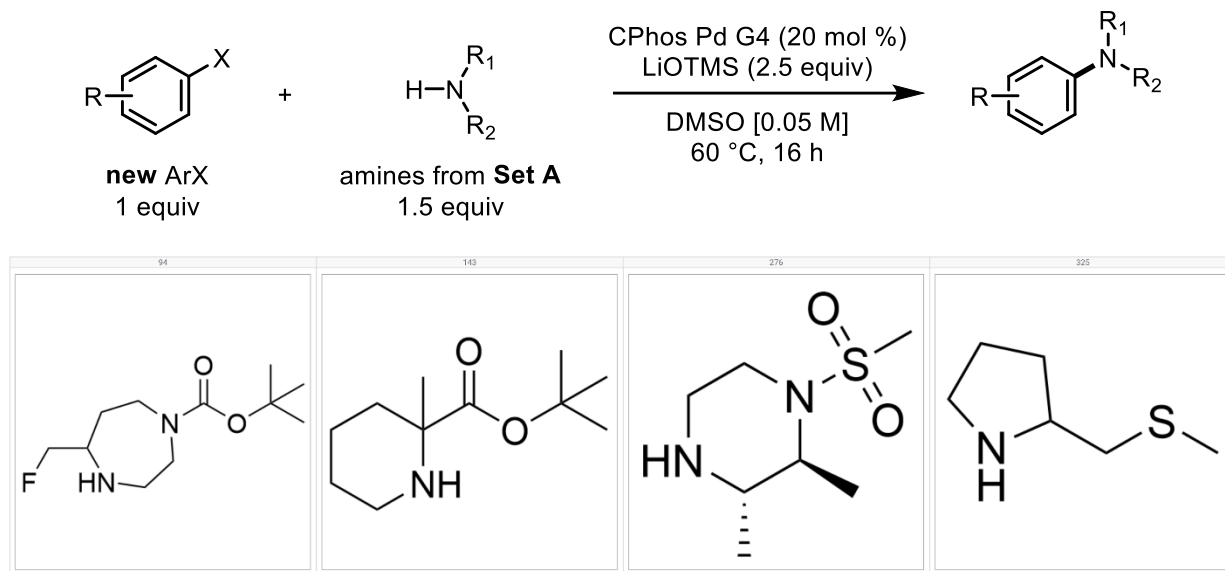


Figure S32. Amine structures for the ArX OOS <20% LCAP library.

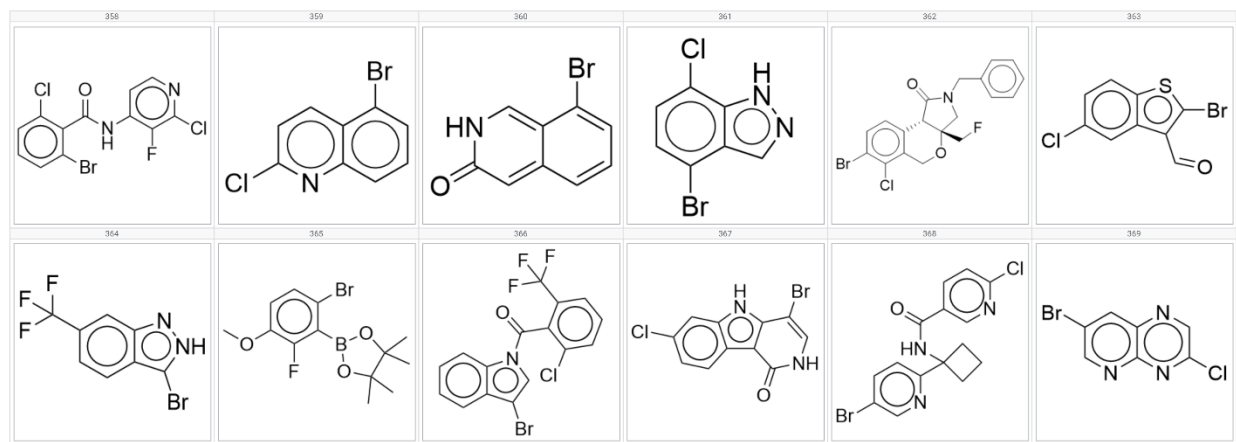


Figure S33. Aryl halide structures for the ArX OOS <20% LCAP library.

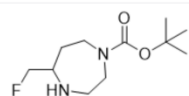
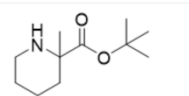
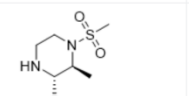
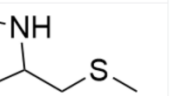
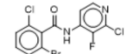
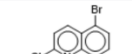
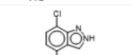
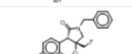
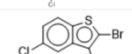
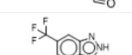
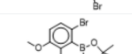
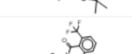
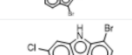
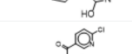
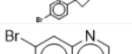
		94 	143 	276 	325 
358		0.00	0.00	0.00	0.00
359		0.00	0.00	2.85	0.00
360		0.00	0.00	6.10	52.67
361		0.00	0.00	0.00	0.00
362		0.00	0.00	0.00	0.00
363		0.00	0.00	5.01	55.10
364		0.00	0.00	0.00	0.00
365		0.00	0.00	0.00	0.00
366		0.00	0.24	0.00	0.21
367		0.00	0.00	0.00	0.00
368		0.00	0.00	0.00	0.00
369		0.00	0.00	0.00	0.00

Figure S34. Results for the ArX OOS <20% LCAP library. Numbers are average product LCAPs (UV254 and U210), blue cells indicating <20% LCAP and green cells indicating >20% LCAP.

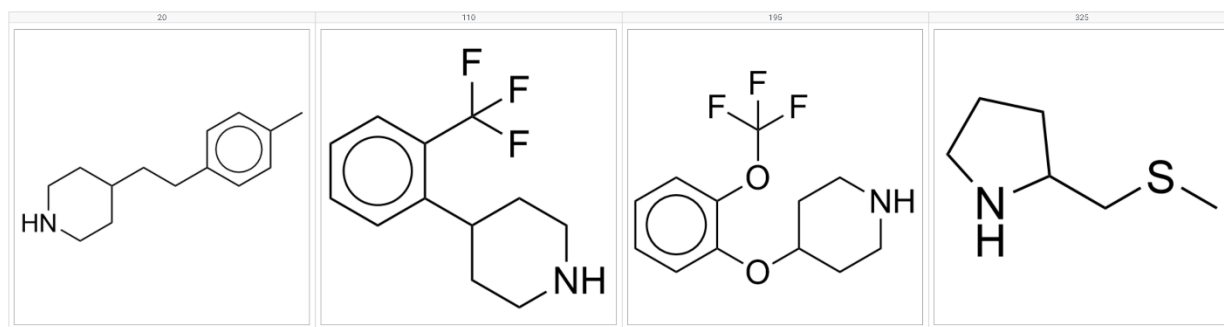


Figure S35. Amine structures for the ArX OOS >20% LCAP library.

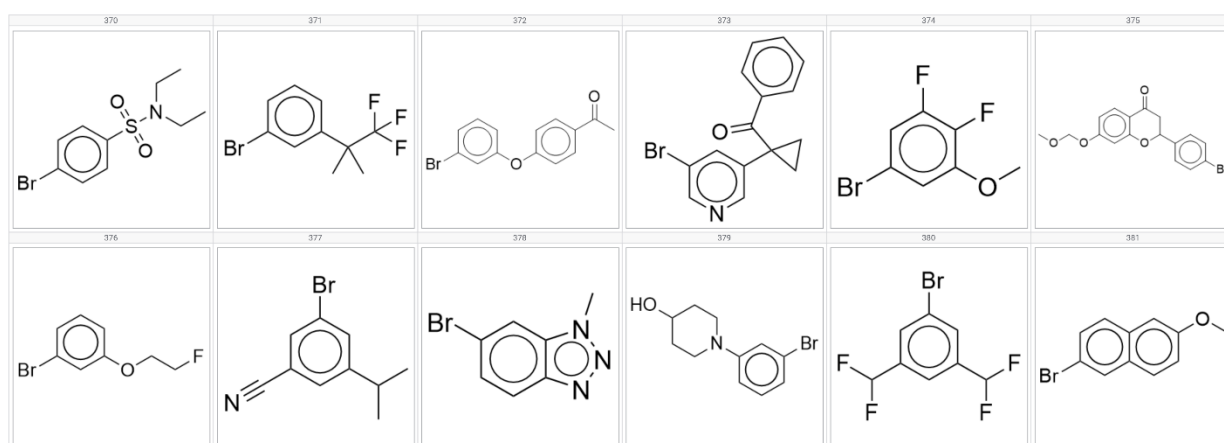


Figure S36. Aryl halide structures for the ArX OOS >20% LCAP library.

		20	110	195	325
370		45.93	56.62	49.03	47.11
371		16.09	35.50	74.77	58.71
372		36.68	41.48	40.00	25.99
373		58.77	71.76	67.75	67.70
374		39.79	54.06	15.96	66.01
375		10.83	66.42	59.05	64.24
376		47.09	54.53	41.18	59.99
377		32.94	50.72	54.24	53.99
378		64.38	66.61	59.68	63.19
379		58.64	66.21	63.74	66.76
380		52.32	64.74	63.67	64.88
381		47.94	57.37	57.03	64.77

Figure S37. Results for the ArX OOS >20% LCAP library. Numbers are average product LCAPs (UV254 and U210), blue cells indicating <20% LCAP and green cells indicating >20% LCAP.

		<i>Actual</i>		
<i>Predicted</i>	ArX OOS (MPNN)	<20% LCAP	>20% LCAP	
	<20% LCAP	46 (TN)	2 (FN)	
	>20% LCAP	3 (FP)	45 (TP)	

accuracy = 0.9479
precision = 0.9375
recall = 0.9574

Figure S38. Confusion matrix with the results of the ArX OOS validation library (TN = true negative; FN = false negative; FP = false positive; TP = true positive).

As summarized in Figure S38 above, the model correctly predicted the performances for the vast majority of substrates in both the <20% LCAP library (46 TN, 2 FN) and >20% LCAP library (45

TP, 3 FP), demonstrating high predictivity for new aryl halides (95% accuracy for the entire ArX OOS library). Notably, amine **325** was used in both the <20% LCAP and >20% LCAP libraries (Figure S34 and Figure S37, respectively) and the model was able to correctly predict the contrasting performance in a vast majority of its reactions.

Both OOS validation library

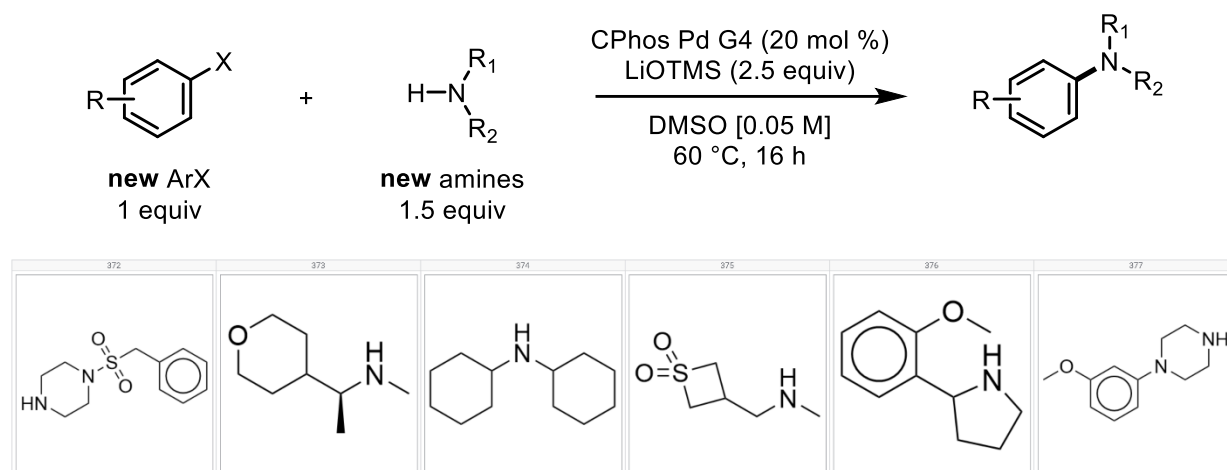


Figure S39. Amine structures for the Both OOS <20% LCAP library.

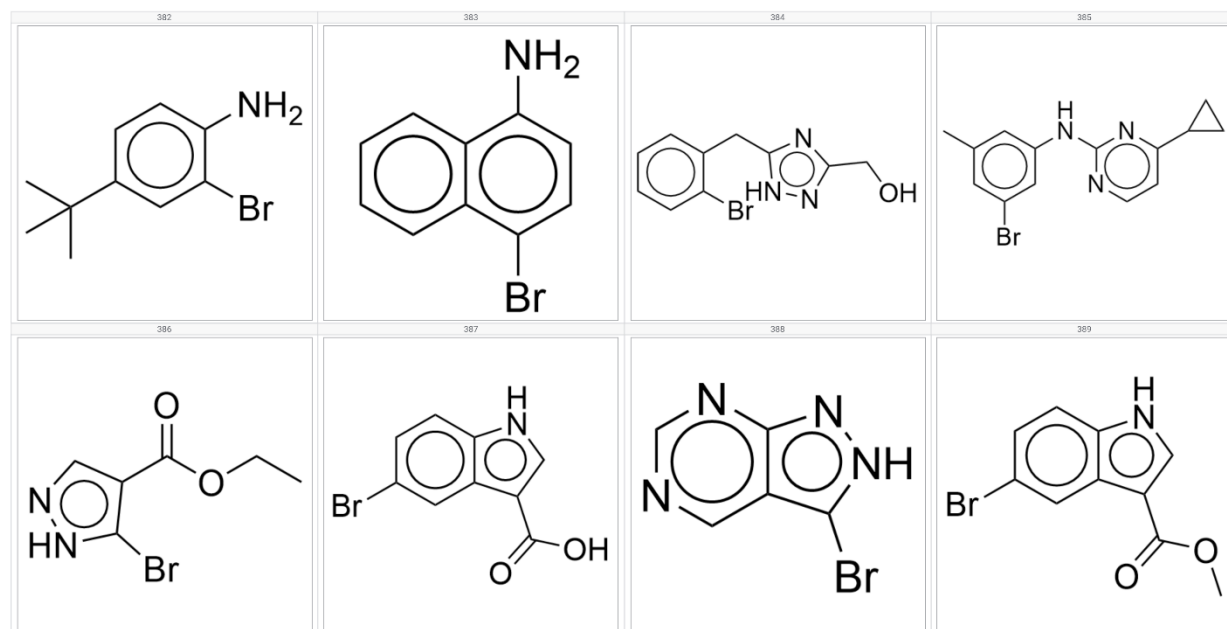


Figure S40. Aryl halide structures for the Both OOS <20% LCAP library.

		372	373	374	375	376	377
382		0.00	0.00	0.00	0.00	0.00	0.00
383		0.00	0.00	0.00	0.00	2.15	2.31
384		0.00	0.00	0.00	0.00	1.29	0.00
385		0.00	0.00	0.00	0.60	0.77	8.02
386		0.00	0.00	0.00	0.00	0.00	0.00
387		0.00	0.00	0.00	0.00	0.00	0.00
388		0.00	0.00	0.00	0.00	0.00	0.00
389		4.91	0.00	0.00	0.00	0.00	0.97

Figure S41. Results for the Both OOS <20% LCAP library. Numbers are average product LCAPs (UV254 and U210), blue cells indicating <20% LCAP and green cells indicating >20% LCAP.

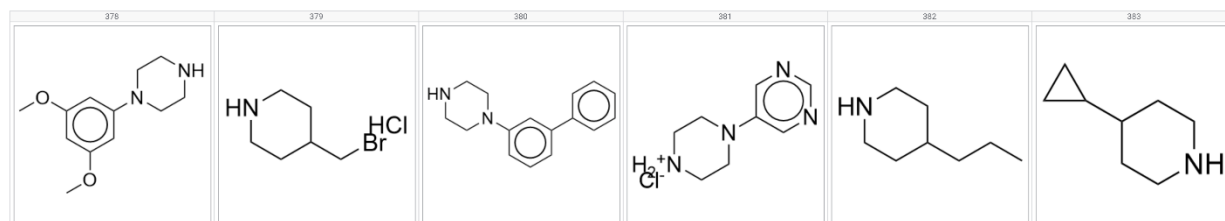


Figure S42. Amine structures for the Both OOS >20% LCAP library.

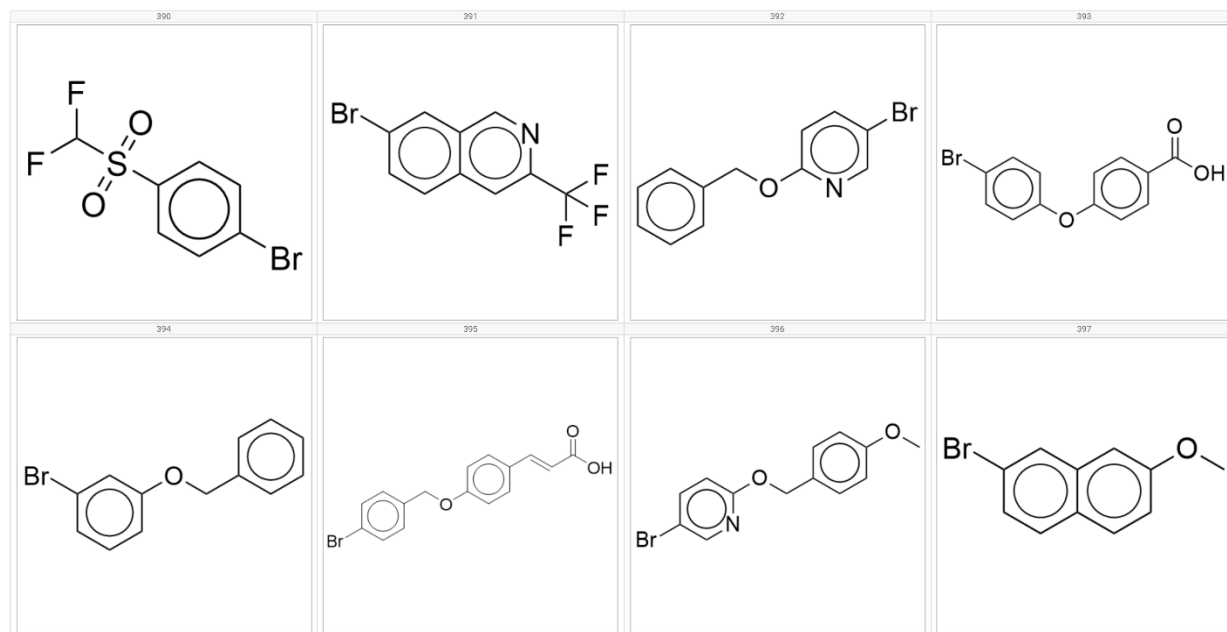


Figure S43. Aryl halide structures for the Both OOS >20% LCAP library.

		378	379	380	381	382	383
390		14.06	0.00	24.89	0.00	1.39	2.88
391		68.48	0.00	60.94	0.00	43.40	44.56
392		52.25	0.00	49.26	0.00	31.46	26.52
393		73.78	0.00	34.54	0.00	41.88	26.36
394		63.87	0.00	59.83	0.00	35.67	23.65
395		60.47	0.00	44.10	0.00	28.46	27.95
396		31.30	0.00	45.58	0.00	21.54	26.49
397		71.33	0.00	61.04	0.00	26.00	31.22

Figure S44. Results for the Both OOS >20% LCAP library. Numbers are average product LCAPs (UV254 and U210), blue cells indicating <20% LCAP and green cells indicating >20% LCAP.

		<i>Actual</i>		
Both OOS (RF)		<20% LCAP	>20% LCAP	
<i>Predicted</i>	<20% LCAP	48 (TN)	0 (FN)	
	>20% LCAP	19 (FP)	29 (TP)	

accuracy = 0.8021
precision = 0.6042
recall = 1.0000

Figure S45. Confusion matrix with the results of the Both OOS validation library (TN = true negative; FN = false negative; FP = false positive; TP = true positive).

As summarized in Figure S45 above, the model correctly predicted the performances of all the substrates in the <20% LCAP library (48 TN, 0 FN). For the >20% LCAP library, 29 predictions were correct (TP) and 19 were incorrect (FP). Notably, the model was unable to correctly predict the outcome of all couplings using amine **379** or amine **381** (Figure S44). Overall, 80% accuracy was calculated for the entire Both OOS library. The higher amount of FP vs FN is consistent with the model being less confident in predicting higher yielding reactions as discussed in the manuscript.

DRS validation library

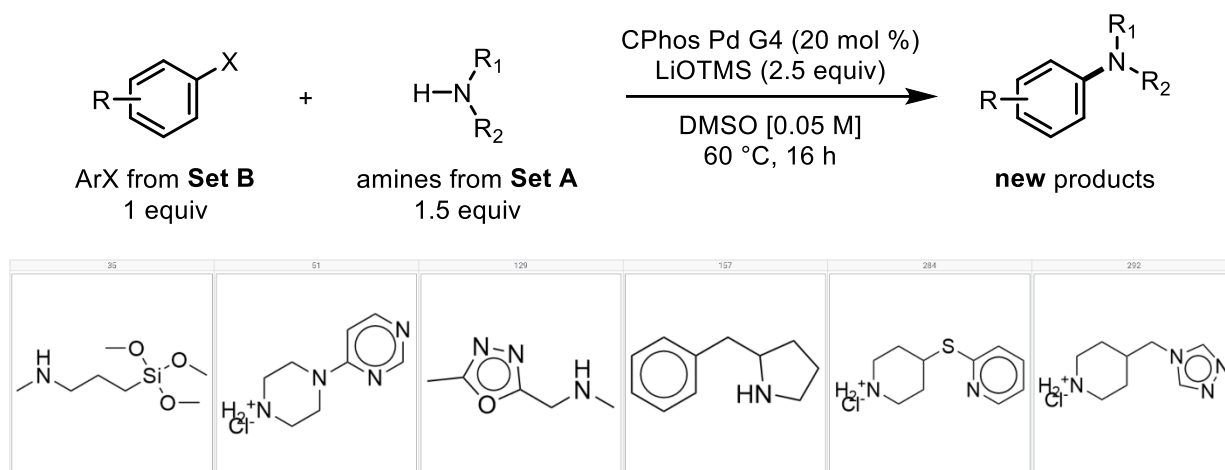


Figure S46. Amine structures for the DRS <20% LCAP library.

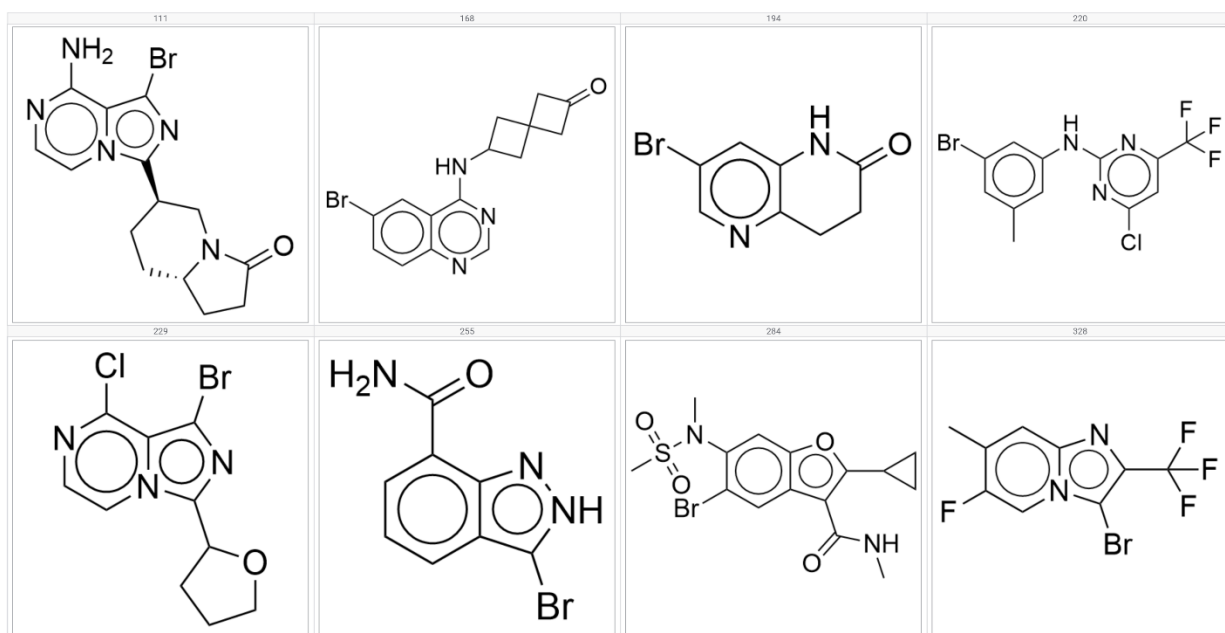


Figure S47. Aryl halide structures for the DRS <20% LCAP library.

		35	51	129	157	284	292
111		0.00	0.00	0.00	0.00	0.00	0.00
168		0.00	0.00	10.67	0.00	0.00	29.47
194		0.00	0.00	0.87	4.58	0.00	0.00
220		0.00	0.00	0.00	0.00	0.00	0.00
229		0.00	0.00	0.00	0.00	0.00	0.79
255		0.00	0.00	0.00	0.00	0.00	0.00
284		0.00	0.00	0.00	0.00	0.00	0.00
328		0.00	0.00	8.05	1.06	0.00	0.74

Figure S48. Results for the DRS <20% LCAP library. Numbers are average product LCAPs (UV254 and U210), blue cells indicating <20% LCAP and green cells indicating >20% LCAP.

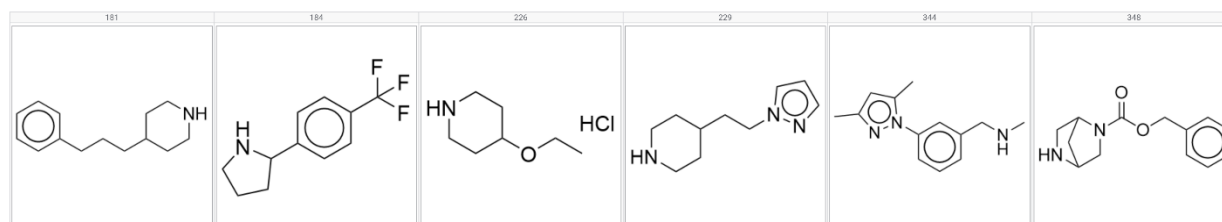


Figure S49. Amine structures for the DRS >20% LCAP library.

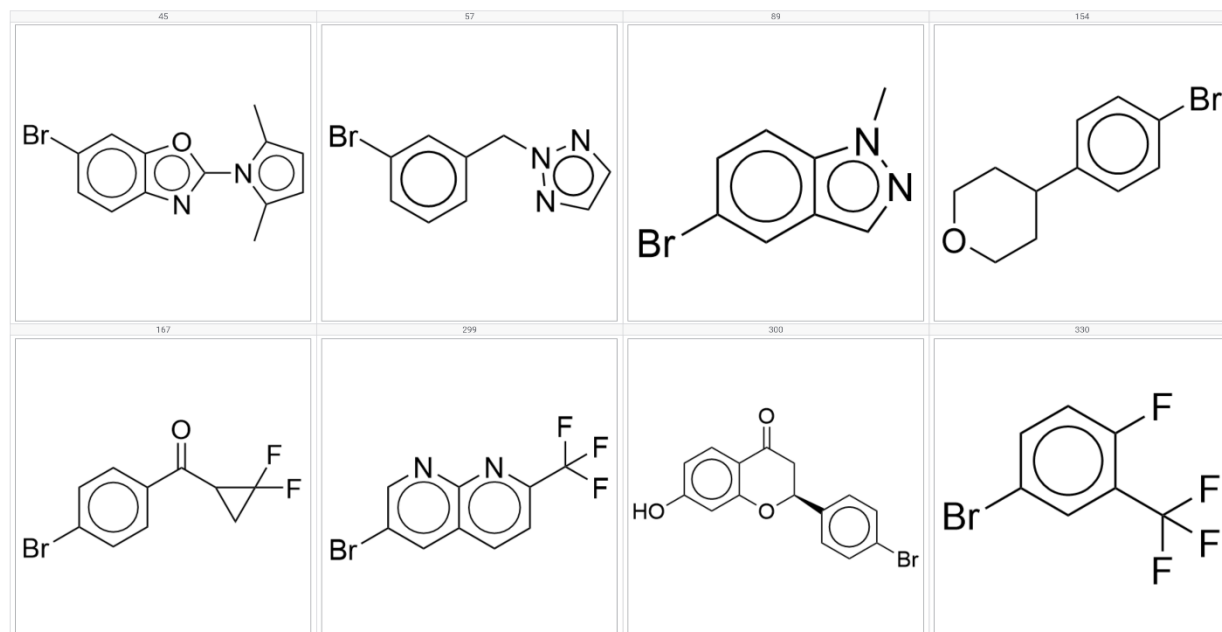


Figure S50. Aryl halide structures for the DRS >20% LCAP library.

		181	184	226	229	344	348
45		79.19	48.29	72.09	70.83	66.08	60.92
57		41.83	74.81	32.94	38.11	79.21	78.99
89		54.59	61.79	51.23	59.13	57.46	77.88
154		40.34	66.60	29.10	22.88	55.28	69.93
167		31.41	27.73	27.19	34.13	33.90	33.43
299		53.48	59.75	72.53	71.27	42.29	59.86
300		66.37	61.48	23.27	22.52	52.03	28.35
330		21.91	48.97	27.81	44.15	52.81	59.50

Figure S51. Results for the DRS >20% LCAP library. Numbers are average product LCAPs (UV254 and U210), blue cells indicating <20% LCAP and green cells indicating >20% LCAP.

		<i>Actual</i>		
DRS25 (RF)		<20% LCAP	>20% LCAP	
<i>Predicted</i>	<20% LCAP	47 (TN)	1 (FN)	
	>20% LCAP	0 (FP)	48 (TP)	<i>accuracy</i> = 0.9896 <i>precision</i> = 1.0000 <i>recall</i> = 0.9796

Figure S52. Confusion matrix with the results of the DRS validation library (TN = true negative; FN = false negative; FP = false positive; TP = true positive).

As summarized in Figure S52 above, the model correctly predicted the performances for the vast majority of substrates in the <20% LCAP library (47 TN, 1 FN) and the entire >20% LCAP library (48 TP, 0 FP), demonstrating high predictivity for new product combinations of tested substrates (99% accuracy for the entire DRS library). The high accuracy indicates that predictions on a full factorial space can be achieved using a fraction of a full factorial data set.

VI. Product Scale-up and Characterization

Product calibrations using UPLC/MS

Isolated products were used for calibrations in order to calculate assay yields from the crude reaction mixtures. The difference in UV response for the product and internal standard (4,4'-bis(trifluoromethyl)-1,1'-biphenyl) at 254 nm was taken into account using the following equation, where R_f is the response factor specific to each product.¹

$$\frac{UV\ Standard}{UV\ Product} \times R_f = \frac{Standard\ (mmol)}{Product\ (mmol)}$$

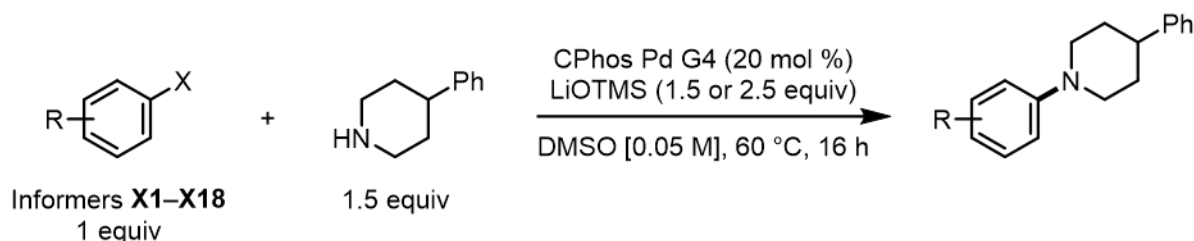
The response factor (R_f) for each product was determined by mixing the product and the internal standard together in DMSO- d_6 and collecting a 1H NMR spectrum. The NMR sample was then diluted with DMSO and directly injected on the UPLC/MS. The ratio of standard to product, determined from the 1H NMR and UV signals from UPLC/MS, were substituted into the equation to solve for R_f .

$$R_f = \frac{Standard\ (mmol)}{Product\ (mmol)} \times \frac{UV\ Product}{UV\ Standard}$$

The equation can then be used to solve for the amount of product formed (assay yield) in a crude reaction mixture.

$$Product\ (mmol) = \frac{Standard\ (mmol) \times UV\ Product}{UV\ Standard \times R_f}$$

General microscale procedure for isolation of Informer halides coupled products



In a nitrogen-filled glovebox, to 4-mL vials (Chemglass Life Sciences, 1 Dram, with red pressure relief cap, Product No. CG-4912-01) equipped with stir bars, was added Informer halide (0.4 mL, 0.25 M solution in DMSO, 0.1 mmol, 1.0 equiv), 4-phenylpiperidine (0.43 mL, 0.35 M solution in DMSO, 0.15 mmol, 1.5 equiv), CPhos Pd G4 (0.71 mL, 0.028 M solution in DMSO, 0.02 mmol, 0.2 equiv), and lithium trimethylsilanolate (either 0.54 mL, 0.28 M solution in DMSO, 0.15 mmol, 1.5 equiv or 0.54 mL, 0.47 M solution in DMSO, 0.25 mmol, 2.5 equiv, indicated for each product below) using single channel pipettors. The vials were capped, placed into a preheated 24-well block at 60 °C, and allowed to stir (600 rpm) for 16 h. The vials were cooled to room temperature and brought outside the glovebox for work-up.

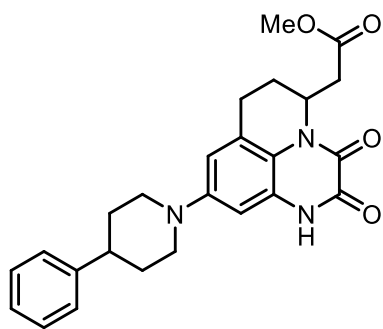
A Waters Oasis PRiME HLB 96-well filtration plate (30 mg sorbent per well, Product No. 186008054) was placed on top of an Analytical Sales 2 mL 96-well V-bottom collection plate (Product No. 59623-23GC) and placed inside a vacuum manifold filtration system. To each well of the filtration plate was added 1.0 mL of pH 7.4 phosphate buffer/Celite mixture (5 g Celite / 1L 1X PBS solution), which was pulled into the collection plate under vacuum. A new collection plate was placed under the filtration plate and to each well of the filtration plate was added an additional 1.0 mL of pH 7.4 phosphate buffer/Celite mixture. On top of the buffer mixture in each well of the filtration plate, 0.13 mL (0.062 mmol) of the crude reaction mixture was added (i.e., the total volume of the crude reaction mixture was portioned across 16 wells). The combined buffer and crude reaction mixture was pulled into the collection plate under vacuum. A new collection plate was placed under the filtration plate and to each well of the filtration plate was added 1.0 mL of 4% aq. MeOH solution, which was pulled into the collection plate under vacuum. A new collection plate was placed under the filtration plate and to each well of the filtration plate was added 1.0 mL of 9:1 MeCN:MeOH solution, which was pulled into the collection plate under vacuum. A new collection plate was placed under the filtration plate and to each well of the filtration plate was

added an additional 1.0 mL of 9:1 MeCN:MeOH solution, which was pulled into the collection plate under vacuum. A 96-well rubber cap mat (Analytical Sales, Product No. 965075) was placed on top of the filtration plate to improve the vacuum seal. The two collection plates containing the 9:1 MeCN:MeOH filtrate were dried under vacuum using a Genevac EZ2 at ambient temperature. The crude solids were suspended in 50 μ L DMSO and the crude product solutions were combined into a single well in a new collection plate (50 μ L $\times$ 32 wells (16 wells/plate)). The crude product solution was submitted for MS-directed purification using Waters XSelect CSH Prep C18 5 μ M OBD Prep Column (Product No. 186005421). Specific purification details for each isolated product are detailed below.

Note: This workflow was used to obtain high purity samples for calibrated assay yields. No attempt was made to optimize the isolation procedure for any product.

For products in which trifluoroacetic acid (TFA) was used during purification, the corresponding amine• n TFA salt was isolated. To determine the number of TFA molecules per product molecule (n), the ^1H NMR and ^{19}F NMR spectra could be compared using 4,4'-bis(trifluoromethyl)-1,1'-biphenyl as the internal standard. The product:standard ratio in the ^1H NMR spectra as well as the TFA:standard ratio in the ^{19}F NMR spectra were determined to calculate the product:TFA ratio.² After determining n , the molecular weight of the amine• n TFA salt is used to calculate the assay yield of the product in crude reaction mixtures.

Product of X1 + 4-phenylpiperidine



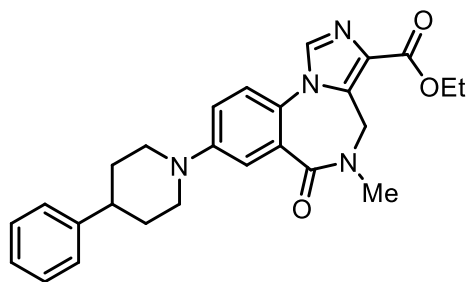
Methyl 2-(2,3-dioxo-9-(4-phenylpiperidin-1-yl)-2,3,6,7-tetrahydro-1H,5H-pyrido[1,2,3-de]quinoxalin-5-yl)acetate was prepared according to the general procedure using LiOTMS (2.5 equiv). The crude mixture was purified (gradient 35–70% CH₃CN in 0.16% NH₄/H₂O over 8 mins with a 25 mL flow rate on a single pump system) to yield the product as a yellow-green solid (assay yield 31%, isolated 3.8 mg, 8.8 μmol, 9% yield).

¹H NMR (500 MHz, DMSO-*d*₆) δ 7.32 – 7.27 (m, 4H), 7.21 – 7.19 (m, 1H), 6.72 (s, 1H), 6.60 (s, 1H), 5.07 (s, 1H), 3.71 (d, *J* = 10.7 Hz, 2H), 3.62 (s, 3H), 2.91 (t, *J* = 13.1 Hz, 1H), 2.78 – 2.71 (m, 3H), 2.70 – 2.57 (m, 3H), 2.10 (d, *J* = 11.8 Hz, 1H), 1.93 – 1.86 (m, 3H), 1.79 – 1.74 (m, 2H).

¹³C NMR (126 MHz, DMSO-*d*₆) δ 170.76, 153.99, 153.39, 147.33, 145.93, 128.39, 126.71, 126.33, 126.12, 125.02, 115.16, 111.81, 100.41, 51.67, 49.65, 46.92, 41.47, 34.90, 32.60, 23.02, 21.34.

HRMS (ESI+) *m/z*: [M+H]⁺ calcd for C₂₅H₂₈N₃O₄ 434.2075; observed 434.2074.

Product of X2 + 4-phenylpiperidine



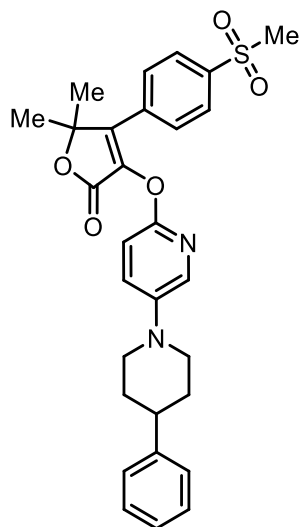
Ethyl 5-methyl-6-oxo-8-(4-phenylpiperidin-1-yl)-5,6-dihydro-4H-benzo[f]imidazo[1,5-a][1,4]diazepine-3-carboxylate was prepared according to the general procedure using LiOTMS (1.5 equiv). The crude mixture was purified (gradient 58–73% CH₃CN in 0.16% NH₄/H₂O over 6 mins with a 25 mL flow rate on a dual pump system) to yield the product as a tan solid (assay yield 90%, isolated 9.3 mg, 20.9 μmol, 21% yield).

¹H NMR (500 MHz, DMSO-*d*₆) δ 8.32 (s, 1H), 7.56 (d, *J* = 8.8 Hz, 1H), 7.39 – 7.25 (m, 6H), 7.20 (t, *J* = 7.0 Hz, 1H), 4.94 (s, 1H), 4.45 (s, 1H), 4.31 (s, 2H), 3.96 (d, *J* = 11.4 Hz, 2H), 3.09 (s, 3H), 2.91 (t, *J* = 11.8 Hz, 2H), 2.74 (t, *J* = 12.1 Hz, 1H), 1.90 (d, *J* = 12.5 Hz, 2H), 1.75 (q, *J* = 12.2 Hz, 2H), 1.33 (t, *J* = 7.1 Hz, 3H).

¹³C NMR (126 MHz, DMSO-*d*₆) δ 166.51, 162.80, 150.62, 146.30, 136.50, 135.66, 129.73, 128.90, 127.43, 127.18, 126.65, 124.14, 123.23, 119.86, 117.02, 60.52, 49.12, 42.61, 41.81, 35.53, 32.89, 14.71.

HRMS (ESI+) *m/z*: [M+H]⁺ calcd for C₂₆H₂₉N₄O₃ 445.2234; observed 445.2236.

Product of X3 + 4-phenylpiperidine



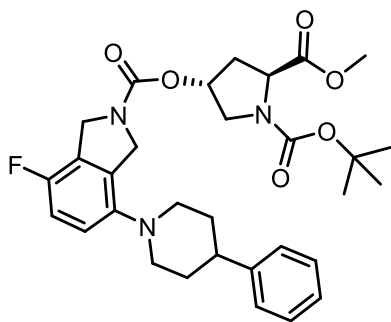
5,5-dimethyl-4-(4-(methylsulfonyl)phenyl)-3-((5-(4-phenylpiperidin-1-yl)pyridin-2-yl)oxy)furan-2(5H)-one•2TFA was prepared according to the general procedure using LiOTMS (1.5 equiv). The crude mixture was purified (gradient 38–63% CH₃CN in 0.16% TFA/H₂O over 6 mins with a 25 mL flow rate on a dual pump system) to yield the product as a brown solid (assay yield 51%, isolated 15.3 mg, 20.5 μ mol, 21% yield).

¹H NMR (500 MHz, DMSO-*d*₆) δ 8.02 (d, *J* = 8.4 Hz, 2H), 7.87 (d, *J* = 8.4 Hz, 2H), 7.84 (d, *J* = 2.8 Hz, 1H), 7.58 (dd, *J* = 9.0, 2.9 Hz, 1H), 7.35 – 7.25 (m, 4H), 7.20 (t, *J* = 6.9 Hz, 1H), 7.03 (d, *J* = 9.0 Hz, 1H), 3.75 (d, *J* = 12.1 Hz, 2H), 3.25 (s, 3H), 2.77 (t, *J* = 11.3 Hz, 2H), 2.66 (t, *J* = 12.1 Hz, 1H), 1.86 (d, *J* = 12.0 Hz, 2H), 1.80 – 1.72 (m, 2H), 1.70 (s, 6H).

¹³C NMR (126 MHz, DMSO-*d*₆) δ 165.42, 154.09, 148.04, 145.91, 144.70, 141.54, 137.06, 134.21, 133.75, 129.08, 128.88, 128.42, 127.58, 126.72, 126.15, 110.75, 84.27, 49.68, 43.28, 41.18, 32.58, 25.95.

HRMS (ESI+) *m/z*: [M+H]⁺ calcd for C₂₉H₃₁N₂O₅S 519.1948; observed 519.1949.

Product of X4 + 4-phenylpiperidine



1-(tert-butyl 2-methyl (2S,4R)-4-((4-fluoro-7-(4-phenylpiperidin-1-yl)isoindoline-2-carbonyl)oxy)pyrrolidine-1,2-dicarboxylate•4TFA was prepared according to the general procedure using LiOTMS (1.5 equiv). The crude mixture was purified (gradient 54–79% CH₃CN in 0.16% TFA/H₂O over 6 mins with a 25 mL flow rate on a dual pump system) to yield the product as a brown oil (assay yield 58%, isolated 19.3 mg, 18.9 μmol, 19% yield).

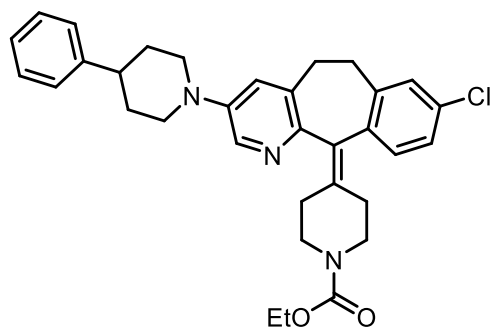
¹H NMR (500 MHz, DMSO-*d*₆) δ 7.38 – 7.25 (m, 4H), 7.22 – 7.19 (m, 1H), 7.11 (t, *J* = 8.6 Hz, 1H), 7.05 – 7.02 (m, 1H), 5.21 – 5.14 (m, 1H), 4.72 – 4.59 (m, 4H), 4.40 – 4.29 (m, 1H), 3.69 – 3.65 (m 3H), 3.63 – 3.51 (m, 2H), 3.32 – 3.20 (m, 2H), 2.85 – 2.74 (m, 2H), 2.66 (t, *J* = 11.9 Hz, 1H), 2.44 – 2.42 (m, 1H), 2.17 – 2.12 (m, 1H), 1.89 – 1.85 (m, 2H), 1.83 – 1.70 (m, 2H), 1.41 – 1.30 (m, 9H).

¹³C NMR (126 MHz, DMSO-*d*₆) δ 173.29, 173.26, 172.89, 172.86, 154.43, 154.34, 154.16, 154.12, 153.73, 153.71, 153.46, 153.43, 152.54, 152.44, 152.40, 146.38, 146.30, 146.25, 144.90, 144.80, 133.61, 133.58, 133.01, 132.97, 130.46, 128.88, 128.85, 127.23, 127.15, 127.12, 126.63, 125.10, 124.94, 124.57, 124.43, 119.60, 117.11, 115.26, 115.21, 115.10, 115.05, 114.79, 114.34, 114.19, 79.96, 79.79, 73.81, 73.24, 58.01, 57.68, 52.56, 52.50, 52.45, 52.05, 51.38, 49.82, 49.43, 41.76, 41.63, 40.46, 40.30, 40.13, 39.96, 39.80, 39.63, 39.46, 36.56, 35.63, 33.82, 28.45, 28.29.

HRMS (ESI+) *m/z*: [M+H]⁺ calcd for C₃₁H₃₉FN₃O₆ 568.2818; observed 568.2825.

The sample is a mixture of diastereomers; some resonances are doubled due to rotamers

Product of X6 + 4-phenylpiperidine



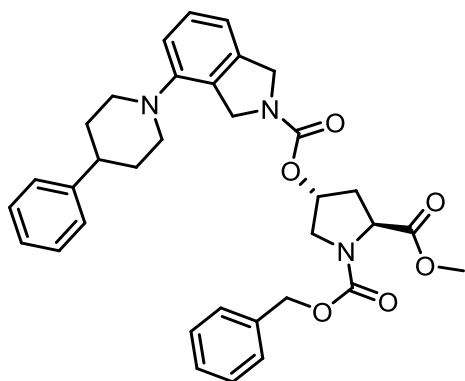
Ethyl 4-(8-chloro-3-(4-phenylpiperidin-1-yl)-5,6-dihydro-11H-benzo[5,6]cyclohepta[1,2-b]pyridin-11-ylidene)piperidine-1-carboxylate was prepared according to the general procedure using LiOTMS (1.5 equiv). The crude mixture was purified (gradient 55–90% CH₃CN in 0.16% NH₄/H₂O over 12 mins with a 25 mL flow rate on a single pump system) to yield the product as a light brown solid (assay yield 30%, isolated 11.9 mg, 22.0 μmol, 22% yield).

¹H NMR (500 MHz, DMSO-*d*₆) δ 8.11 (d, *J* = 2.2 Hz, 1H), 7.34 – 7.24 (m, 5H), 7.21 – 7.17 (m, 2H), 7.12 (d, *J* = 2.1 Hz, 1H), 7.06 (d, *J* = 8.1 Hz, 1H), 4.03 (q, *J* = 7.0 Hz, 2H), 3.84 (d, *J* = 11.4 Hz, 2H), 3.67 – 3.55 (m, 2H), 3.30 – 3.20 (m, 2H), 3.18 – 3.16 (m, 2H), 2.84 – 2.72 (m, 4H), 2.67 (t, *J* = 12.1 Hz, 1H), 2.40 – 2.31 (m, 1H), 2.26 – 2.22 (m, 2H), 2.19 – 2.10 (m, 1H), 1.85 (d, *J* = 12.0 Hz, 2H), 1.72 (q, *J* = 12.1 Hz, 2H), 1.17 (t, *J* = 7.1 Hz, 3H).

¹³C NMR (126 MHz, DMSO-*d*₆) δ 154.54, 146.21, 145.91, 145.67, 140.56, 139.09, 135.63, 134.61, 133.39, 132.84, 131.27, 130.21, 128.56, 128.40, 126.72, 126.14, 125.60, 123.47, 60.67, 48.53, 44.47, 41.38, 32.48, 31.27, 30.80, 30.43, 14.63.

HRMS (ESI+) *m/z*: [M+H]⁺ calcd for C₃₃H₃₇ClN₃O₂ 542.2569; observed 542.2570.

Product of **X8** + 4-phenylpiperidine



1-benzyl **2-methyl** **(2S,4R)-4-(((4-(4-phenylpiperidin-1-yl)isoindoline-2-carbonyl)oxy)pyrrolidine-1,2-dicarboxylate•3TFA** was prepared according to the general procedure using LiOTMS (1.5 equiv). The crude mixture was purified (gradient 45–70% CH₃CN in 0.16% TFA/H₂O over 6 mins with a 25 mL flow rate on a dual pump system) to yield the product as a brown oil (assay yield 61%, isolated 23.5 mg, 25.4 μ mol, 25% yield).

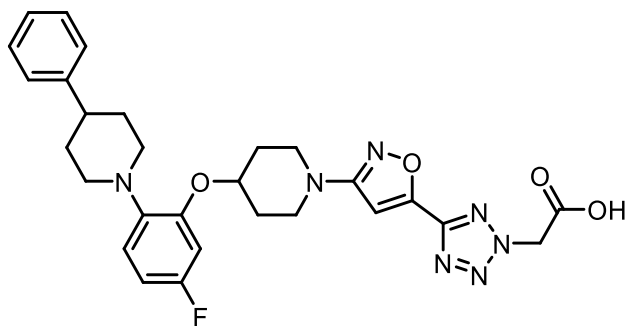
¹H NMR (500 MHz, DMSO-*d*₆) δ 7.39 – 7.18 (m, 11H), 7.05 – 6.95 (m, 2H), 5.20 (s, 1H), 5.15 – 4.90 (m, 2H), 4.69 – 4.51 (m, 4H), 4.51 – 4.39 (m, 1H), 3.67 (s, 3H), 3.56 (s, 2H), 3.38 – 3.31 (m, 2H), 2.86 – 2.83 (m, 2H), 2.74 – 2.59 (m, 1H), 2.49 – 2.40 (m, 1H), 2.26 – 2.17 (m, 1H), 1.95 – 1.70 (m, 4H).

¹³C NMR (126 MHz, DMSO-*d*₆) δ 172.48, 172.17, 172.13, 154.20, 154.11, 153.93, 153.65, 153.63, 153.38, 153.35, 153.22, 153.17, 147.64, 145.92, 145.85, 138.42, 138.40, 137.88, 137.85, 136.70, 136.61, 136.50, 136.44, 129.55, 129.02, 128.91, 128.85, 128.45, 128.42, 128.38, 128.35, 127.90, 127.86, 127.43, 127.41, 127.36, 127.24, 126.79, 126.74, 126.19, 117.01, 116.85, 116.80, 73.28, 73.22, 72.57, 72.50, 66.44, 66.39, 66.32, 57.69, 57.65, 57.26, 52.73, 52.57, 52.31, 52.28, 52.15, 52.13, 51.83, 51.29, 50.66, 41.33, 36.31, 36.13, 35.28, 35.10, 33.28, 33.13.

HRMS (ESI+) *m/z*: [M+H]⁺ calcd for C₃₄H₃₈N₃O₆ 584.2755; observed 584.2762.

The sample is a mixture of diastereomers; some resonances are doubled due to rotamers

Product of X9 + 4-phenylpiperidine



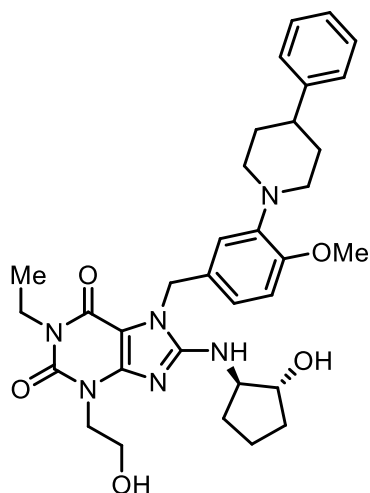
2-(5-(3-(4-(5-fluoro-2-(4-phenylpiperidin-1-yl)phenoxy)piperidin-1-yl)isoxazol-5-yl)-2H-tetrazol-2-yl)acetic acid was prepared according to the general procedure using LiOTMS (2.5 equiv). The crude mixture was purified (gradient 30–65% CH₃CN in 0.16% NH₄/H₂O over 8 mins with a 25 mL flow rate on a single pump system) to yield the product as a light brown solid (assay yield 38%, isolated 6.6 mg, 12.1 μmol, 12% yield).

¹H NMR (500 MHz, DMSO-*d*₆) δ 7.23 – 7.19 (m, 2H), 7.18 – 7.10 (m, 4H), 7.03 (dd, *J* = 10.8, 2.7 Hz, 1H), 6.95 (dd, *J* = 8.7, 6.4 Hz, 1H), 6.73 (td, *J* = 8.4, 2.7 Hz, 1H), 5.09 (s, 2H), 4.79 – 4.72 (m, 1H), 3.67 – 3.58 (m, 2H), 3.46 – 3.37 (m, 4H), 2.69 – 2.54 (m, 3H), 2.03 – 1.93 (m, 2H), 1.84 – 1.69 (m, 6H).

¹³C NMR (126 MHz, DMSO-*d*₆) δ 167.06, 165.31, 158.97, 158.03, 157.07, 154.54, 150.69 (d, *J* = 10.1 Hz), 146.11, 140.12 (d, *J* = 2.9 Hz), 127.55 (d, *J* = 260.0 Hz), 126.07, 119.38 (d, *J* = 9.7 Hz), 107.12 (d, *J* = 21.1 Hz), 104.00 (d, *J* = 25.4 Hz), 96.19, 71.98, 56.92, 51.56, 43.52, 41.67, 33.62, 29.01.

HRMS (ESI⁺) *m/z*: [M+H]⁺ calcd for C₂₈H₃₁FN₇O₄ 548.2416; observed 548.2418.

Product of X12 + 4-phenylpiperidine



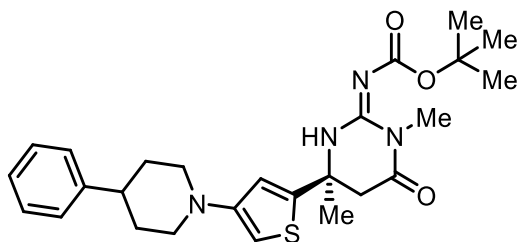
1-ethyl-8-(((1R,2R)-2-hydroxycyclopentyl)amino)-3-(2-hydroxyethyl)-7-(4-methoxy-3-(4-phenylpiperidin-1-yl)benzyl)-3,7-dihydro-1H-purine-2,6-dione was prepared according to the general procedure using LiOTMS (1.5 equiv). The crude mixture was purified (gradient 45–80% CH₃CN in 0.16% NH₄/H₂O over 8 mins with a 25 mL flow rate on a single pump system) to yield the product as a white solid (assay yield 55%, isolated 21.5 mg, 35.7 μmol, 36% yield).

¹H NMR (500 MHz, DMSO-*d*₆) δ 7.33 – 7.26 (m, 4H), 7.22 – 7.18 (m, 1H), 7.06 (s, 1H), 6.99 (d, *J* = 6.6 Hz, 1H), 6.90 (d, *J* = 8.3 Hz, 1H), 6.86 (d, *J* = 8.4 Hz, 1H), 5.22 (s, 2H), 4.80 (t, *J* = 5.8 Hz, 1H), 4.77 (d, *J* = 4.2 Hz, 1H), 4.00 – 3.93 (m, 3H), 3.92 – 3.85 (m, 3H), 3.75 (s, 3H), 3.61 (q, *J* = 6.3 Hz, 2H), 3.44 (d, *J* = 11.1 Hz, 2H), 2.65 – 2.55 (m, 3H), 2.06 (dq, *J* = 13.6, 7.7 Hz, 1H), 1.90 – 1.73 (m, 5H), 1.70 – 1.58 (m, 2H), 1.53 – 1.43 (m, 2H), 1.08 (t, *J* = 6.9 Hz, 3H).

¹³C NMR (126 MHz, DMSO-*d*₆) δ 153.44, 152.58, 151.40, 150.41, 148.71, 146.18, 141.58, 129.44, 128.37, 126.76, 126.06, 121.46, 118.35, 111.69, 101.16, 76.18, 61.43, 57.76, 55.41, 51.09, 44.95, 44.59, 41.83, 35.13, 33.29, 32.49, 29.98, 20.60, 13.31.

HRMS (ESI+) *m/z*: [M+H]⁺ calcd for C₃₃H₄₃N₆O₅ 603.3290; observed 603.3289.

Product of X13 + 4-phenylpiperidine



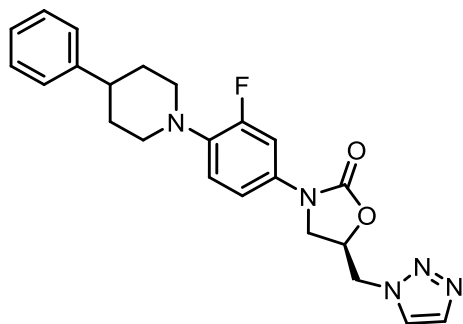
tert-butyl (S,Z)-(1,4-dimethyl-6-oxo-4-(4-(4-phenylpiperidin-1-yl)thiophen-2-yl)tetrahydropyrimidin-2(1H)-ylidene)carbamate was prepared according to the general procedure using LiOTMS (2.5 equiv). The crude mixture was purified (gradient 56–86% CH₃CN in 0.16% NH₄/H₂O over 12 mins with a 25 mL flow rate on a single pump system) to yield the product as a pale yellow solid (assay yield 68%, isolated 9.9 mg, 20.5 μmol, 21% yield).

¹H NMR (500 MHz, DMSO-*d*₆) δ 10.00 (s, 1H), 7.34 – 7.24 (m, 4H), 7.20 (t, *J* = 7.0 Hz, 1H), 7.03 (d, *J* = 1.4 Hz, 1H), 6.22 (d, *J* = 1.4 Hz, 1H), 3.60 (t, *J* = 12.1 Hz, 2H), 3.21 (d, *J* = 16.2 Hz, 1H), 3.11 (d, *J* = 16.1 Hz, 1H), 3.04 (s, 3H), 2.68 – 2.55 (m, 3H), 1.87 – 1.79 (m, 2H), 1.78 – 1.68 (m, 2H), 1.66 (s, 3H), 1.43 (s, 9H).

¹³C NMR (126 MHz, DMSO-*d*₆) δ 167.73, 163.15, 156.95, 152.23, 147.47, 146.02, 128.42, 126.73, 126.16, 118.09, 98.32, 78.61, 53.09, 50.35, 50.21, 44.16, 41.26, 32.60, 29.49, 27.94.

HRMS (ESI+) *m/z*: [M+H]⁺ calcd for C₂₆H₃₅N₄O₃S 483.2425; observed 483.2424.

Product of X14 + 4-phenylpiperidine



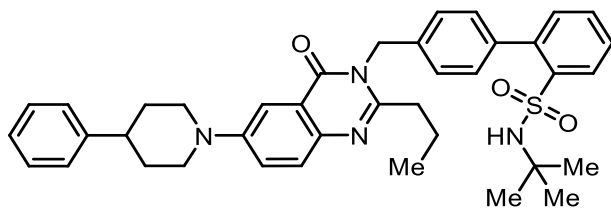
(R)-5-((1H-1,2,3-triazol-1-yl)methyl)-3-(3-fluoro-4-(4-phenylpiperidin-1-yl)phenyl)oxazolidin-2-one was prepared according to the general procedure using LiOTMS (1.5 equiv). The crude mixture was purified (gradient 40–75% CH₃CN in 0.16% NH₄/H₂O over 8 mins with a 25 mL flow rate on a single pump system) to yield the product as a white solid (assay yield 80%, isolated 15.2 mg, 36.1 μ mol, 36% yield).

¹H NMR (500 MHz, DMSO-*d*₆) δ 8.17 (s, 1H), 7.77 (s, 1H), 7.40 (dd, *J* = 14.8, 1.9 Hz, 1H), 7.33 – 7.27 (m, 4H), 7.22 – 7.17 (m, 1H), 7.16 – 7.06 (m, 2H), 5.15 – 5.08 (m, 1H), 4.83 (d, *J* = 5.1 Hz, 2H), 4.21 (t, *J* = 9.1 Hz, 1H), 3.86 (dd, *J* = 9.3, 5.7 Hz, 1H), 3.41 (d, *J* = 11.6 Hz, 2H), 2.81 – 2.71 (m, 2H), 2.69 – 2.59 (m, 1H), 1.93 – 1.75 (m, 4H).

¹³C NMR (126 MHz, DMSO-*d*₆) δ 154.59 (d, *J* = 243.7 Hz), 153.50, 146.00, 136.54 (d, *J* = 9.0 Hz), 133.40, 132.71 (d, *J* = 10.6 Hz), 128.40, 126.76, 126.12, 125.87, 119.67 (d, *J* = 4.1 Hz), 114.29 (d, *J* = 2.9 Hz), 106.76 (d, *J* = 26.2 Hz), 70.76, 51.73, 51.38, 47.12, 41.45, 33.13.

HRMS (ESI+) *m/z*: [M+H]⁺ calcd for C₂₃H₂₅FN₅O₂ 422.1987; observed 422.1985.

Product of X15 + 4-phenylpiperidine



***N*-(*tert*-butyl)-4'-((4-oxo-6-(4-phenylpiperidin-1-yl)-2-propylquinazolin-3(4*H*)-yl)methyl)-[1,1'-biphenyl]-2-sulfonamide** was prepared according to the general procedure using LiOTMS (1.5 equiv). The crude mixture was purified (gradient 55–90% CH₃CN in 0.16% NH₄/H₂O over 8 mins with a 25 mL flow rate on a single pump system) to yield the product as an off-white solid (assay yield 88%, isolated 7.7 mg, 11.9 μmol, 12% yield).

¹H NMR (500 MHz, DMSO-*d*₆) δ 8.03 (d, *J* = 7.9 Hz, 1H), 7.61 (t, *J* = 7.2 Hz, 2H), 7.57 – 7.50 (m, 3H), 7.38 (d, *J* = 8.0 Hz, 2H), 7.34 – 7.25 (m, 5H), 7.23 – 7.16 (m, 3H), 6.57 (s, 1H), 5.44 (s, 2H), 3.95 (d, *J* = 12.3 Hz, 2H), 2.88 (t, *J* = 11.5 Hz, 2H), 2.79 – 2.67 (m, 3H), 1.91 (d, *J* = 11.9 Hz, 2H), 1.85 – 1.68 (m, 4H), 0.99 – 0.88 (m, 12H).

¹³C NMR (126 MHz, DMSO-*d*₆) δ 161.65, 153.85, 149.59, 145.92, 142.06, 139.91, 139.61, 138.74, 136.02, 132.60, 131.76, 129.63, 128.42, 128.06, 127.79, 127.74, 126.72, 126.15, 125.51, 124.65, 120.39, 108.68, 53.35, 49.15, 45.33, 41.42, 35.58, 32.52, 29.31, 19.54, 13.60.

HRMS (ESI+) *m/z*: [M+H]⁺ calcd for C₃₉H₄₅N₄O₃S 649.3207; observed 649.3204.

VII. Selection of Secondary Amines and Aryl Halides for Scope Experiments

Selection of amines for amine scope experiment (Manuscript Figure 4B, Part I)

For each molecule in the MBBC amine inventory, a set of relevant functional groups was compiled and the presence or absence of the functional group was determined using the rdkit library function `rdkit.Chem.rdchem.GetSubstructMatches`. The functional group definitions are given below. The molecules with exactly one unprotected secondary amine, no alcohols, no thiols, no carboxylic acids, no sulfinic or sulfonic acids, no phosphinic acids, no aromatic N–H bonds, no boron atoms, no malemides, no primary imines, no amides, no aryl halides, and a molecular weight in the range of 115–400 were separated out to compose the secondary amine set (~3405). Next, 383 amines were randomly selected from this set. Finally, the amine used for reaction condition optimization (4-phenylpiperidine) was also included, resulting in a total of 384 selected amines. Due to commercial availability, only 348 amines were procured and used in the experiment (**Set A**, Manuscript Figure 4B, **Part I**).

Functional groups relevant to the secondary amine criteria

```
secondary_amine = Chem.MolFromSmarts('[NX3;H1]')
alcohol = Chem.MolFromSmarts('[c,C](-[O;H1])')
thiol = Chem.MolFromSmarts('[S;H1]')
acid = Chem.MolFromSmarts('[C,S,P](=O)[O;H1]')
aromatic_nH = Chem.MolFromSmarts('[n;H1]')
boron = Chem.MolFromSmarts('[B]')
malemide = Chem.MolFromSmarts('[N;H1](C(=O))C(=O)')
primary_imine = Chem.MolFromSmarts('[C]=[N;H1]')
amide = Chem.MolFromSmarts('C(=O)[N;H1,H2]')
aryl_halide = Chem.MolFromSmarts('[c;H0](-[F,Cl,Br,I])')
```

Selection of aryl halides for amine scope experiment (Manuscript Figure 4B, Part I)

6-bromoquinoline (**Br-1**) and Informer halides **X2**, **X3**, and **X13** were selected as coupling partners for the amine scope experiment because these substrates were previously used for the optimization screens.

Selection of aryl halides for aryl halide scope experiment (Manuscript Figure 4B, Part II)

For each molecule in the MBBC aryl halide inventory, a set of relevant functional groups was compiled and the presence or absence of the functional group was determined using the rdkit library function `rdkit.Chem.rdchem.GetSubstructMatches`. The functional group

definitions are given below. The molecules with exactly one aryl bromide, no alkyl bromides, no 6 membered aromatic rings with a bromide ortho or para to an aromatic nitrogen, no aryl iodides, no thionyl or sulfonyl halides, no silyl halides, no α -halo carbonyls, no acyl or thioacyl halides, and a molecular weight in the range of 195–480 were separated out to compose the aryl halide set (~5350). Next, 380 aryl halides were selected randomly from this set. Aryl halides previously used in the optimization of reaction conditions were also manually included. Using expert knowledge, any reactive/incompatible functional groups (e.g., oximes, isocyanates, etc.) were further identified and filtered out, resulting in a set of 365 aryl halides. Finally, Informer halides **X1–X15** were also included, resulting in a total of 380 selected aryl halides. Due to commercial availability, only 357 aryl halides were procured and used in the experiment (**Set B**, Manuscript Figure 4B, **Part II**).

Functional groups relevant to the aryl halide criteria

```
aryl_bromide = Chem.MolFromSmarts('[c;H0](-[Br])')
alkyl_br = Chem.MolFromSmarts('[C;H0](-[Br])')
ortho_to_6_member_aromatic_nitrogen = Chem.MolFromSmarts('[n]1:[c](-[Br])[a]:[a]:[a]:[a]1')
para_to_6_member_aromatic_nitrogen = Chem.MolFromSmarts('[n]1:[c]:[a]:[a](-[Br]):[a]:[a]1')
aryl_iodide = Chem.MolFromSmarts('[c;H0](-[I])')
thionyl_or_sulfonyl_halide = Chem.MolFromSmarts('[S](=O)[F,Cl,Br,I]')
silyl_halide = Chem.MolFromSmarts('[Si][F,Cl,Br,I]')
alpha_halo_carbonyl = Chem.MolFromSmarts('[C](=O)C[F,Cl,Br,I]')
acyl_or_thioacyl_halide = Chem.MolFromSmarts('C(=[O,S])[F,Cl,Br,I]')
```

Selection of amines for aryl halide scope experiment (Manuscript Figure 4B, Part II)

The selection of amines **A1–A8** (Figure S14) as coupling partners for the aryl halide scope experiment was based on the mean and standard deviation of the reaction performance for all 348 amines used in the earlier amine scope experiment (Manuscript Figure 4B, **Part I**). Amines with appreciable reactivities were selected with the criterion of a mean product UV absorbance being between 30000–110000 UV254 AreaAbs units. The final set of structurally diverse amines, **A1–A8**, was selected from this list following expert review ensuring varying reactivities. Among 152 amines with 30000 < UV254 AreaAbs < 110000, **A1–A8** were ranked between 1st and 141st based on mean UV254 AreaAbs and ranked between 17th and 149th based on standard deviation of UV254 AreaAbs, highlighting diverse selection of amines with varying reactivities.

VIII. Details of Featurization, Pistachio data curation, t-SNE, and UMAP

Generation of Morgan Fingerprints

Morgan Fingerprints were generated with the rdkit library function `rdkit.Chem.AllChem.GetMorganFingerprintAsBitVect()` with radius = 2.

Generation of machine learned “Quantum Mechanical” (QM) descriptor fingerprints

QM descriptors were predicted using a trained graph neural network (<https://github.com/yanfeiguan/qmdesc>).³ Predicted properties included atom-level descriptors (partial charge, Fukui indices, NMR shifts) and bond-level properties (bond order and bond distance). The reactive C–X, N–H, and C–N bonds were identified for each aryl halide, amine, and product respectively. Each of the atom level properties and the bond-level properties for each molecule were converted to a vector of length 25 via Gaussian Expansion (see below for details), and were concatenated to form a vector of length 250, which was used as the QM fingerprint vector for following studies.

Gaussian Expansion. In studies for which QM features were used, each QM descriptor was converted to a vector using a Gaussian expansion prior to use.⁴ Gaussian expansion converts a scalar QM descriptor value μ to a vector $\mathbf{d}$ with the following equation.

$$d_k = e^{-\frac{1}{2}\left(\frac{\mu - (r_{min} + k\Delta\mu)}{\sigma}\right)^2} \quad (0 \leq k \leq \frac{r_{max} - r_{min}}{\Delta\mu})$$

In this work, the Gaussian expansion was performed with the following setup.

$$\begin{aligned}\Delta\mu &= \sigma = 0.05 \times (\mu_{max} - \mu_{min}) \\ r_{max} &= \mu_{max} + 0.1 \times (\mu_{max} - \mu_{min}) \\ r_{min} &= \mu_{min} - 0.1 \times (\mu_{max} - \mu_{min})\end{aligned}$$

Generation of one-hot features. The one-hot encoded features were generated by the sklearn library function `sklearn.preprocessing.OneHotEncoder()`.

Curation of Pistachio data set (Manuscript Figure 4A)

Below summarizes the curation process to obtain ~48k Buchwald-Hartwig amination reaction data set that is relevant to the scope of this work.

1. Using NextMove software, query Pistachio data set (2023Q3) to get 'bromo Buchwald-Hartwig amination' reactions with yield information available.
2. Remove reactions with the following criteria:
 - i. SMILES cannot be parsed
 - ii. Multiple products were recorded
 - iii. Palladium catalyst was not used
 - iv. Intramolecular coupling reactions
 - v. Amine is not a primary or secondary amine
 - vi. Halide is not directly connected with an aromatic carbon in a ring.
3. Determine and annotate the type of amine (primary vs secondary).
4. Confirm that aryl bromide reactant contains 'aryl bromide' structure around the C atom that forms C–N bond in the product and amine reactant contains 'primary amine' or 'secondary amine' structure around the N atom that forms C–N bond in the product.

Construction of t-SNE plots (Manuscript Figure 4A and 8C). t-distributed stochastic neighbor embedding (t-SNE) plots visualizing the distribution of chosen molecules were constructed for the aryl halides and amines using `sklearn.manifold.TSNE`. t-SNE maps high-dimensional data onto a lower dimensional space for visualization. To calculate the t-SNE components, first, 2048-bit, radius-2 Morgan fingerprints of molecules were calculated for each molecule. Then, the t-SNE decomposition was performed with the cosine distance metric. The randomly chosen molecules appear well distributed in the larger pool.

Construction of UMAP plot (Manuscript Figure 1C). Uniform Manifold Approximation and Projection (UMAP) plot visualizing the distribution of coupling products and approved drugs was constructed using a Python package `umap-learn`. UMAP projects high-dimensional data onto a lower dimensional space for visualization. Using the DrugBank database,⁵ a list of 2623 approved small molecule drugs was compiled and 502 molecules with a bond between an aromatic carbon and a non-aromatic nitrogen (SMARTS pattern: [c;H0][NX3;+0;H0,H1]) with molecular weight <1000 were identified for comparison with our C–N coupling products. These 502 small molecule drugs, the 4236 products from **Part I** and **Part II** scope experiments in this work, the 123 products from Denmark's report,⁶ and the 5 products from Doyle & Dreher's report⁷ were used as the data set for this analysis.

To calculate the UMAP components, 14 physical chemical properties that are often used to assess the drug-like nature of small molecules were calculated using RDKit and scaled by empirical cumulative distribution functions.⁸ Those properties include molecular weight, number of rotatable bonds, number of hydrogen bond donors/acceptors, fraction of carbon atoms that are sp³ hybridized, number of rings, number of aromatic rings, number of atom stereo centers, topological polar surface area, Crippen logP, number of atoms, number of positive/negative atoms, and number of ring assemblies. Then, the UMAP decomposition was performed with the Euclidean distance metric.

IX. Additional Analyses

Distribution of binary reactivity class labels

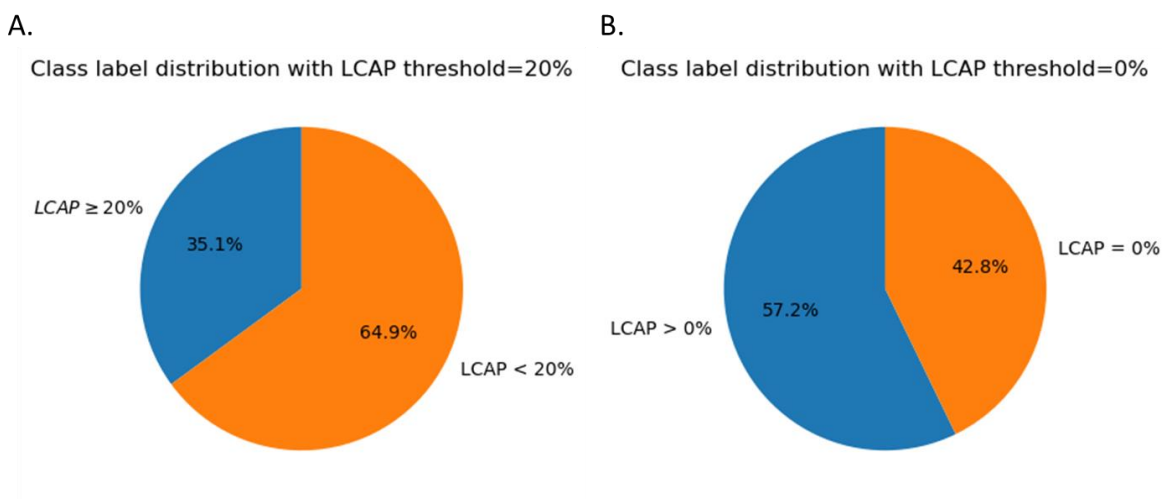


Figure S53. Binary reactivity class label distribution with LCAP thresholds of (A) 20% and (B) 0%

Generation of reactivity fingerprints and comparing reactivity trends

The reaction outcomes from **Part I** and **Part II** were separately used to construct a "reactivity fingerprint" (RFP) for each amine and aryl halide. The RFP of a molecule is a list of its reaction outcomes (LCAPs). This list of reaction outcomes is consistently ordered of by coupling partner for every RFP within the same category (e.g., amines or aryl halides). For example, the RFP of amine A1 from **Part I** would be: [LCAP_{A1+X1}, LCAP_{A1+X2}, LCAP_{A1+X3}, ...] and the RFP of amine A2 would be: [LCAP_{A2+X1}, LCAP_{A2+X2}, LCAP_{A2+X3}, ...]. The Pearson Correlation Coefficient (PCC) is then calculated for every pair of RFPs and assembled into the correlation matrices shown below. In the matrices below, the x and y axes correspond to the species label and each axes intersection corresponds to the calculated PCC for that pair of RFPs. A higher PCC value indicates that the two species are positively correlated and exhibit similar reactivity.

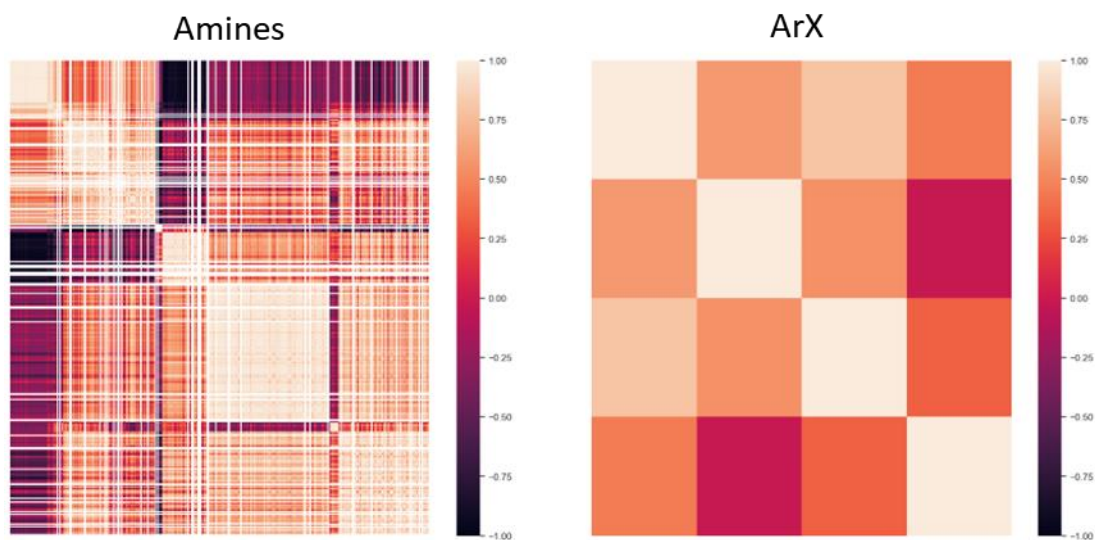


Figure S54. Correlation matrices for amine scope **Part I**. (Left) Correlation matrix of the amine reactivity fingerprints with amines 1–348 on the x and y axes. (Right) Correlation matrix of the aryl halide reactivity fingerprints with aryl halides **Br-1**, **X2**, **X3**, **X13** on the x and y axes.

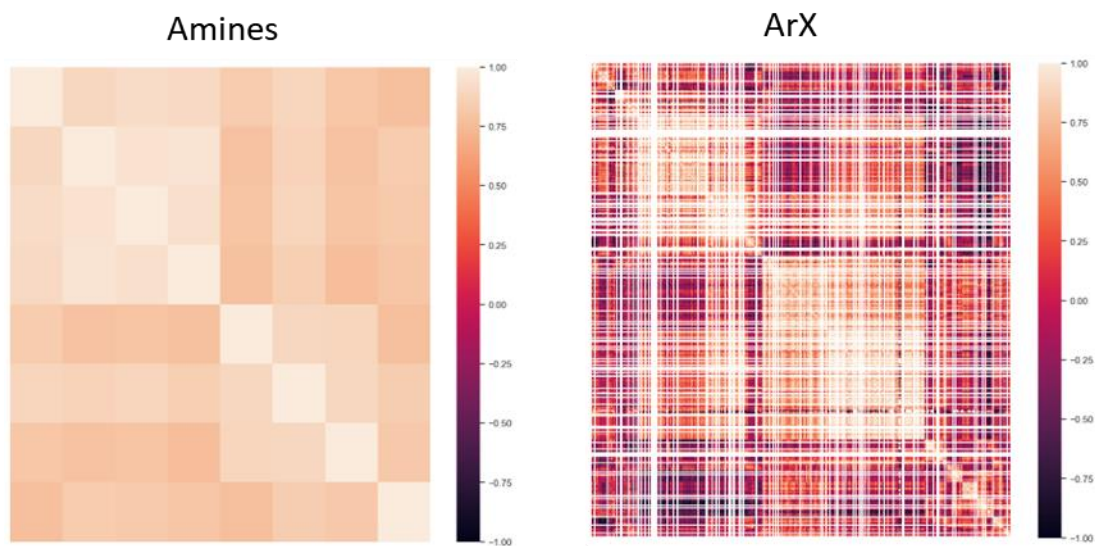


Figure S55. Correlation matrices for aryl halide scope **Part II**. (Left) Correlation matrix of the amine reactivity fingerprints with amines **A1–A8** on the x and y axes. (Right) Correlation matrix of the aryl halide reactivity fingerprints with aryl halides 1–357 on the x and y axes.

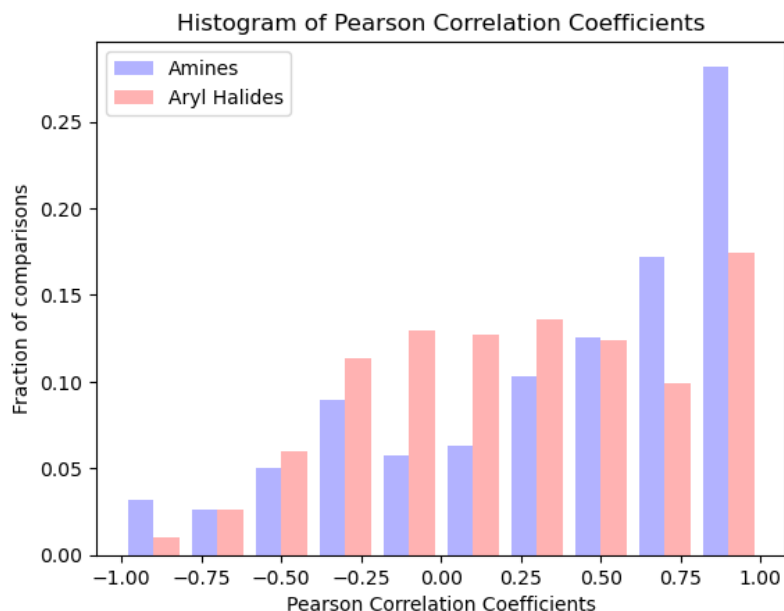


Figure S56. A fraction histogram of pairwise Pearson Correlation Coefficient (PCC) values for reactivity fingerprint comparison of each aryl halide and amine used in scope experiments **Part I** and **Part II**.

Figure S56 shows that the distribution of Pearson Correlation Coefficient (PCC) values of reactivity fingerprints for pairs of aryl halides and amines. Mathematically, the PCC between a non-reactive species (whose reactivity fingerprint is a null vector) and other species cannot be calculated, but for a more complete analysis we treated the correlation between two non-reactive species (two null vectors) as equal to 1 and included them in the histogram shown in Figure S56. PCC values for pairs of amines are generally higher than the PCC values for pairs of aryl halides, suggesting that the amines have overall more similar reactivity patterns compared to the aryl halides.

Correlation between similarity of structures and similarity of reactivities

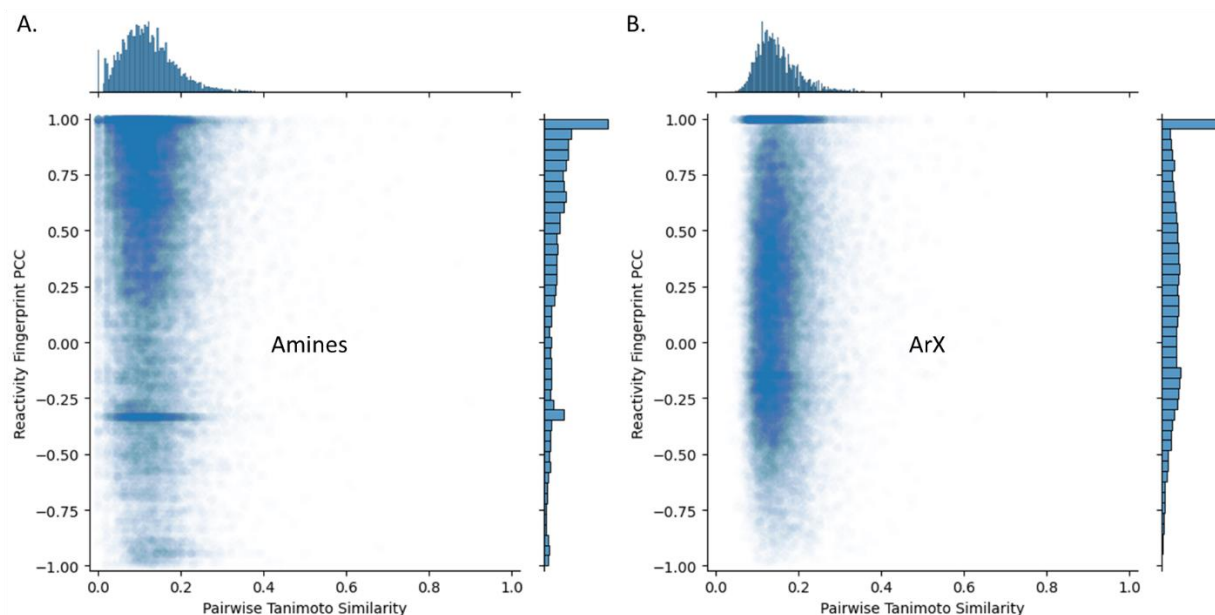


Figure S57. Scatter plot of pairwise Tanimoto similarity vs pairwise reactivity fingerprint Pearson Correlation Coefficient (PCC) for (A) amines and (B) aryl halides. Transparency was adjusted to show the density of data points.

Figure S57 plots pairwise Tanimoto similarity vs pairwise PCC of reactivity fingerprints for all amines and aryl halides in the reaction data. Correlation coefficients of the data points in Figure S57 were determined to be -0.045 for amines (A) and 0.025 for aryl halides (B), showing that there is no significant correlation between similarity of molecular structures (Tanimoto similarity) and similarity of reactivities (reactivity fingerprint PCC) in our data set. It indicates that the structural similarity of two molecules does not guarantee similar reactivities.

Comparison of overlaps between C–N coupling products and approved drugs (Manuscript Figure 1C)

In Figure 1C where C–N coupling products are visualized in the approved small molecule drug space using UMAP, there exist regions where C–N coupling products from Denmark's report⁶ and Doyle and Dreher's report⁷ seemingly overlap more with approved drugs compared to the C–N coupling products in our data set (Figure S58).

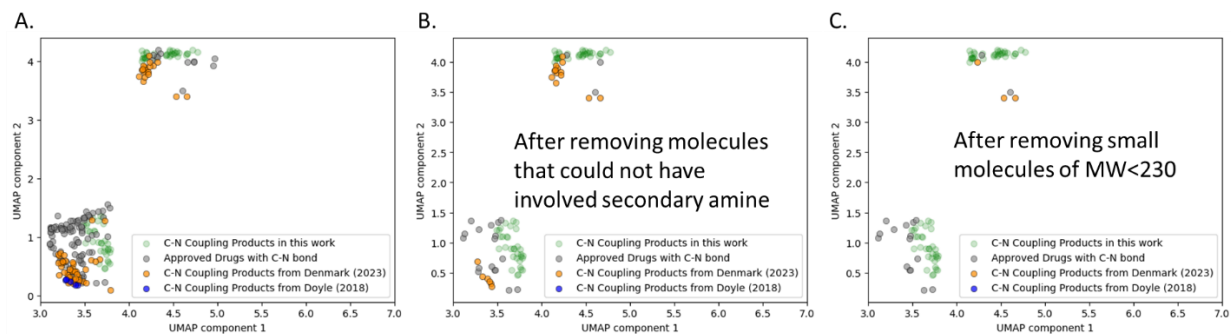


Figure S58. UMAP visualization of selected regions from Manuscript Figure 1C where C–N coupling products from previous reports overlap more with approved drugs compared to the C–N coupling products in this work. (A) Selected regions from Figure 1C. (B) Selected regions after filtering out molecules without a bond between an aromatic carbon and a non-aromatic tertiary nitrogen (C) Selected regions after filtering out small molecules with MW <230.

We investigated those regions in Figure S58A above and found that ~82% of approved drugs and ~72% of coupling products in previous reports did not have a bond between an aromatic carbon and a non-aromatic tertiary nitrogen, implying that those substrates are not the products of C–N couplings using secondary amines and thus are not comparable with our data set (Figure S58B). Also, among the remaining molecules, 15% of approved drugs and ~84% of coupling products in previous reports were small molecules (MW <230), which would also not be comparable with our data set due to the MW filters implemented in this work (amine MW >115 and aryl halide MW >195, see **Selection of Secondary Amines and Aryl Halides for Scope Experiments** section). After removing molecules that are not comparable to our data set, there is no significant disparity in the extent of overlap between approved drugs, C–N coupling products in previous reports, and C–N coupling products in this work (Figure S58C).

Investigation of amines in Pistachio data that are different from the amines in MBBC library (Manuscript Figure 4A)

In Manuscript Figure 4A amine chemical space visualization, there are a significant number of secondary amines from the Pistachio data set with TSNE1 >60 while almost all amines from the MBBC library show TSNE1 <60. Figure S59 visualizes a random subset of amines from the Pistachio data set in TSNE1 >60 region.

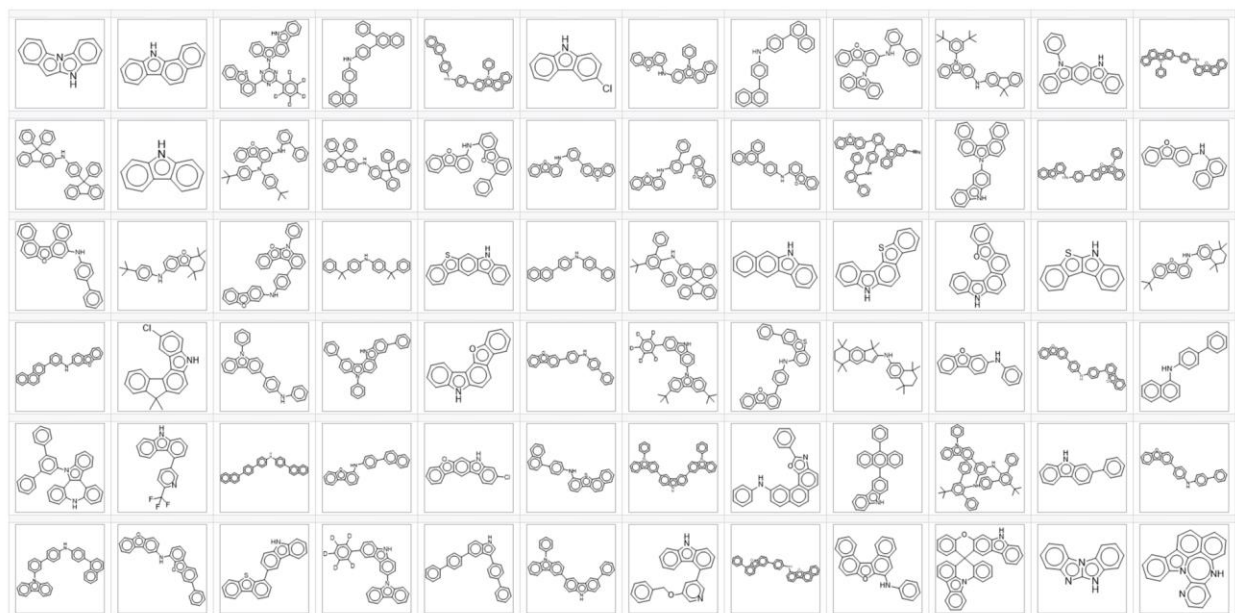


Figure S59. Random subset of amines from the Pistachio data with TSNE1 >60 in Manuscript Figure 4A (left).

It was found that most of the amines from the Pistachio data set in this region (TSNE1 >60) are considered out of the scope of our data set as 55% of them were large molecules with MW >400 (secondary amines in our data set were filtered to have MW between 115–400). Among the remaining 45%, more than half of them contained an aromatic N–H bond which was also filtered out in the secondary amine selection process in this work. Figure S60 below shows the amine space after removing large molecules and amines with aromatic N–H bonds. When compared to Manuscript Figure 4A (left), it can be clearly seen that most of the amines in the Pistachio data set that did not overlap with the amines in our data set have been removed. We believe that most of the amines in the Pistachio data that still do not overlap with the amines in our data set in Figure S60 would be further removed by applying the same remaining functional group filters that were applied to the amines in our selection process (see **Selection of Secondary Amines and Aryl Halides for Scope Experiments** section).

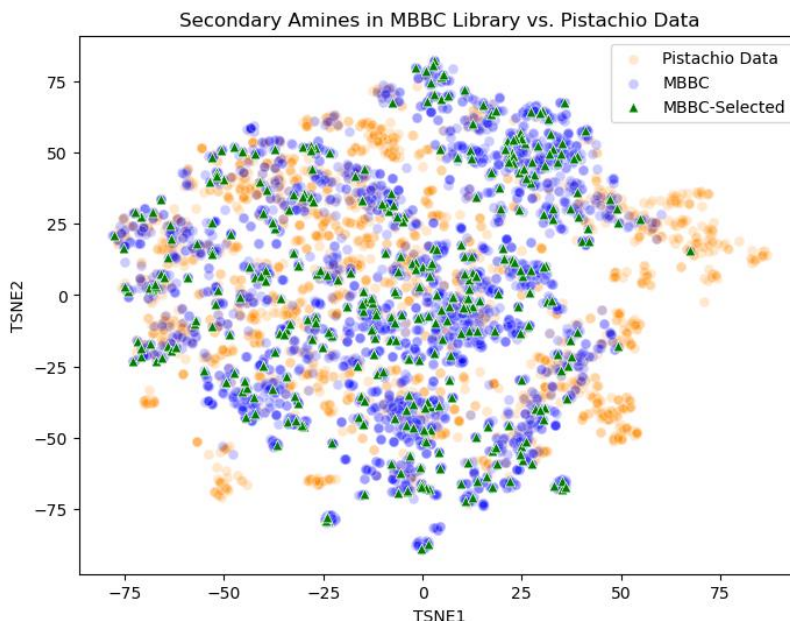


Figure S60. Amine chemical space visualization after removing large amines and amines with aromatic N–H bond.

Clustering analysis of amines and aryl halides in MBBC library

K-means clustering analysis was performed based on t-SNE coordinates of amines and aryl halides to evaluate representative selection from the MBBC library. The number of clusters was determined to be 250 by elbow method and silhouette score analysis, which was the maximum number of clusters explored. Figures S61 and S62 shows the distribution of all substrates from the amine and aryl halide MBBC libraries, respectively, in 250 clusters as well as the distribution of selected substrates used in building our data set. As briefly mentioned in the manuscript, these results confirm that diverse and representative selection of amines and aryl halides from the MBBC library can be achieved using a random selection strategy.

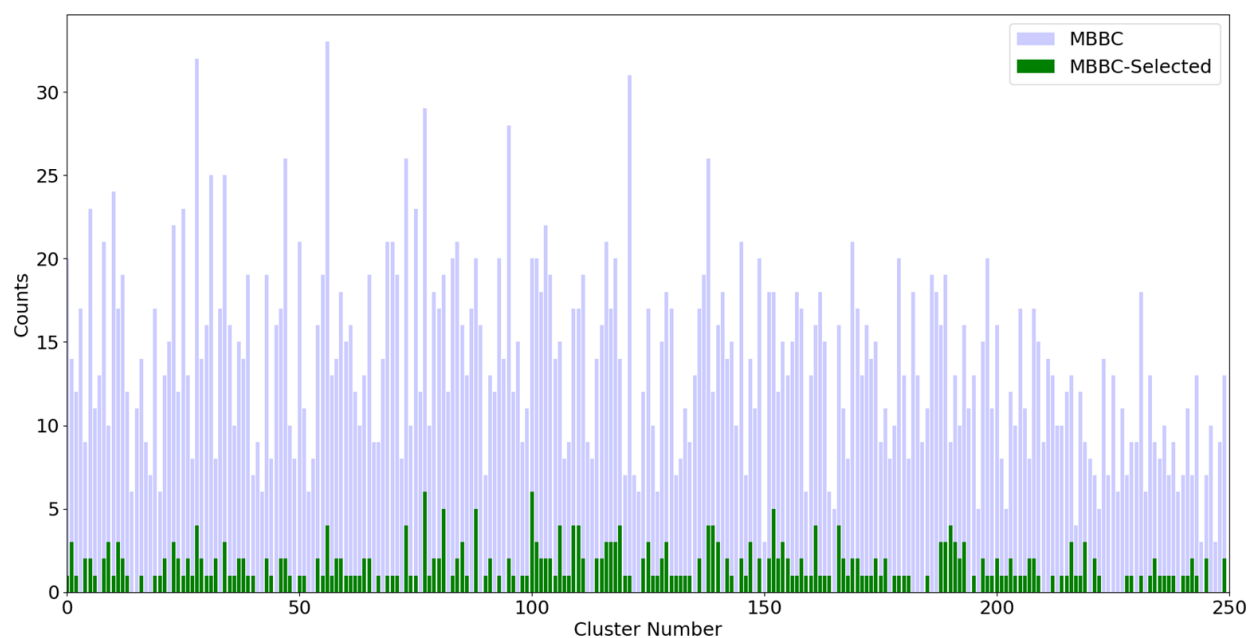


Figure S61. K-means clustering analysis of amines in the MBBC chemical space illustrating that 191/250 clusters are represented by the randomly selected amines used in **Part I**.

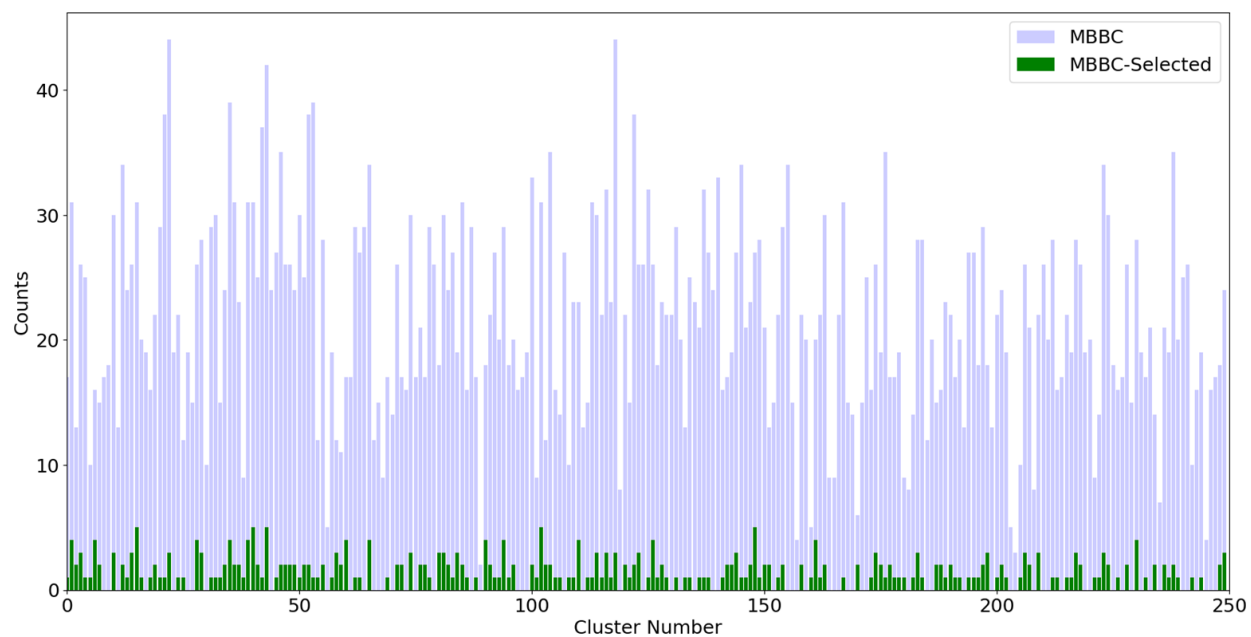


Figure S62. K-means clustering analysis of aryl halides in the MBBC chemical space illustrating that 180/250 clusters are represented by the randomly selected aryl halides used in **Part II**.

X. Data Splitting

The data was split as a series of N-fold cross validations. Due to the structure of diverse data set generation, the data set contained far more data for amines **A1–A8** and aryl halides **Br-1**, **X2**, **X3**, and **X13**. Thus, during the splitting procedure, the reactions involving these 12 molecules were assigned to separate splits (see below for more details).

Random split. The data set was split randomly into 10 subsets S_0 – S_9 . For fold i ($i = 0$ – 9), subset S_i and $S_{(i-1) \bmod 10}$ were held out as the test and validation set respectively. The remaining 8 folds were used as training data. This resulted in 10 random splits of the data (Figure S63).

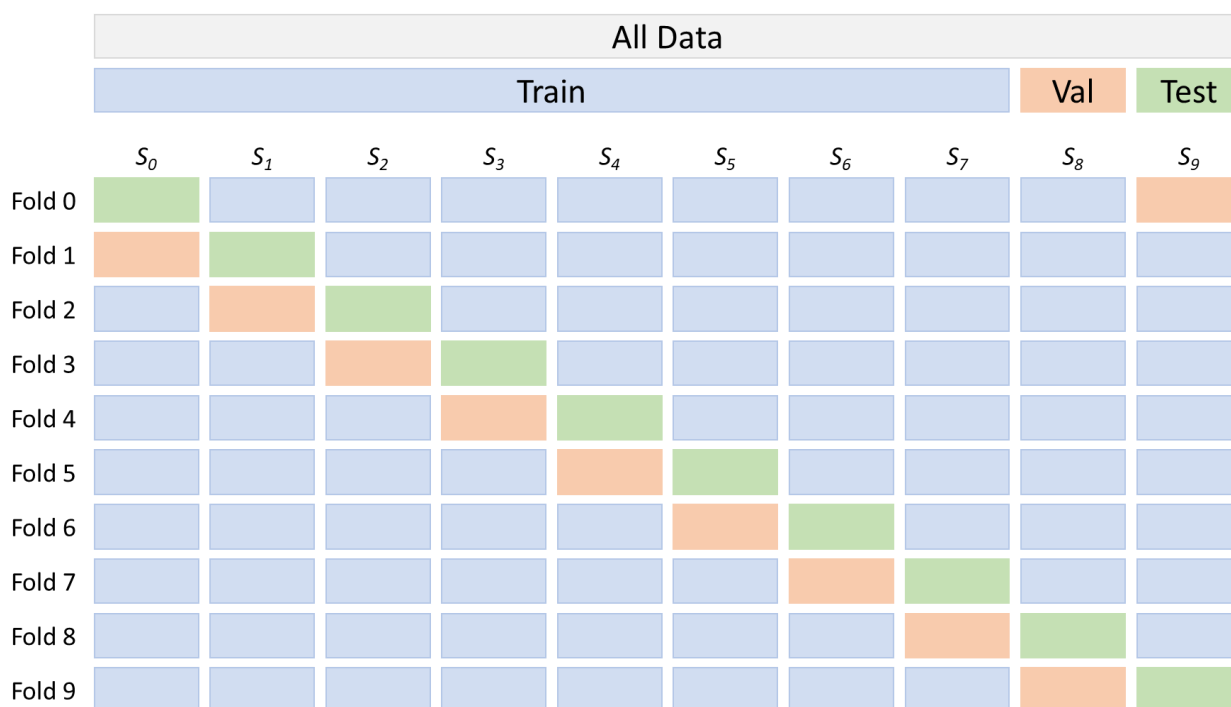


Figure S63. Visualization of the 10 splits created for the Random data split cross validation.

Amine out-of-scope split (Amine OOS): The amines in the data set were extracted, then all the amines except **A1–A8** were partitioned randomly into 10 subsets A_0 – A_9 . Then the eight amines **A1–A8** with the most data were added to amine subsets A_0 – A_7 respectively. For fold i ($i = 0$ – 9), any reactions that had an amine in A_i or $A_{(i-1) \bmod 10}$ were held out as the test and validation set respectively. The other reactions were used as training data. This resulted in 10 Amine OOS splits of the data (Figure S64).

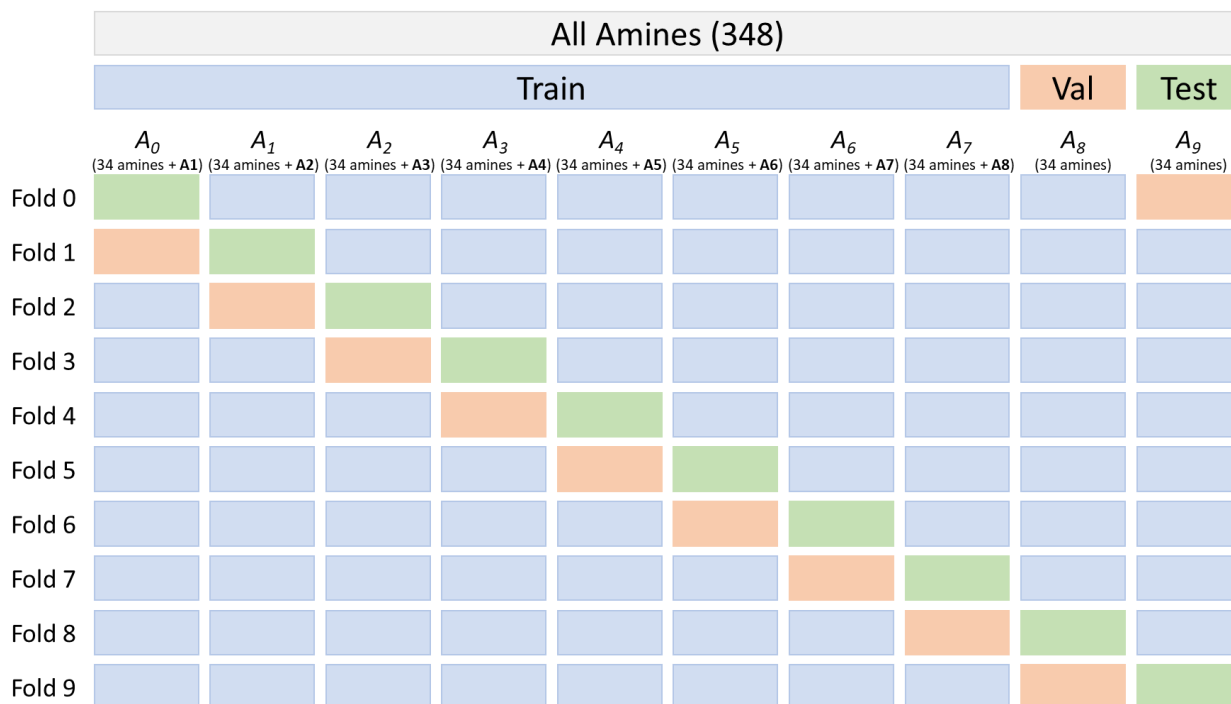


Figure S64. Visualization of the 10 splits created for the Amine OOS data split cross validation.

Aryl halide out-of-scope split (ArX OOS): The aryl halides in the data set were extracted, then all the aryl halides except **Br-1**, **X2**, **X3**, and **X13** were partitioned randomly into 10 subsets X_0 – X_9 . Then the four aryl halides **Br-1**, **X2**, **X3**, and **X13** with the most data were added to X_0 – X_3 respectively. For fold i ($i = 0$ – 9), any reactions that had an aryl halide in X_i or $X_{(i-1) \bmod 10}$ were held out as the test and validation set respectively. The other reactions were used as training data. This resulted in 10 ArX OOS splits of the data (Figure S65).

All ArX (357)										
Train								Val	Test	
	X_0 (35 ArX + Br-1)	X_1 (35 ArX + X2)	X_2 (35 ArX + X3)	X_3 (35 ArX + X13)	X_4 (36 ArX)	X_5 (36 ArX)	X_6 (36 ArX)	X_7 (35 ArX)	X_8 (35 ArX)	X_9 (35 ArX)
Fold 0										
Fold 1										
Fold 2										
Fold 3										
Fold 4										
Fold 5										
Fold 6										
Fold 7										
Fold 8										
Fold 9										

Figure S65. Visualization of the 10 splits created for the ArX OOS data split cross validation.

Both out-of-scope split (Both OOS): The subsets A_0 – A_7 and X_0 – X_3 that were utilized for the Amine OOS and ArX OOS splits were used to form eight sets of pairs B_0 – B_7 as follows: (A_0 , X_0), (A_1 , X_1), (A_2 , X_2), (A_3 , X_3), (A_4 , X_0), (A_5 , X_1), (A_6 , X_2), (A_7 , X_3). For Both OOS fold i ($i = 0$ – 7), all reactions that involved an amine in B_i AND an aryl halide in B_i were held out as the test set. Of the remaining reactions, the ones that involved an amine in $B_{(i-1) \bmod 8}$ AND an aryl halide in $B_{(i-1) \bmod 8}$ were held out as the validation set. The training set for Both OOS fold i was constructed by taking the full data set and removing all the reactions that involved any amine OR aryl halide in the validation or test sets. This resulted in 8 Both OOS splits of the data (Figure S66).

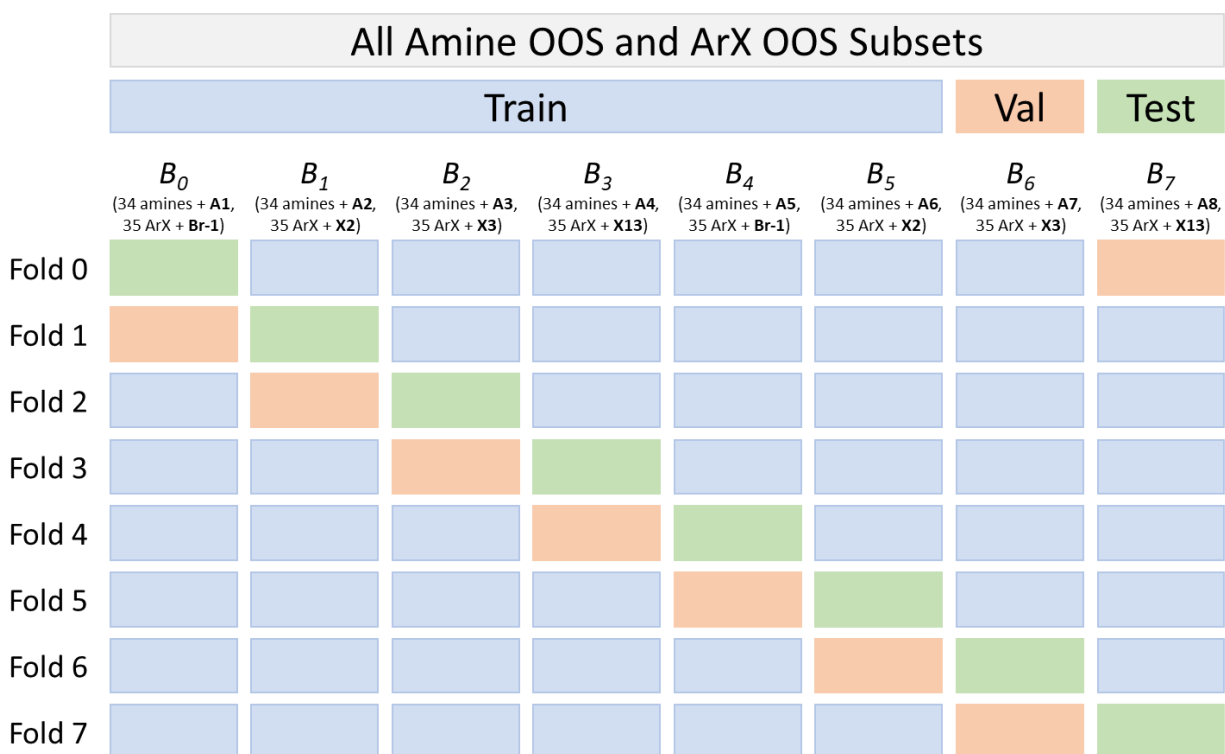


Figure S66. Visualization of the 8 splits created for the Both OOS data split cross validation.

Dimensionality Reduction Split (DRS): The eight amines (**A1–A8**) and four aryl halides (**Br-1**, **X2**, **X3**, and **X13**) with the most data formed four sets D_0 – D_3 as follows: (**A1**, **A2**, **X2**), (**A3**, **A4**, **X3**), (**A5**, **A6**, **X13**), (**A7**, **A8**, **Br-1**). For DRS25 fold i ($i = 0$ – 3), all the reactions involving amines OR aryl halide in D_i were used as the training set. For DRS75 fold i ($i = 0$ – 3), all the reactions except the ones involving amines OR aryl halides in D_i were used as the training set. For DRS50 folds, six different sets of amines and aryl halides (DD_0 – DD_5) were formed by combining two sets among D_0 – D_3 (e.g., $DD_0 = D_0 + D_1$, $DD_1 = D_0 + D_2$, etc.). For DRS50 fold i ($i = 0$ – 5), all the reactions involving amines OR aryl halides in DD_i were used as the training set. For all three types of DRS splits, validation reactions and test reactions were randomly selected without replacement among the reactions that were not included in the training set, while keeping the same ratio of Training:Validation:Test=8:1:1. These resulted in 4 DRS25 splits (Figure S67), 4 DRS75 splits (Figure S68), and 6 DRS50 splits (Figure S69).

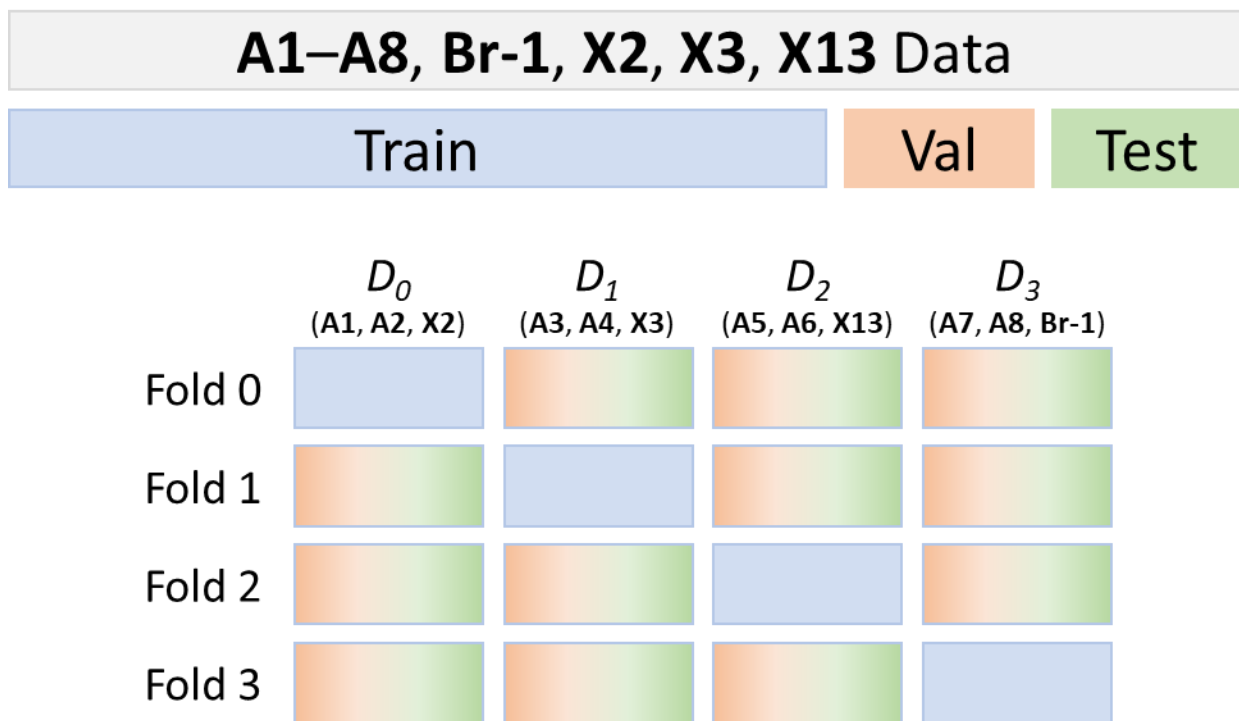


Figure S67. Visualization of the 4 splits created for the DRS25 data split cross validation.

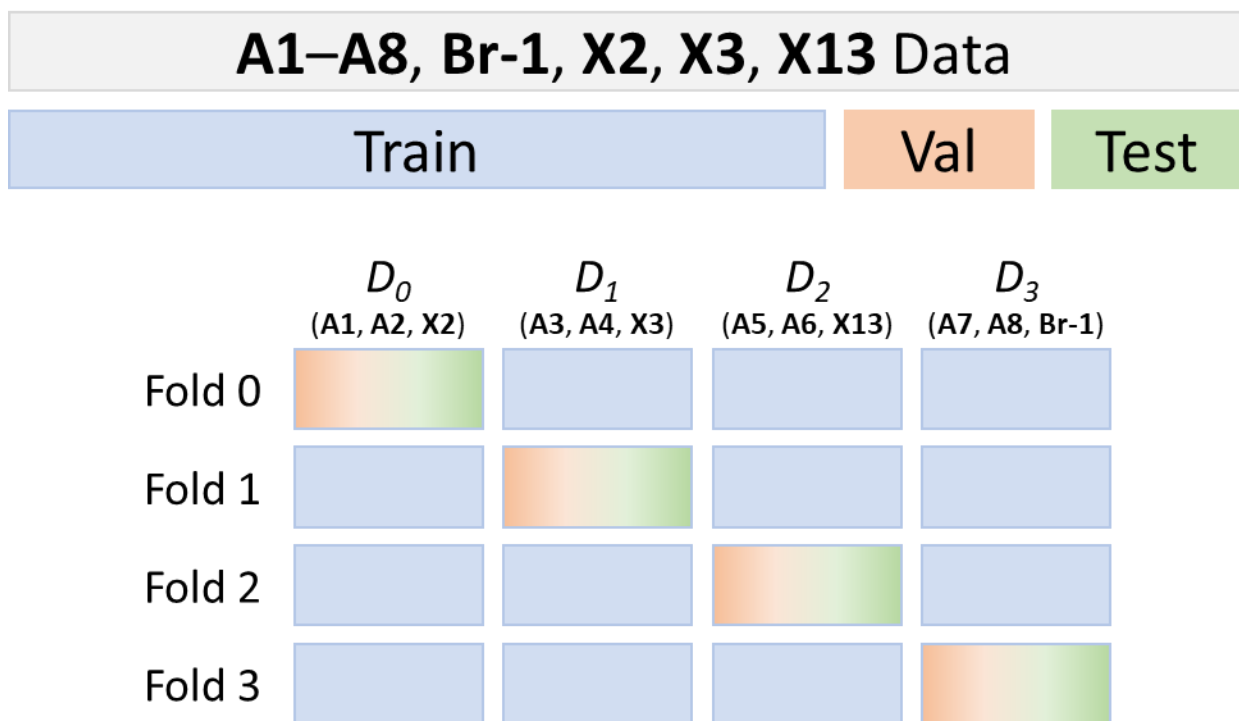


Figure S68. Visualization of the 4 splits created for the DRS75 data split cross validation.

A1–A8, Br-1, X2, X3, X13 Data						
Train				Val	Test	
	DD_0 (A1, A2, A3, A4, X2, X3)	DD_1 (A1, A2, A5, A6, X2, X13)	DD_2 (A1, A2, A7, A8, X2, Br-1)	DD_3 (A3, A4, A5, A6, X3, X13)	DD_4 (A3, A4, A7, A8, X3, Br-1)	DD_5 (A5, A6, A7, A8, X13, Br-1)
Fold 0						
Fold 1						
Fold 2						
Fold 3						
Fold 4						
Fold 5						

Figure S69. Visualization of the 6 splits created for the DRS50 data split cross validation.

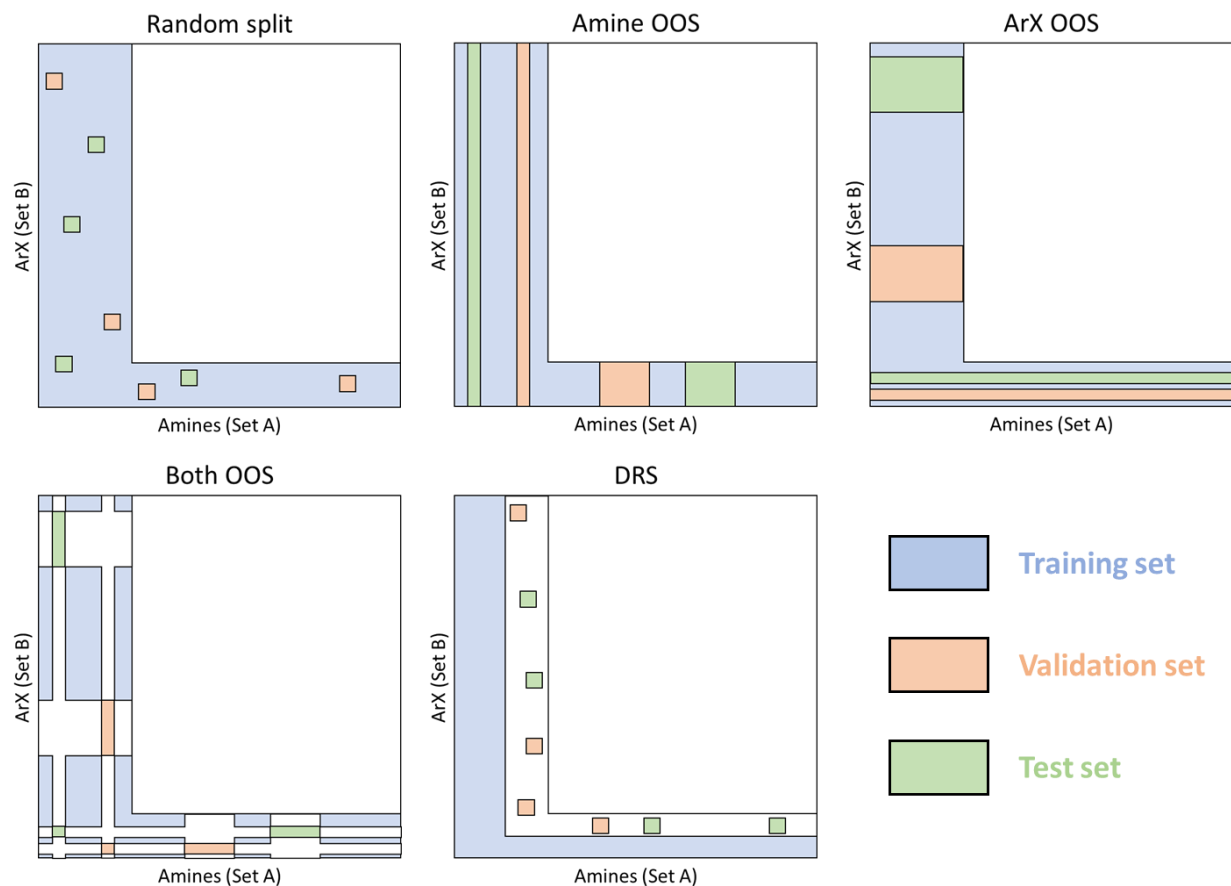


Figure S70. Splitting strategies including both validation and test sets.

Split Index	Random	ArX OOS	Amine OOS	Both OOS	DRS50	DRS25	DRS75
0	[3364 420 420]	[2952 624 628]	[3221 493 490]	[2248 70 70]	[2110 262 262]	[1057 131 131]	[3159 394 394]
1	[3364 420 420]	[2952 628 624]	[3224 490 490]	[2250 70 69]	[2110 262 262]	[1057 131 131]	[3159 394 394]
2	[3364 420 420]	[2952 624 628]	[3221 490 493]	[2248 69 70]	[2110 262 262]	[1057 131 131]	[3159 394 394]
3	[3364 420 420]	[3288 628 288]	[3221 493 490]	[2248 70 69]	[2110 262 262]	[1057 131 131]	[3159 394 394]
4	[3364 420 420]	[3632 288 284]	[3221 490 493]	[2248 69 70]	[2110 262 262]		
5	[3364 420 420]	[3632 284 288]	[3221 493 490]	[2248 70 69]	[2110 262 262]		
6	[3364 420 420]	[3636 288 280]	[3221 490 493]	[2248 69 70]			
7	[3364 420 420]	[3644 280 280]	[3575 493 136]	[2246 70 70]			
8	[3360 420 424]	[3644 280 280]	[3932 136 136]				
9	[3360 424 420]	[3300 280 624]	[3575 136 493]				

Table S1. Split sizes. Training, validation, and test set sizes are given as a tuple. Following the construction details above, the splits used for the modeling were constructed as reflected in this table.

XI. Modeling Details

Model Optimization Workflow

To find the model with the best performance, different model types, input features, and model hyperparameters were optimized. To avoid combinatorial explosion in grid search, model optimization was split into two phases: ‘Model Type & Input Feature Optimization’ (Phase 1) and ‘Model Hyperparameter Optimization’ (Phase 2). In Phase 1, default model hyperparameters were used and only model types, input features, and feature scaler on/off (Empirical cumulative distribution function (ECDF) scaler) were varied. Model performance metrics were given as an average across the folds described earlier in the data splitting details section and the best-performing model & input feature combination was identified for each data split (Random split, Amine OOS split, ArX OOS split, Both OOS split, DRS25 split, DRS50 split, DRS75 split) based on validation Accuracy@10% as described in the manuscript. In Phase 2, model hyperparameters were varied and optimized while keeping the same model type, input feature, and scaler on/off previously identified for each data split. After Phase 2, two additional runs of training and inference were performed for the best-performing models so that model performance uncertainties (standard error) could be estimated (see error bars in Manuscript Figure 7A).

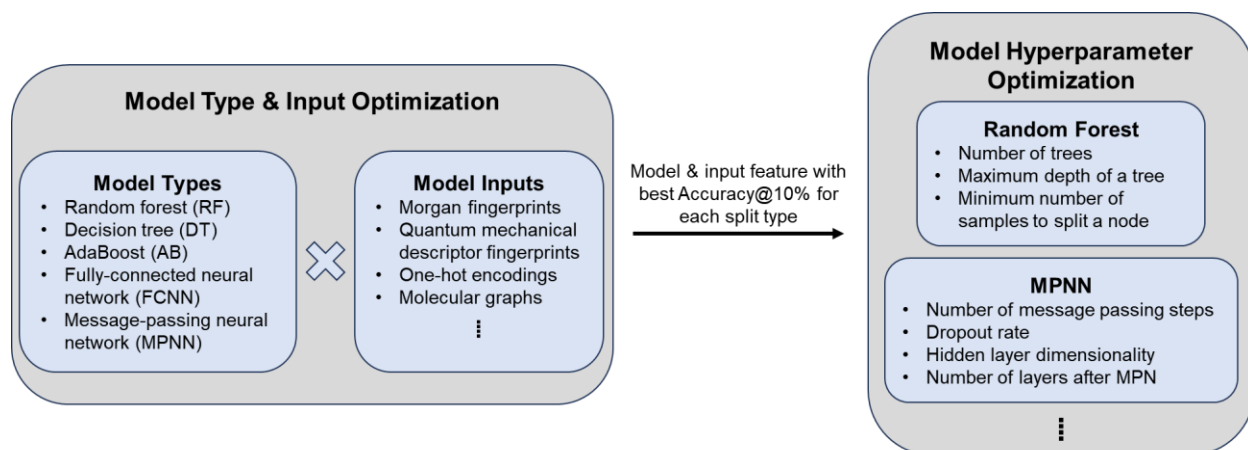


Figure S71. Model optimization workflow.

Model Implementation Details

Below are the lists of models, input features, and hyperparameters explored.

- *Classification models*: Random Forest, Decision Tree, AdaBoost, Fully Connected Neural Network (FCNN), Message-Passing Neural Network (MPNN)
- *Regression models*: Random Forest, Decision Tree, Lasso, Ridge, FCNN, MPNN
- *Input features*: 128-, 256-, 512-bit Morgan fingerprints (FP128, FP256, FP512), Concatenated vector of Morgan fingerprints and quantum mechanical fingerprints (FP128+QMFP, FP256+QMFP, FP512+QMFP), Concatenated vector of one-hot encodings of amine and aryl halide.
 - In Phase 2, 1024-bit Morgan fingerprint (FP1024) and FP1024+QMFP were also explored as input features along with different combinations of model hyperparameters. Those longer input features were not explored in Phase 1 to reduce the time needed for model training/evaluation to a reasonable range.
- *Scaler on/off*: Input features were scaled or not scaled using empirical cumulative distribution function.
- *Random Forest model hyperparameters*: number of trees (`n_estimators` = 100, 300), maximum tree depth (`max_depth` = None, 10, 20, 30), minimum number of samples to split a node (`min_samples_split` = 2, 3, 4). Default parameters are underlined.
- *MPNN model hyperparameters*: number of message passing steps (`depth` = 3, 4), dropout rate (`dropout` = 0, 0.3), hidden layer dimensionality (`hidden_size` = 300, 500), hidden layer dimensionality of feed-forward neural network part of the model (`ffn_hidden_size` = 300, 500), number of hidden layers in feed-forward neural network part of the model (`ffn_num_layers` = 1, 2, 3). Default parameters are underlined.

Random Forest, Decision Tree, AdaBoost, Lasso and Ridge models were implemented using Python `scikit-learn` library. MPNN model was implemented using `chemprop` package with reaction SMILES as inputs that were converted to graphs in the model.⁹ MPNN model also accepted additional features as inputs along with the default input of reaction graphs, so in Phase 1 of MPNN model exploration, FP256 and QMFP features were explored as additional inputs.

Fully Connected Neural Network model was implemented using PyTorch, with the general architecture shown below.

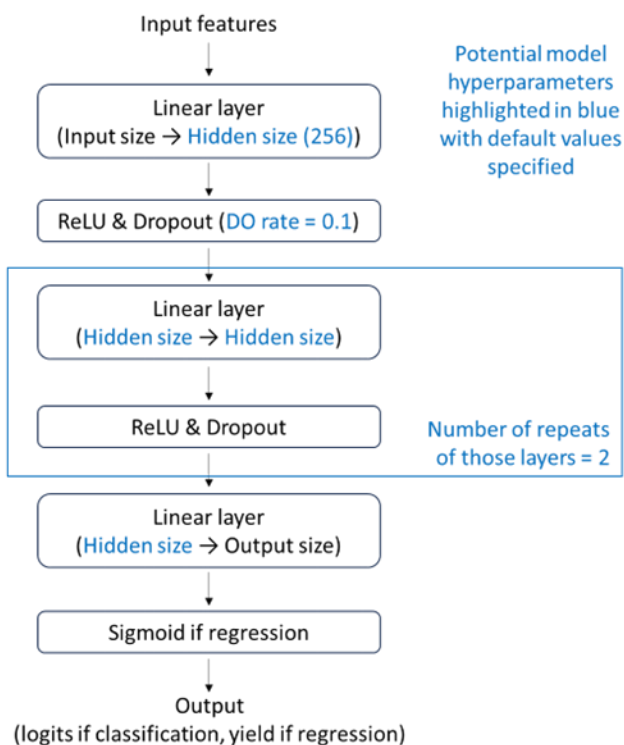


Figure S72. General architecture of Fully Connected Neural Network model.

Definition of Model Confidence

Model predictions of binary labels are based on model scores ranging from 0 to 1. A score closer to 1 indicates a high-yield reaction, while a score closer to 0 indicates a low-yield reaction. We define the model confidence as follows. For high-yield predictions, model confidence equals the model score. For low-yield predictions, model confidence is $1 - \text{model score}$. Thus, a model confidence greater than 0.7 corresponds to a score greater than 0.7 for high-yield reactions or less than 0.3 for low-yield reactions.

XII. Modeling Results and Additional Discussions

A complete table of the results from all model runs can be found in a separate spreadsheet that we provided as Supplementary Information. Some key results are visualized below.

Binary classification with 20% LCAP threshold

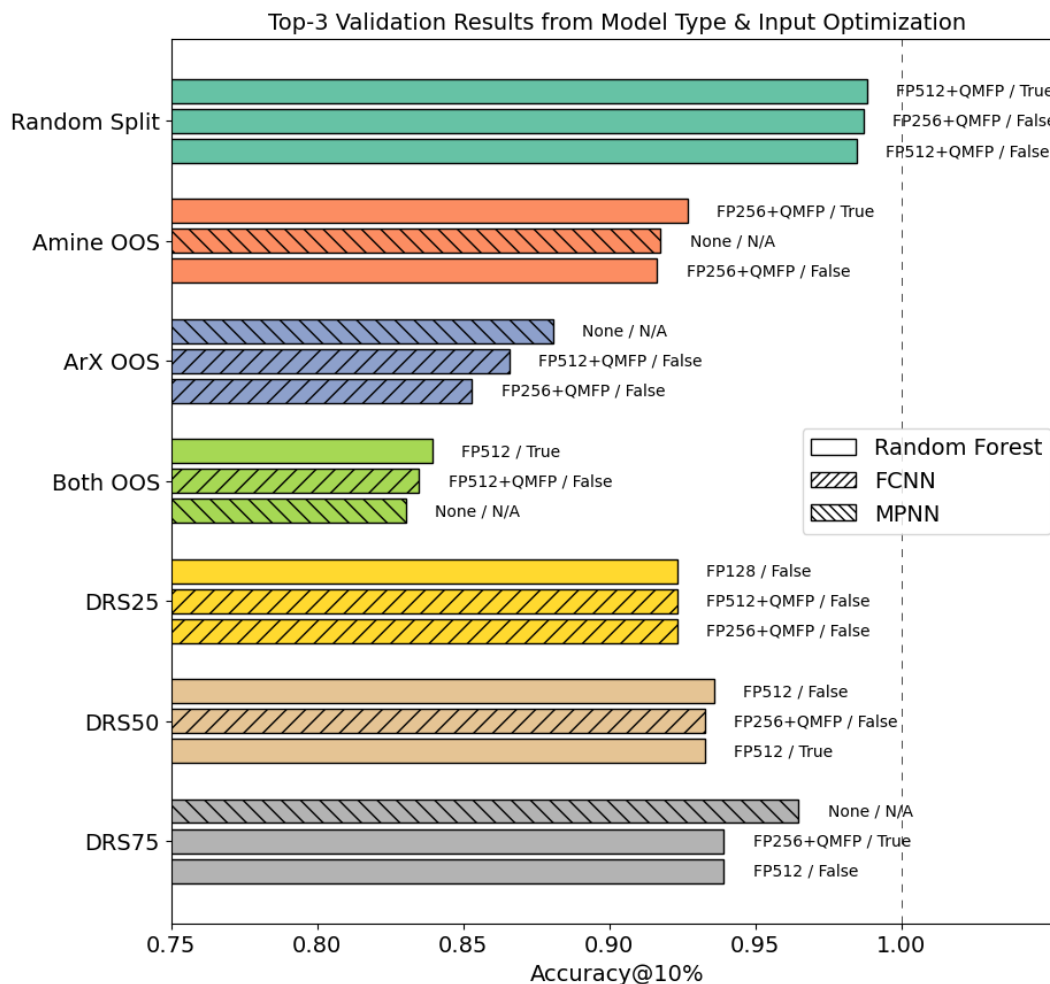


Figure S73. Best-performing model type & input feature combination for each data split in ‘Model Type & Input Feature Optimization’ (Phase 1) based on validation Accuracy@10%. Model types are indicated by hatches and input feature and the use of feature scaler (True/False) are noted in texts.

As briefly mentioned in the manuscript, supplementing Morgan fingerprints with quantum mechanical fingerprints was helpful in many cases as seen in Figure S73. However, there were also cases in which Morgan fingerprints alone without QM fingerprints performed better and there were no clear and generalizable trends.

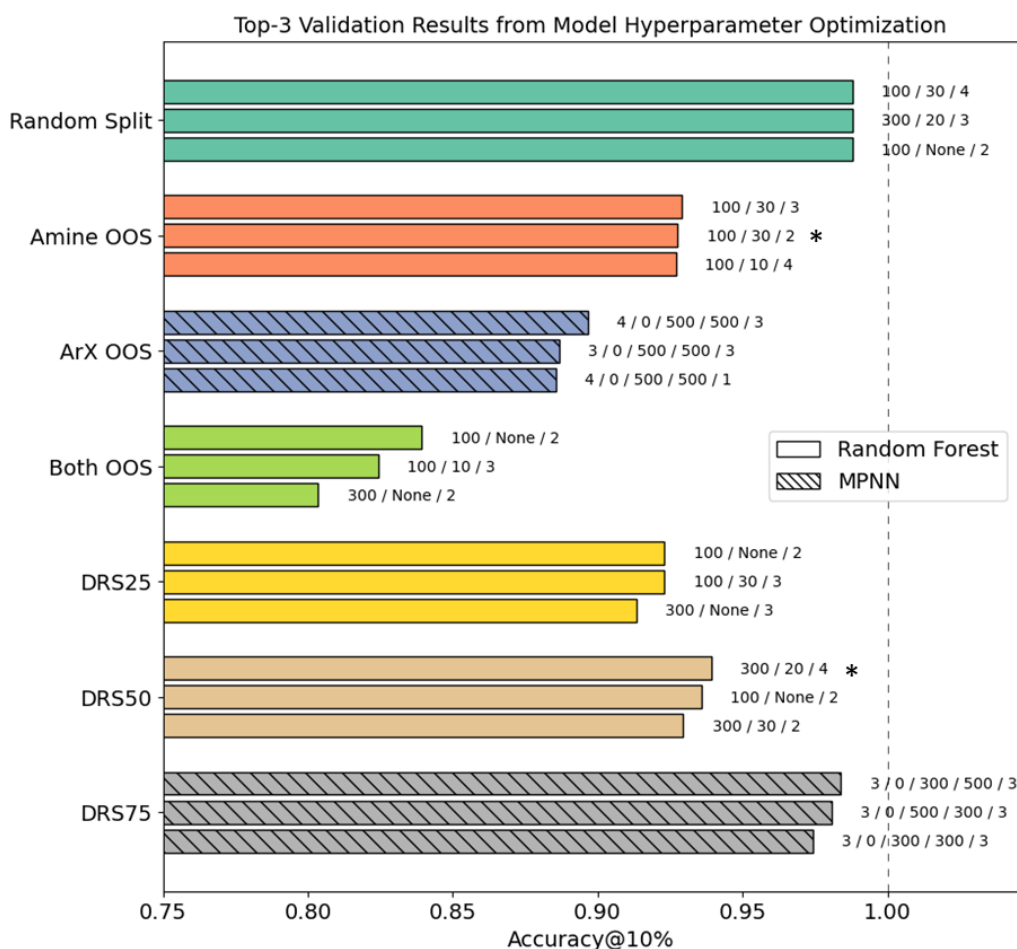


Figure S74. Best-performing model hyperparameters for each data split in ‘Model Hyperparameter Optimization’ (Phase 2) based on validation Accuracy@10%. Model types are indicated by hatches and hyperparameters are noted in texts. For Random Forest models, hyperparameters are listed in the order of `n_estimators` / `max_depth` / `min_samples_split`. For MPNN models, hyperparameters are listed in the order of `depth` / `dropout` / `hidden_size` / `ffn_hidden_size` / `ffn_num_layers`. Asterisk (*) indicates that FP1024+QMFP were the input features.

Since longer Morgan fingerprints (1024-bit) were explored as inputs in Phase 2 for the reasons mentioned earlier, Figure S74 contains some results from using FP1024+QMFP as input features and are marked with asterisks. For the best-performing models in Figure S74 for each data split, training and inference processes were repeated two more times for estimating model uncertainty. In Manuscript Figure 7, final test results with additional model evaluation metrics for the best-performing models are presented as the average of three training/inference runs with error bars representing 1 standard error.

Binary classification with 0% LCAP threshold and regression results

Other than the binary classification with 20% LCAP as the threshold discussed in the manuscript, we also investigated binary classification with 0% LCAP as the threshold and regression in Phase 1. Best-performing models for those analyses from Phase 1 are summarized below.

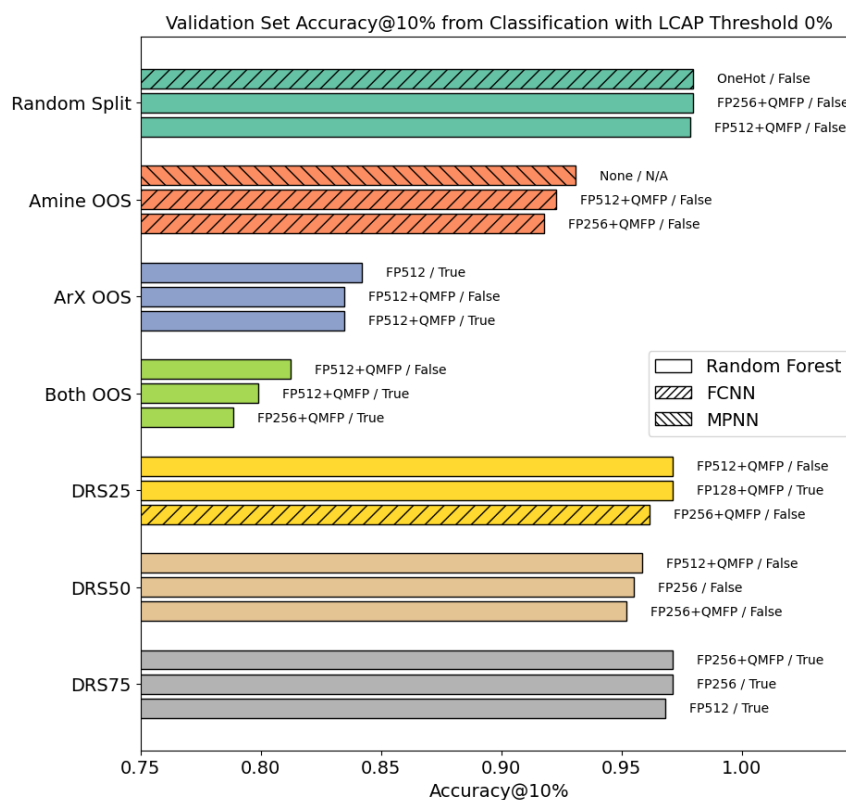


Figure S75. Best-performing model type & input feature combination for each data split (from Phase 1) for binary classification with 0% LCAP threshold based on validation Accuracy@10%. Model types are indicated by hatches and input feature and the use of feature scaler (True/False) are noted in texts.

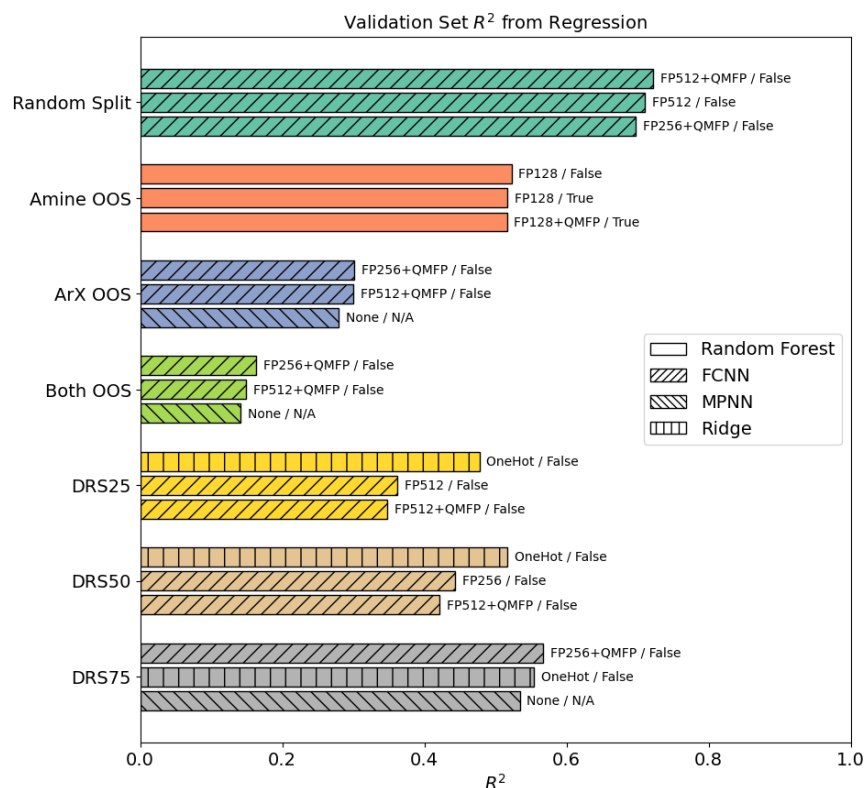


Figure S76. Best-performing model type & input feature combination for each data split (from Phase 1) for regression task based on validation R^2 (Coefficient of Determination). Model types are indicated by hatches and input feature and the use of feature scaler (True/False) are noted in texts.

In some data splits in Figures S75 and S76, one-hot encodings that lack chemical information performed relatively well compared to other input features with rich chemical information. To rationalize this observation, we revisited our reaction data and found that there are many occurrences in the data set where all or most of the reactions associated with certain amines or aryl halide species have exactly 0% LCAP. We hypothesize that these better-than-expected performances with one-hot encodings as inputs are arising from the models learning to predict 0% LCAP whenever it sees those certain amines and/or aryl halides. In these instances, one-hot encodings without any additional chemical information or input are sufficient enough for good predictivity. However, one-hot encodings were not among the best model input features for the more challenging predictions (e.g., out-of-scope splits), which is consistent with the models' inability to extrapolate to new structures using these input features.

Correlation between model confidence and prediction accuracy

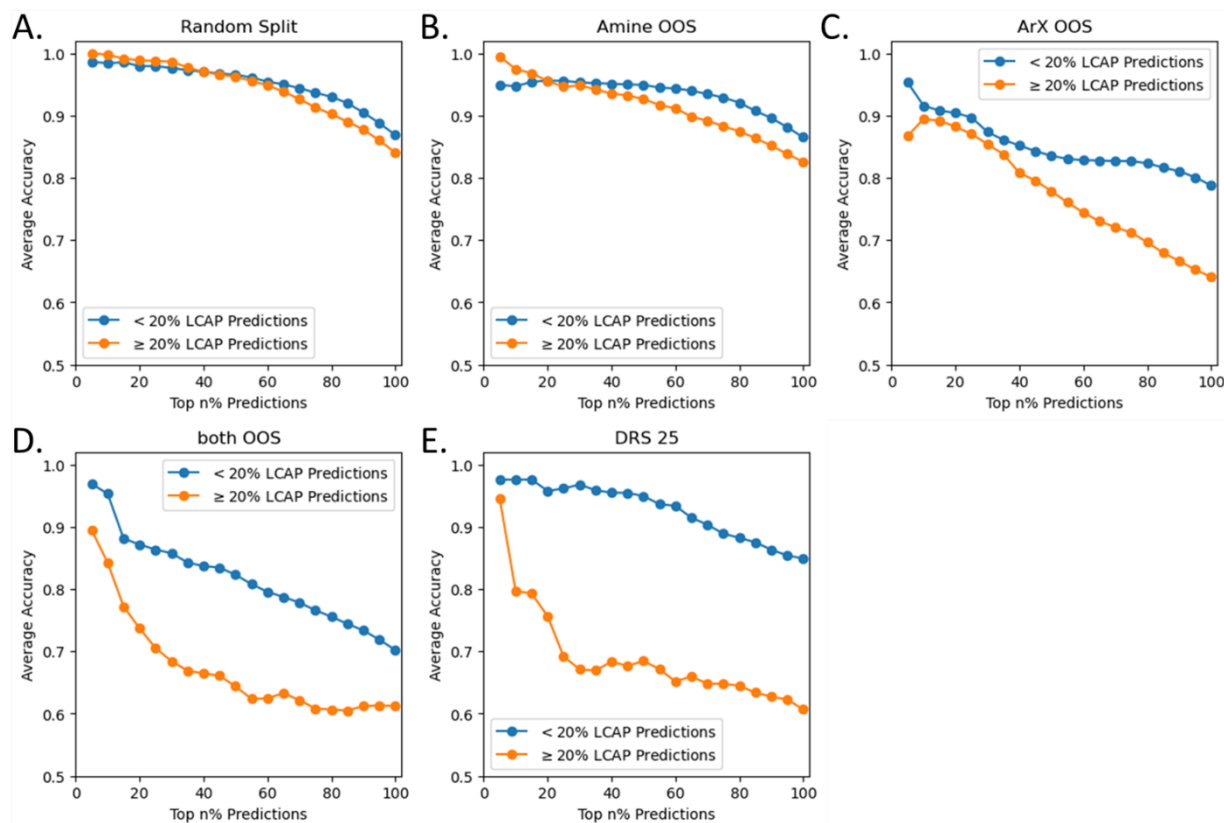


Figure S77. Integral accuracy averaging plots for test set predictions of (A) random split, (B) Amine out-of-scope (OOS) split, (C) ArX OOS split, (D) Both OOS split, and (E) dimensionality reduction split (DRS25).

To quantitatively investigate the correlation between model confidence and model prediction accuracy, we plotted integral accuracy averaging plots for each data split type.¹⁰ In Figure S77, each data point shows the average accuracy (y-axis) of top n% predictions (x-axis) based on model confidence for each classification label (<20% LCAP predictions in blue, $\geq 20\%$ LCAP predictions in orange). The prediction data used in Figure S77 is the same as used in Figure 7B. It clearly shows a positive correlation between model confidence and model accuracy, indicating that our prediction models can be used to identify promising reactions and discard reactions that are unlikely to succeed.

Also, Figure S77 shows that <20% LCAP predictions are generally more accurate than $\geq 20\%$ LCAP predictions. This trend aligns with the models' confidence being generally higher for predicting low-performing reactions, which could partially be attributed to the uneven distribution

of reactivity labels in our data set (<20% LCAP for ~65% of scope reactions), as previously discussed in the manuscript for Figure 7B.

Dummy classifier baseline comparison

Data Split Type	Model Accuracy (Fig 7)	Model Accuracy Standard Error (Fig 7)	Dummy Classifier Accuracy
Random Split	0.859960	0.000913	0.649394
Amine OOS	0.833987	0.002226	0.651140
ArX OOS	0.739686	0.003817	0.653250
Both OOS	0.681496	0.005874	0.629840
DRS 25	0.734733	0.010511	0.654580
DRS 50	0.756573	0.002214	0.649491
DRS 75	0.843909	0.001099	0.654822

Table S2. Prediction accuracies of the models and dummy classifiers across different data split types.

In Figure 7, the model accuracy presented is the average prediction accuracy across N-fold cross-validation splits of varying data sizes and therefore the precise model baseline accuracy for each data split type may slightly differ from the baseline negative rate (64.9%). To investigate further, we implemented DummyClassifier using `scikit-learn` Python package with the strategy set to ‘prior’. The resulting prediction accuracies are summarized in Table S2. As expected, the dummy classifier accuracies differ from the 64.9% majority class fraction and vary slightly across the different data split types. As shown in Table S2, model accuracies are meaningfully higher than those of the dummy classifiers, further demonstrating that the models offer predictive value beyond the simple baseline.

Additional comments on data processing: Gaussian Expansion and empirical cumulative distribution function (ECDF) feature scaling

When applicable, the Gaussian Expansion of features for quantum mechanical fingerprint (QMFP) vector generation and empirical cumulative distribution function (ECDF) scaling of features were executed prior to data splitting rather than separately in training, validation, and test sets to ensure

consistent conversion of scalar QM descriptors to QMFP vectors and the consistent scaling of features.

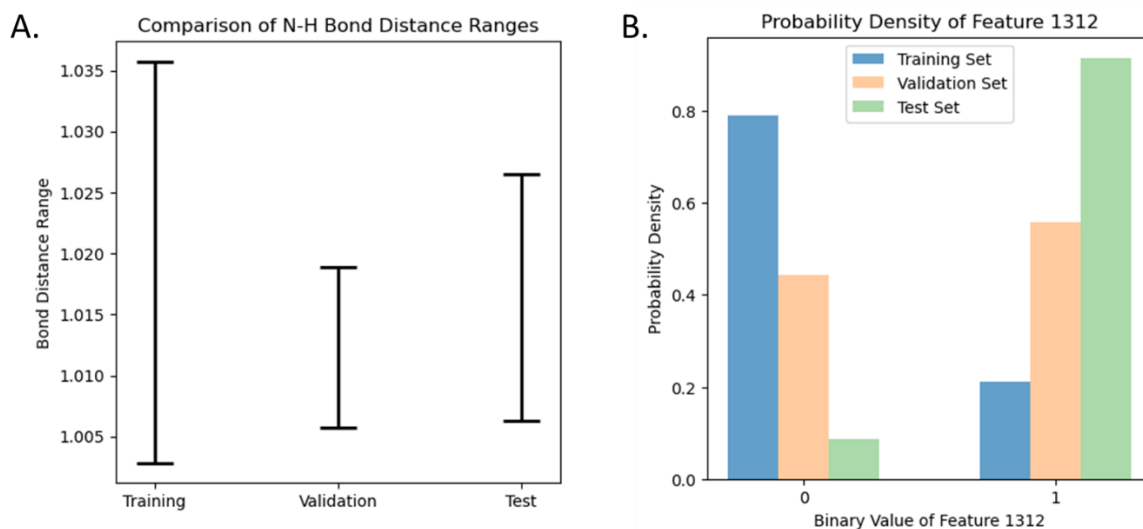


Figure S78. Examples of features with different distributions in training, validation, and test sets. (A) Ranges of amine N-H bond distance values in the training, validation, and test sets of Amine out-of-scope (OOS) split #0 before Gaussian Expansion. (B) Distributions of binary feature 1312 (the 289th element of the 512-bit Morgan fingerprint vector of a product molecule) in training, validation, and test sets of Both OOS split #0.

For example, Figure S78 shows examples of features with vastly different distributions in training, validation, and test sets. As described in the Gaussian Expansion section, minimum and maximum values of a descriptor in a given data set determine the range of Gaussian Expansion and resulting QMFP vector. If the minimum and maximum values of a descriptor differ significantly between training, validation, and test sets, as shown in Figure S78A, executing Gaussian Expansion separately for each set would convert the same descriptor value into vastly different feature vectors across training, validation, and test sets. This would be undesirable as it would introduce unnecessary noise in modeling. Also, Figure S78B shows very different distributions of a binary feature in a Morgan fingerprint vector across the training, validation, and test sets. Since ECDF-scaled feature values are determined by the overall distribution of the feature values in a given data set, executing ECDF scaling separately for each set would cause binary values of 0 and 1 to be converted into different values across the training, validation, and test sets.

This is especially problematic for out-of-scope (OOS) split studies, as the validation and test sets involve completely new species and some descriptors can have significantly different values in the validation and test sets compared to the training set, as shown in Figure S78. Additionally, validation and test sets are much smaller than the training set (e.g., the validation/test sets contain only 70 reactions for Both OOS splits — see Table S1) and have much lower species diversity in general. Those factors could further bias the Gaussian Expansion of QM descriptors or ECDF scaling of features.

To avoid these problems of inconsistent conversion that could introduce error and bias into the modeling analysis, we decided to perform the Gaussian Expansion and ECDF scaling before splitting the training, validation, and test sets. Furthermore, by using a larger data set for Gaussian Expansion or ECDF scaling, we expect the distribution of descriptor values or features to better approximate the true distribution within our chemical space of interest.

Since Gaussian Expansion and ECDF scaling was performed prior to splitting the data set, we considered the potential for data leakage. Although no reactivity information was involved in Gaussian Expansion and ECDF scaling, and thus no reactivity information would leak, we investigated whether potential data leakage had any non-negligible impact using model results from Phase 1 of the modeling workflow. Our assumption was that if there were any meaningful effects from data leakage, prediction accuracies would generally be higher when QM fingerprints (QMFP) were used as input features or when ECDF scaling was applied.

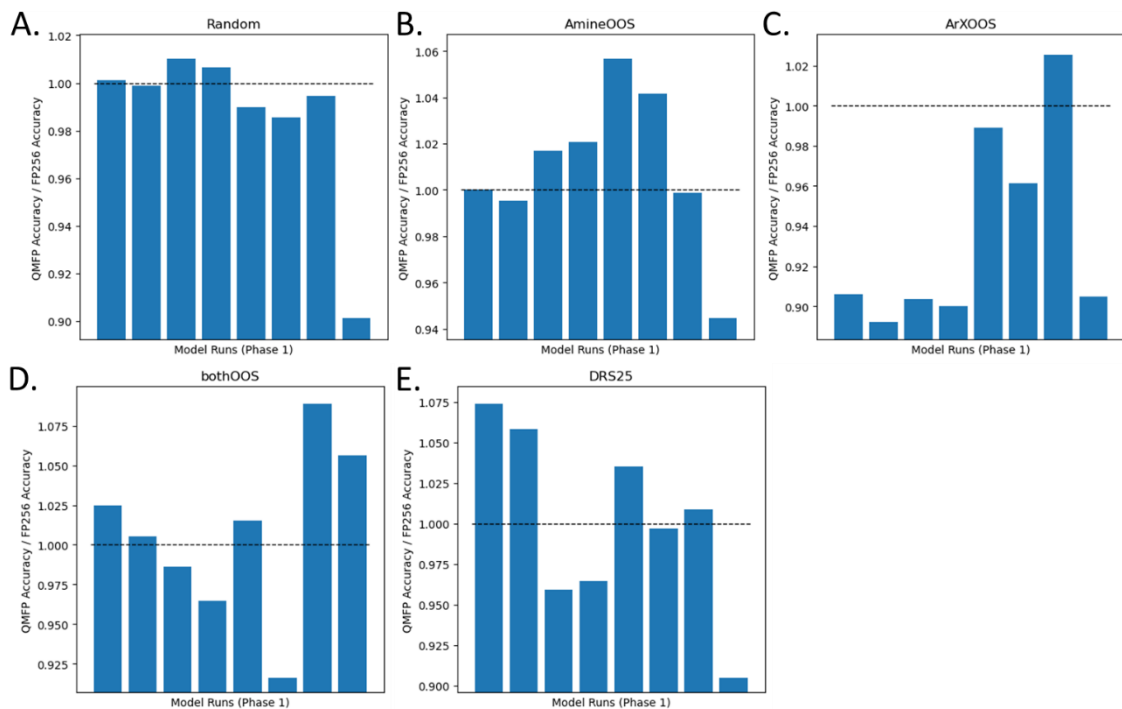


Figure S79. Comparison of model performances when quantum mechanical fingerprints (QMFP) were used as the input feature vs. when 256-bit Morgan fingerprints (FP) were used as the input feature for (A) random split, (B) Amine OOS split, (C) ArX OOS split, (D) Both OOS split, and (E) dimensionality reduction split (DRS25). Each bar represents the ratio of ‘*prediction accuracy with QMFP as the input feature*’ to ‘*prediction accuracy with 256-bit Morgan FP as the input feature*’. For each bar, the model architecture (Random Forest, Decision Tree, AdaBoost, or Fully-Connected Neural Network) and the use of ECDF scaler (on or off) were kept the same, ensuring that any difference in prediction accuracy was solely due to the choice of the input feature.

In Figure S79, each bar represents the ratio of ‘*prediction accuracy with QMFP as the input feature*’ to ‘*prediction accuracy with 256-bit Morgan fingerprints as the input feature*’. We compared QMFP and 256-bit Morgan FP due to their similar lengths (the length of a QMFP vector is 250), in an effort to prevent complications such as differences in the number of model parameters or varying levels of input noise that could stem from using different numbers of features. If model performances were exactly the same, the bar height would be 1 (dashed line). Figure S79 shows that bar heights generally fluctuate around 1 with no discernible trend of consistently being >1 . For some data splits, such as ArX OOS, using QMFP as the input feature generally decreased model performance compared to using the 256-bit Morgan FP. Overall, we found no indication of any impact from potential data leakage.

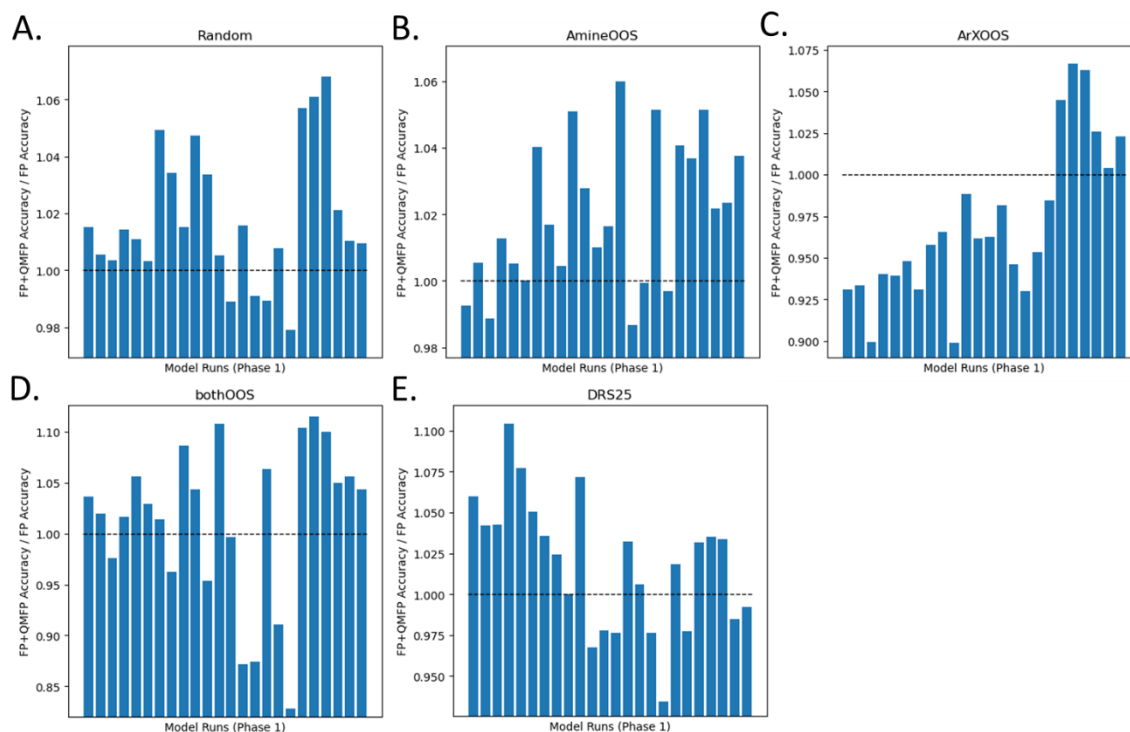


Figure S80. Comparison of model performances when the concatenated vector of xxx-bit Morgan fingerprints (FP) + quantum mechanical fingerprints (QMFP) was used as the input feature vs. when the xxx-bit Morgan FP was used as the input feature for (A) random split, (B) Amine OOS split, (C) ArX OOS split, (D) Both OOS split, and (E) dimensionality reduction split (DRS25). Each bar represents the ratio of ‘*prediction accuracy with xxx-bit Morgan FP + QMFP as the input feature*’ to ‘*prediction accuracy with xxx-bit Morgan FP as the input feature*’. For each bar, the model architecture (Random Forest, Decision Tree, AdaBoost, or Fully-Connected Neural Network), Morgan FP length (128-, 256-, or 512-bit) and the use of ECDF scaler (on or off) were kept the same, with only the input feature being varied.

In Figure S80, we also performed another comparison with each bar representing the ratio of ‘*prediction accuracy with the concatenated vector of xxx-bit Morgan FP + QMFP as the input feature*’ to ‘*prediction accuracy with the xxx-bit Morgan FP as the input feature*’ (xxx=128, 256, 512) to investigate the impact of adding QMFP to Morgan FP as the input feature. For some data splits (e.g., Amine OOS), concatenating QMFP with Morgan FP seemed to improve model performance, but in some other splits (e.g., ArX OOS) the opposite was true. Overall, there was no clear correlation between adding QMFP to the input feature and an increase or decrease in model accuracy. Additionally, it should be noted that there are other factors to consider when interpreting the results in this case, such as the different number of features used or the varying

levels of possible complementarity¹¹ between Morgan FP and QMFP in representing the reactivities of amines and aryl halides, unlike the previous comparison of QMFP vs 256-bit Morgan FP.

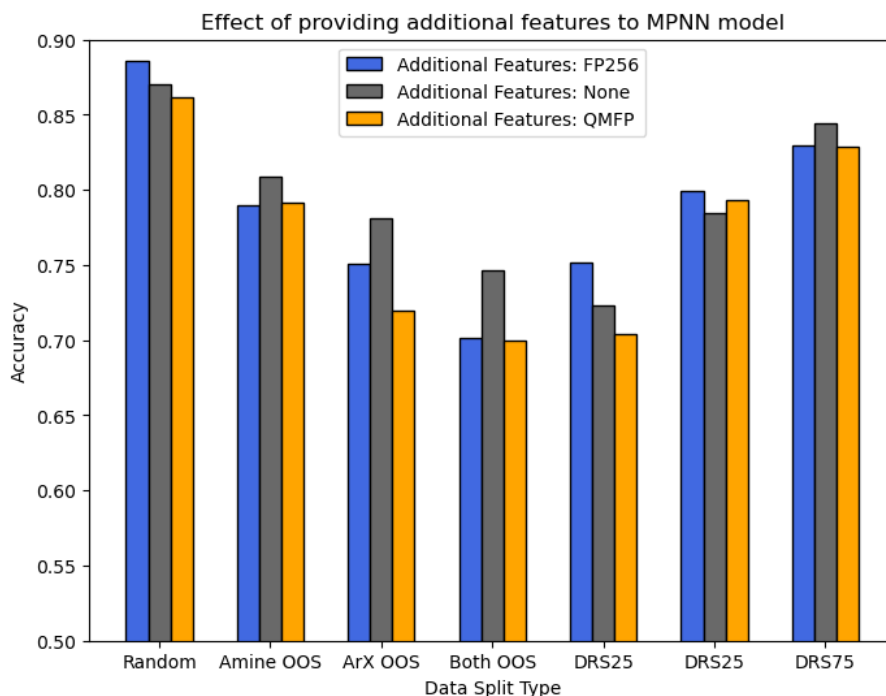


Figure S81. Chemprop Message-Passing Neural Network model performances with and without additional features (256-bit Morgan fingerprints or quantum mechanical fingerprints) provided to the model.

In Figure S81, the effect of providing 256-bit Morgan FP (FP256) or QMFP as additional features to the Chemprop Message-Passing Neural Network (MPNN) model was investigated. The Chemprop MPNN model allows the user to provide additional input features that can potentially complement the default molecular graph input.^{9c} Compared to MPNN model performances without any additional input features, providing QMFP features reduced model performance in most split types (6 out of 7), which is the opposite of what we expect to observe if significant data leakage had occurred. On the other hand, providing 256-bit Morgan FP as additional input features was slightly more helpful, improving model accuracies in 3 out of 7 split types.

Moreover, Figure S73 shows that QMFP features did not necessarily lead to improved model performance in most cases for Phase 1 of the modeling workflow, which could be an indirect

indication that any potential data leakage had no significant impact. Out of the 7 types of data splits, the best-performing models for 5 of them did not involve QMFP features as inputs.

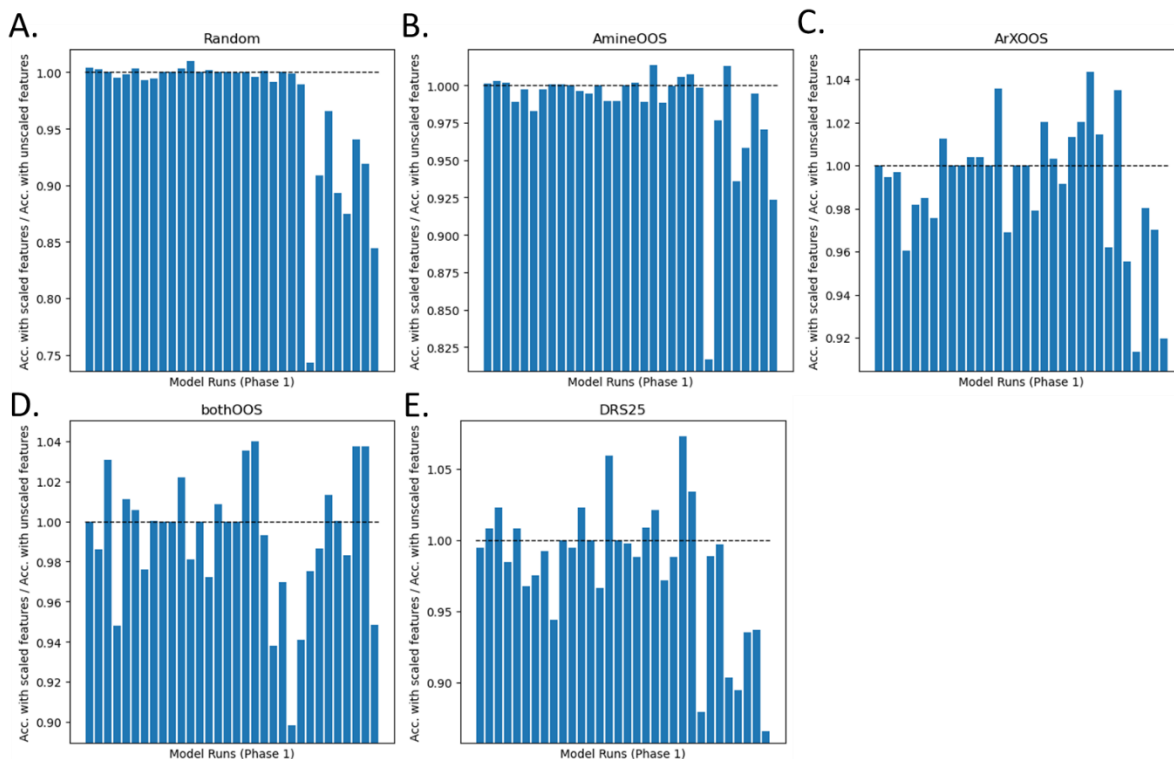


Figure S82. Comparison of model performances when empirical cumulative distribution function (ECDF) scaler was on vs. when ECDF scaler was off for (A) random split, (B) Amine OOS split, (C) ArX OOS split, (D) Both OOS split, and (E) dimensionality reduction split (DRS25). Each bar represents the ratio of ‘*prediction accuracy with ECDF scaler on*’ to ‘*prediction accuracy with ECDF scaler off*’. For each bar, the model architecture (Random Forest, Decision Tree, AdaBoost, or Fully-Connected Neural Network) and input feature ('OneHot', 'QMFP', 'FP128', 'FP128+QMFP', 'FP256', 'FP256+QMFP', 'FP512', 'FP512+QMFP') were kept the same, with only the scaler on/off being varied.

Figure S82 shows another comparison that is similar to Figures S79–S80. Each bar represents the ratio of ‘*prediction accuracy with ECDF scaler on*’ to ‘*prediction accuracy with ECDF scaler off*’. Bars generally fluctuate around 1 and there are several instances where ECDF scaling significantly decreased model performances, especially in the random split and Amine OOS split. Overall, the conclusion remains the same: we found no indication of potential data leakage.

Lastly, we believe that another indirect indication that any potential data leakage had a negligible impact is our successful results from the validation library analysis. If there had been significant data leakage, we would expect the models to overfit to the validation and test sets to some extent, thereby boosting apparent model validation and test set accuracies. In that case, our models would have struggled to provide accurate predictions for the validation library analysis. Based on our successful results from these libraries, we believe this was not the case and supports our conclusion that any potential data leakage had no significant impact.

XIII. References

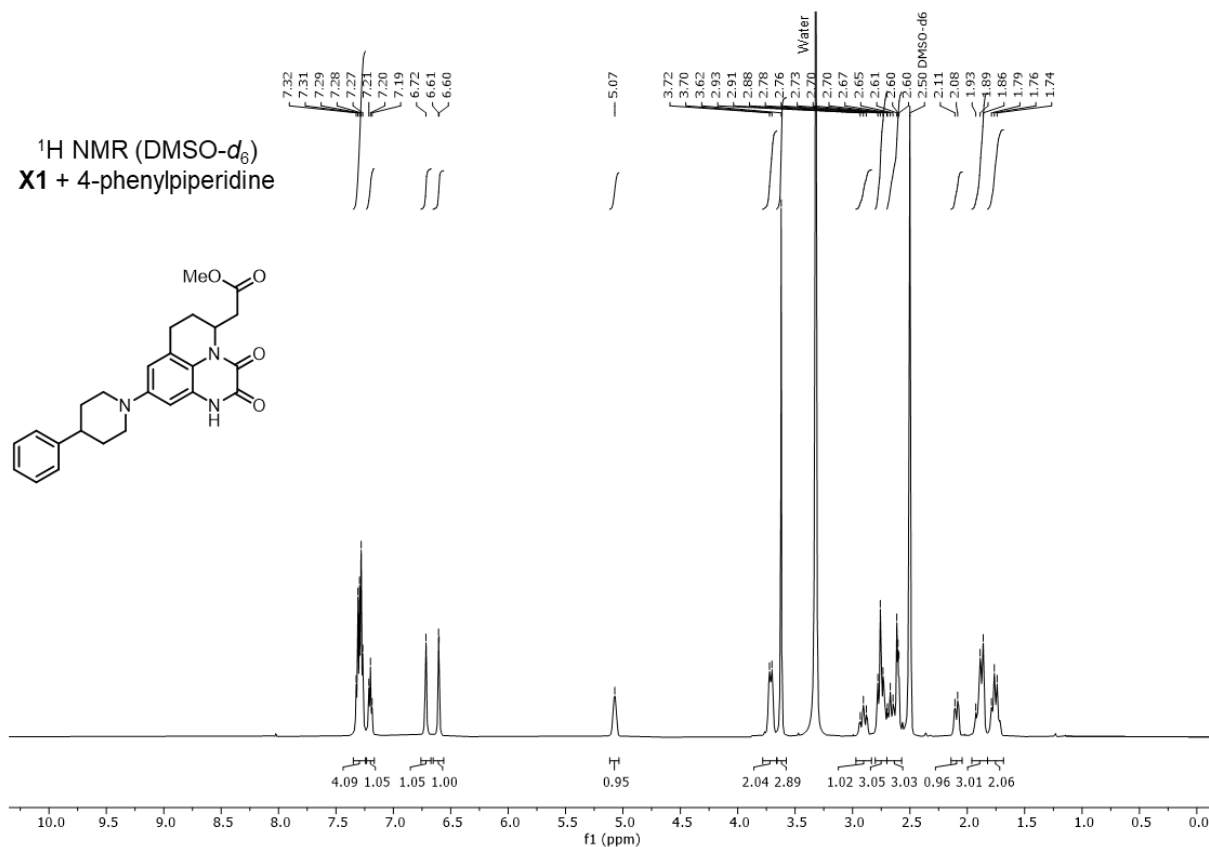
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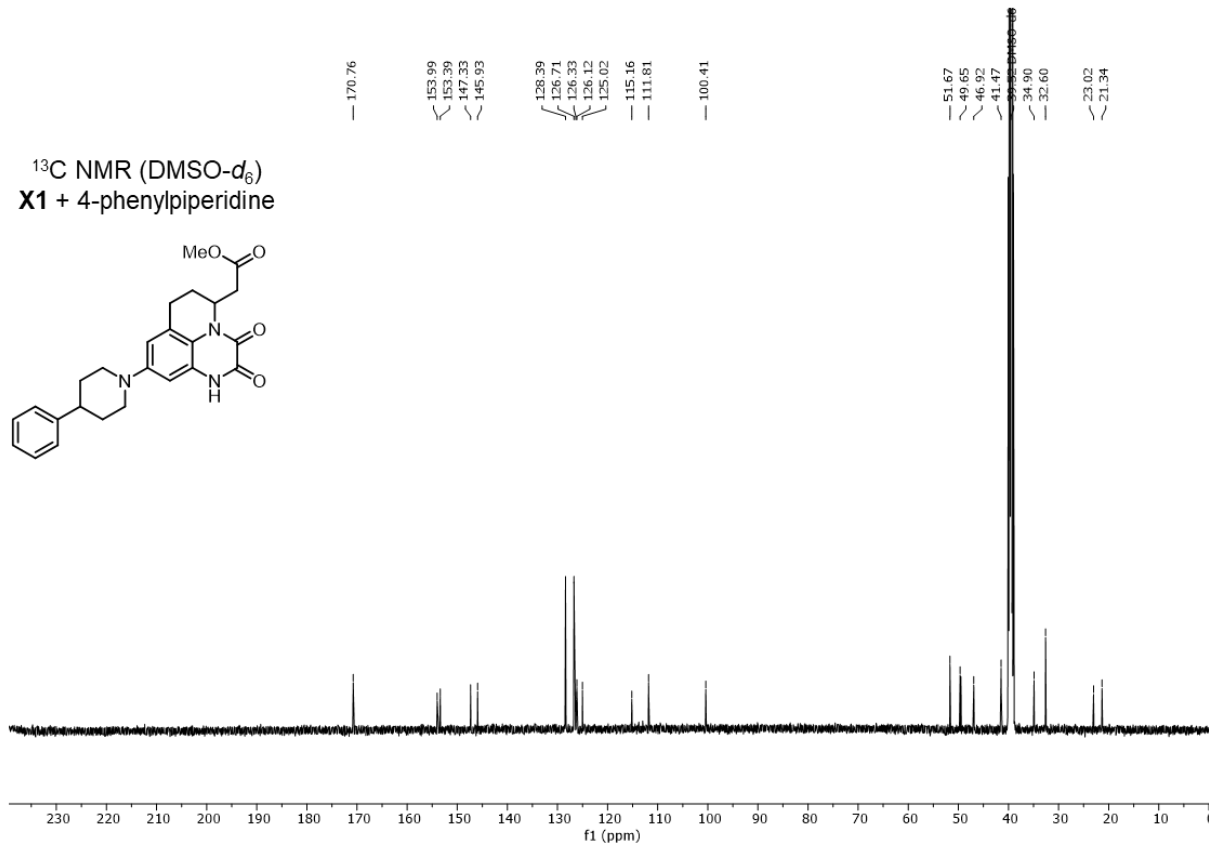
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XIV. NMR Spectra

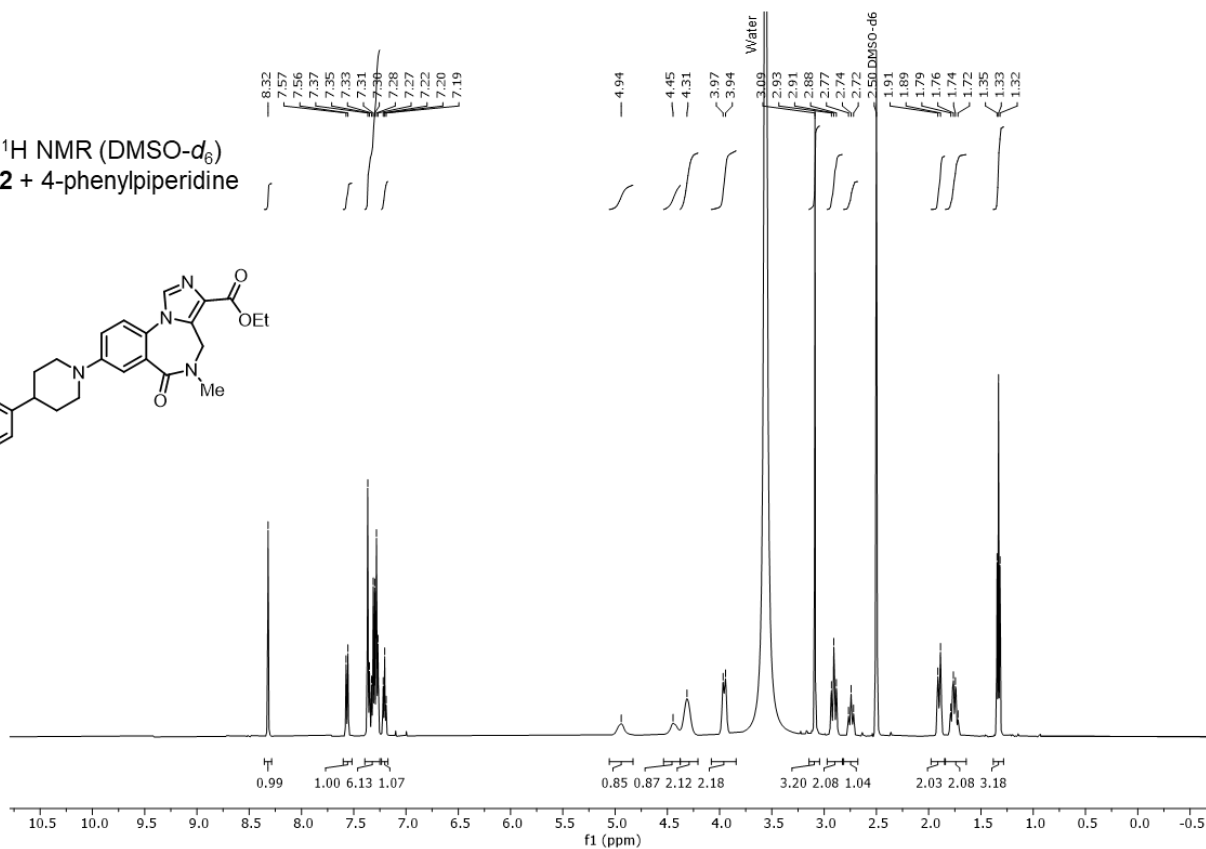
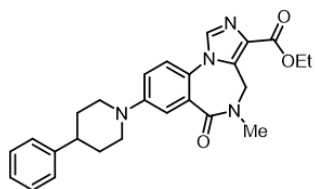
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X1 + 4-phenylpiperidine



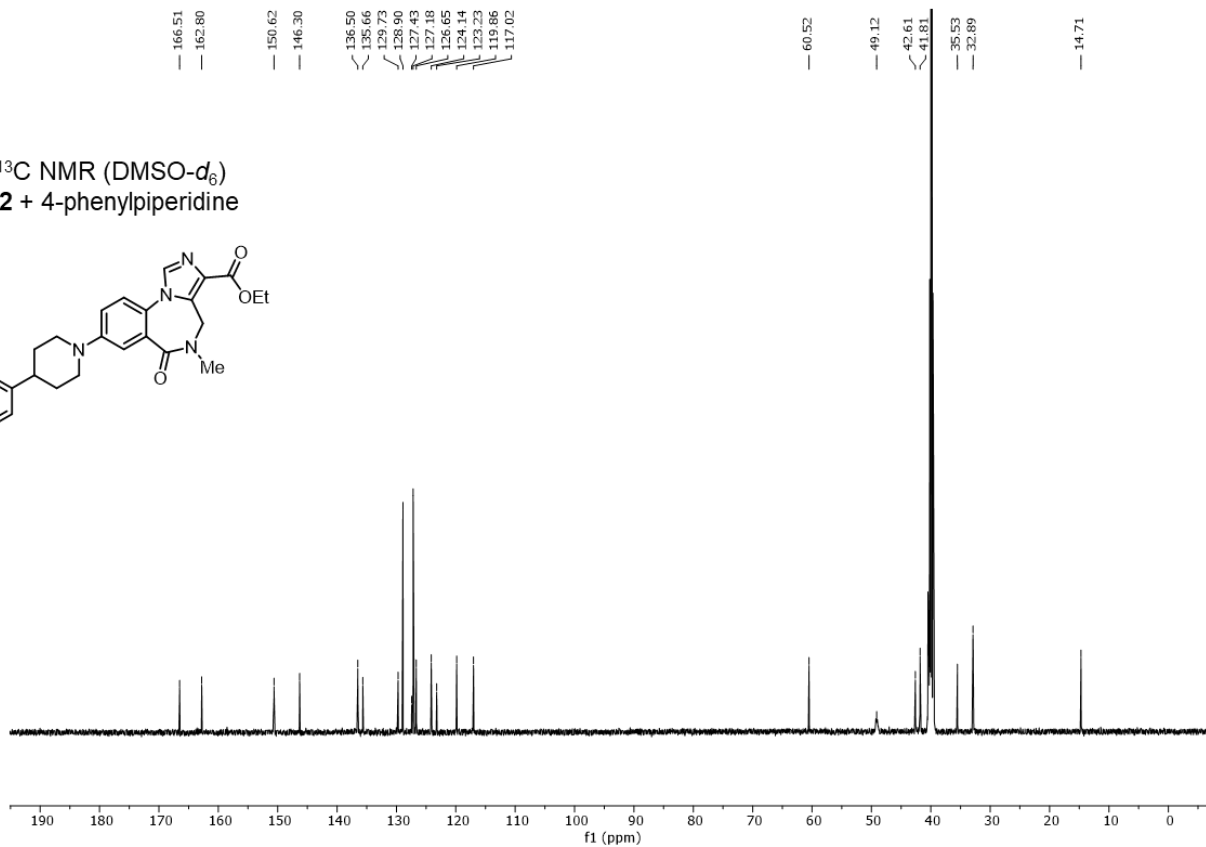
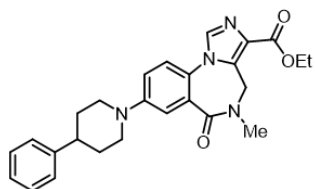
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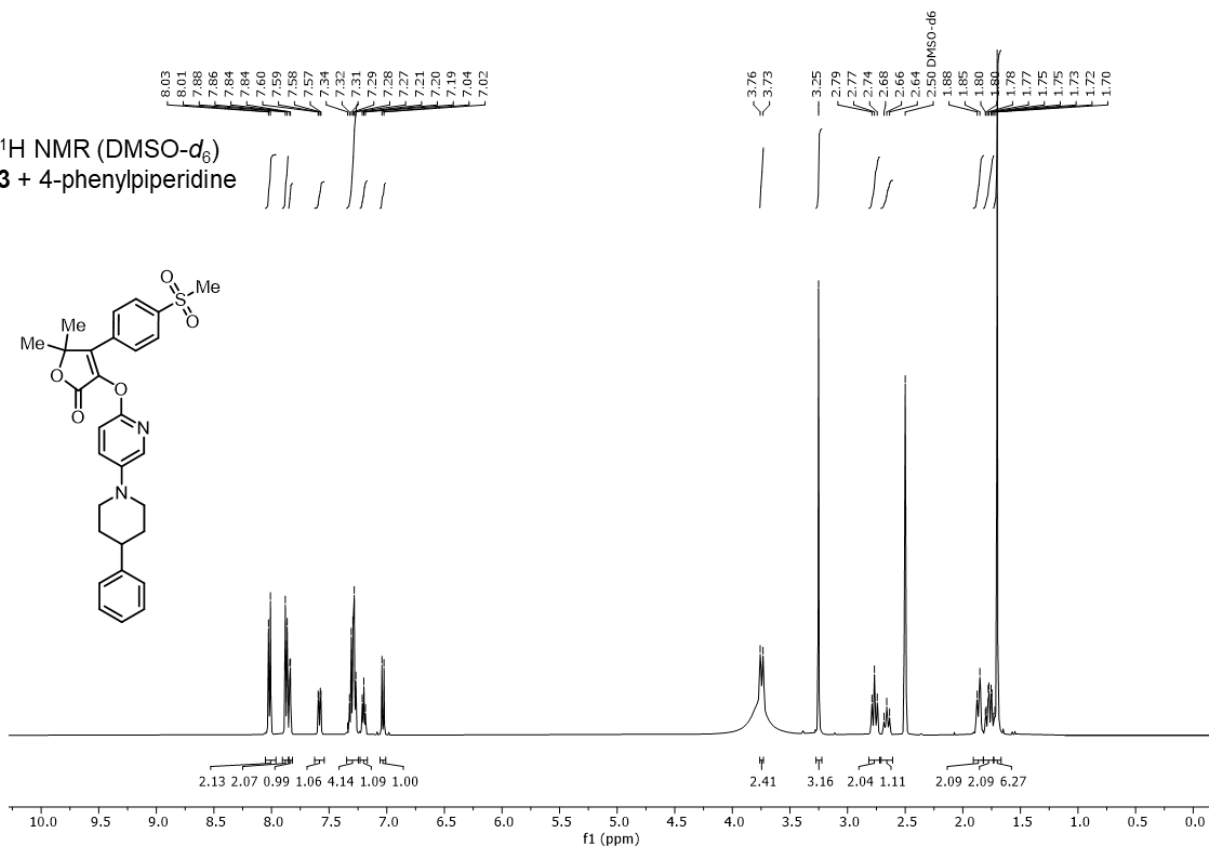
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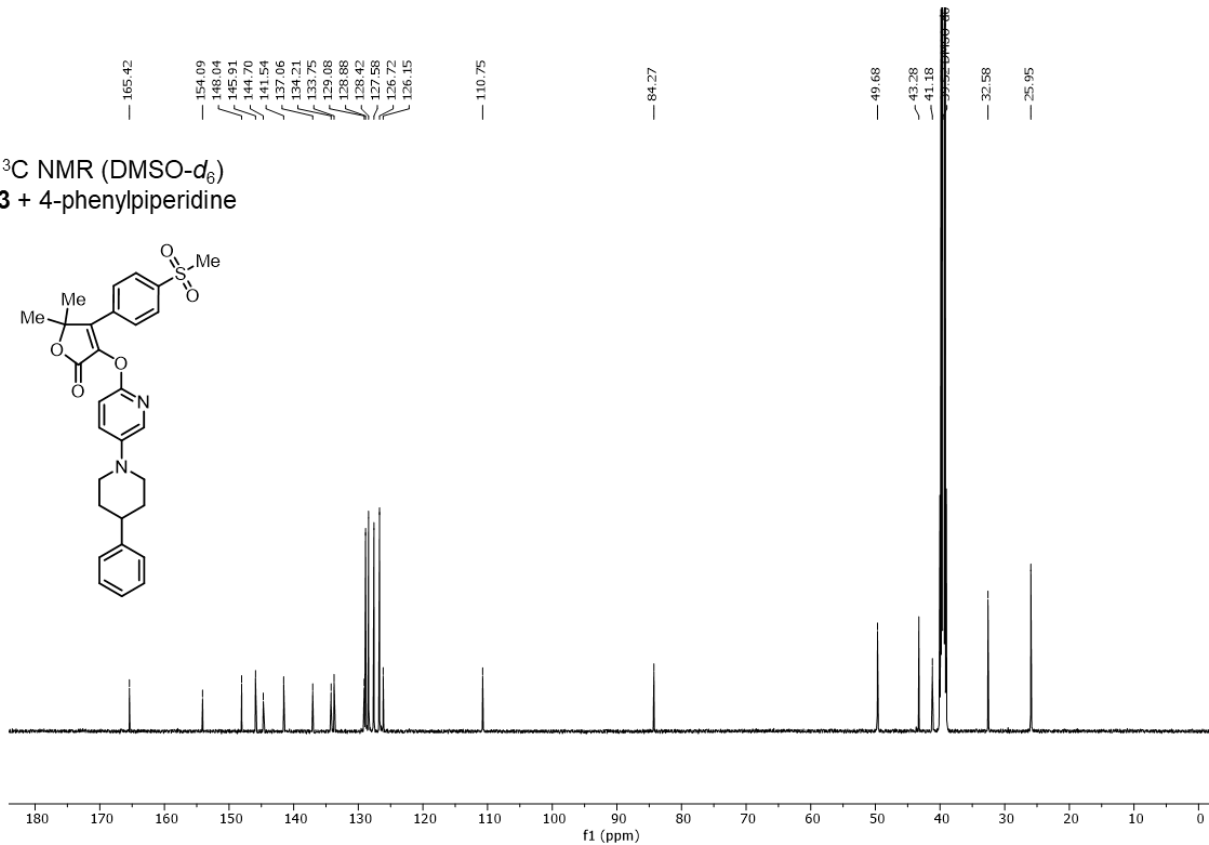
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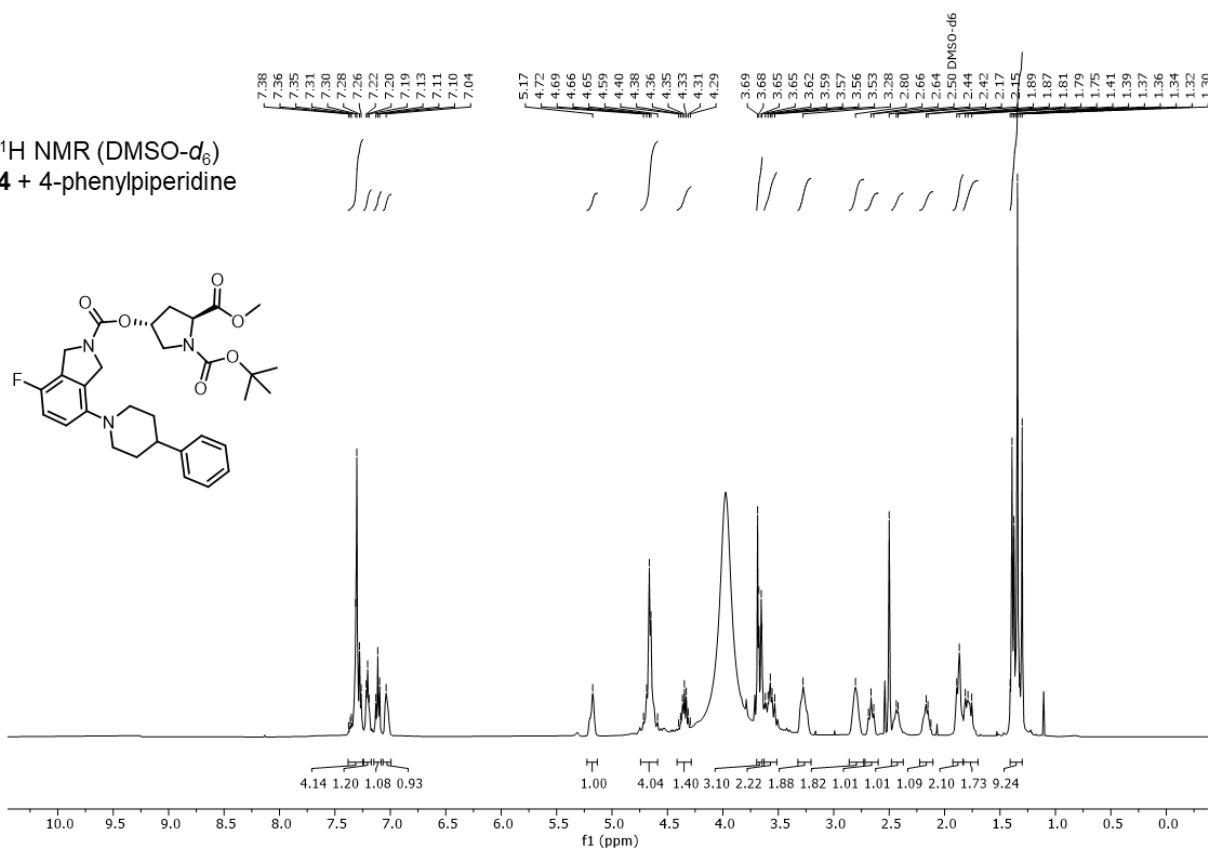
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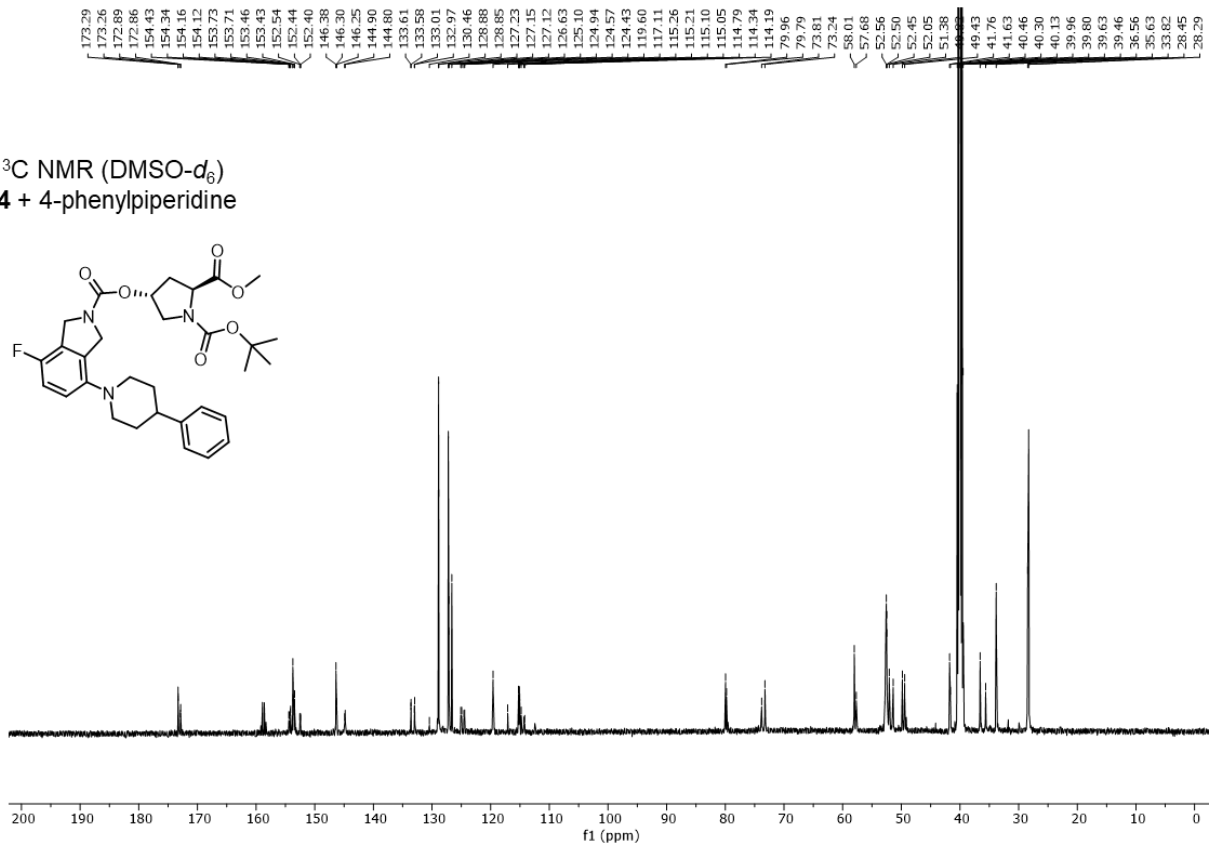
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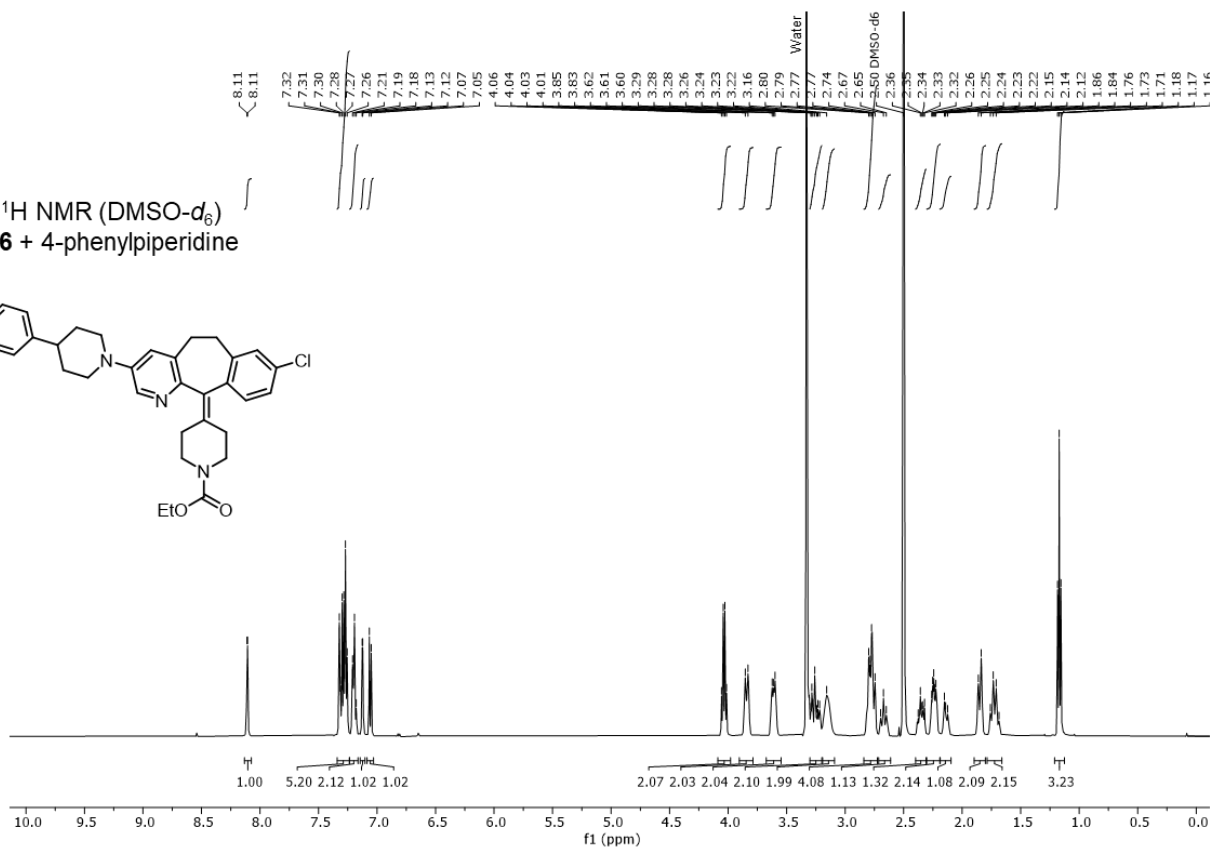
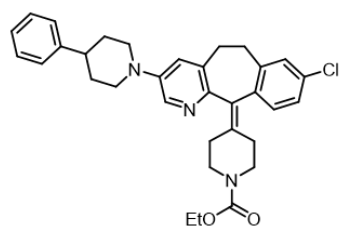
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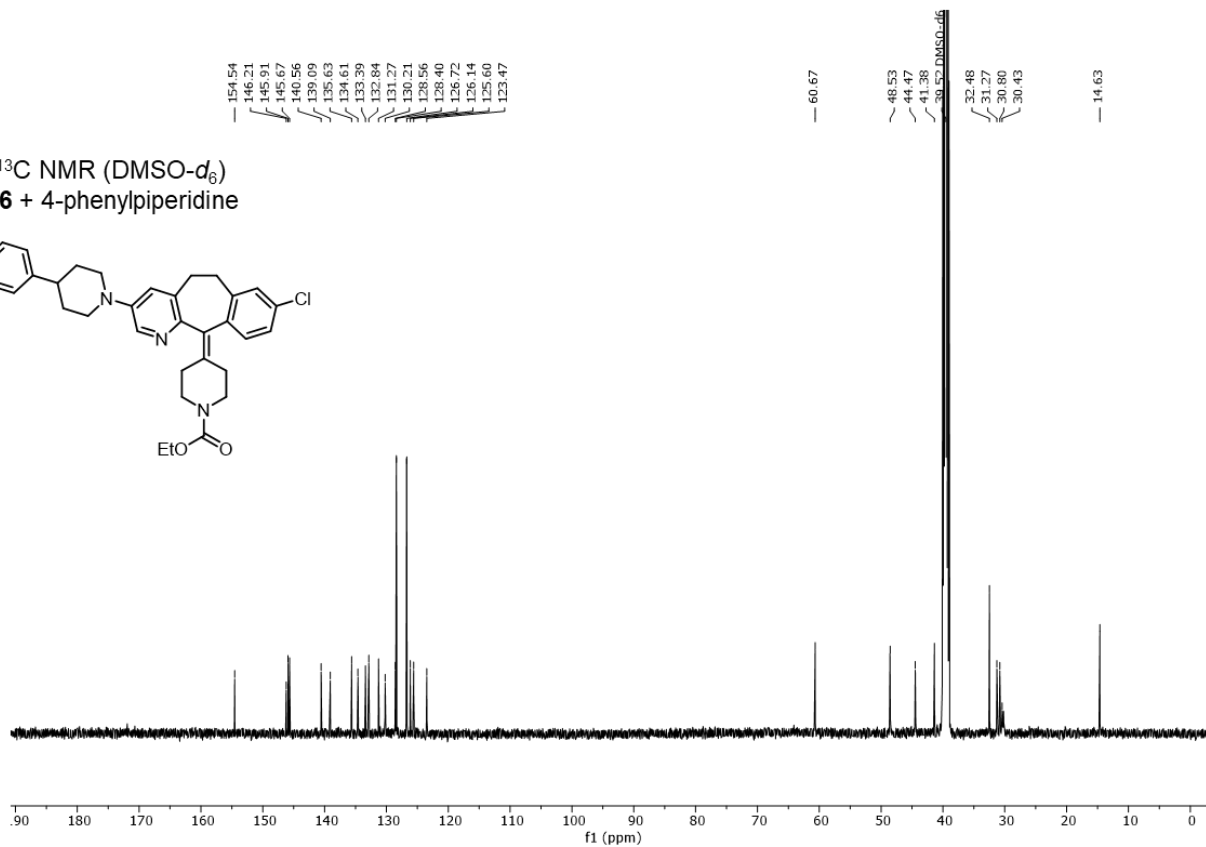
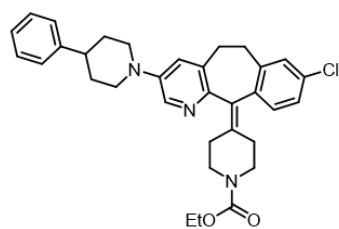
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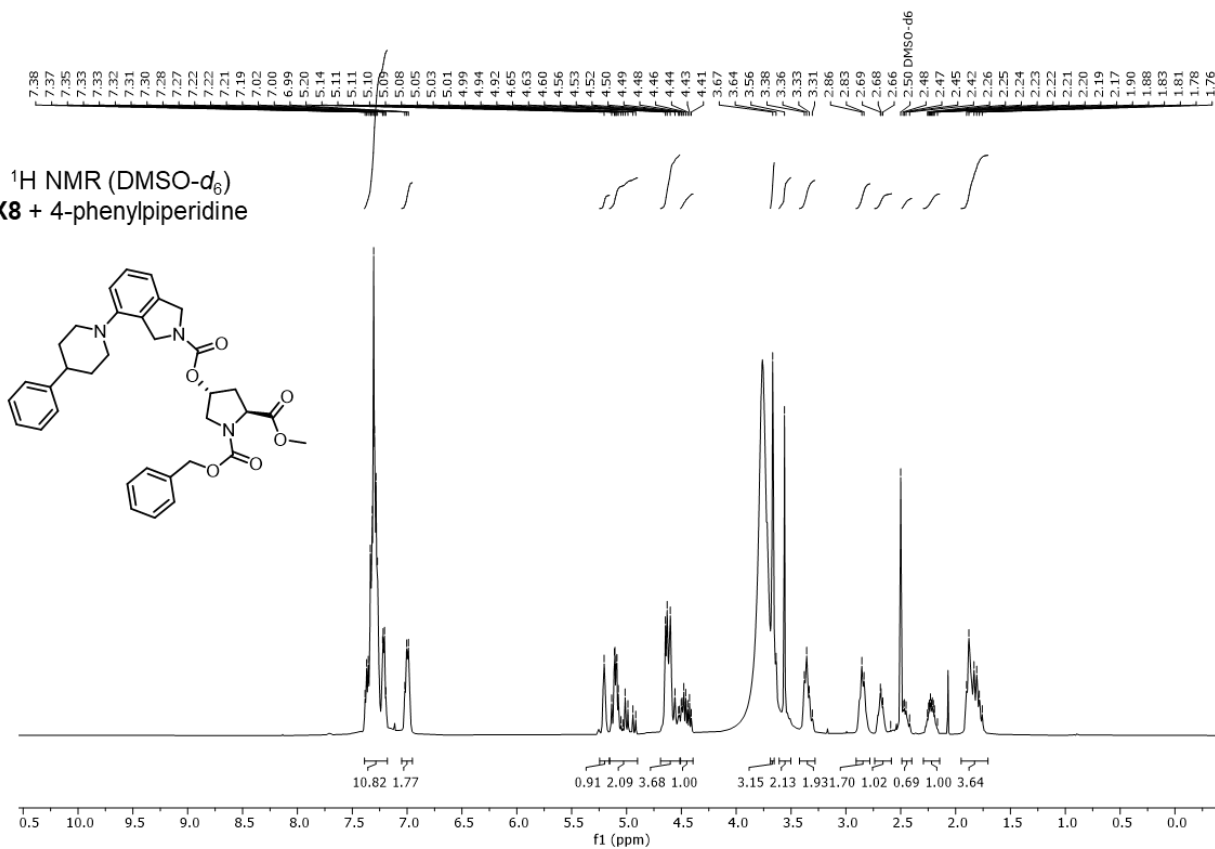
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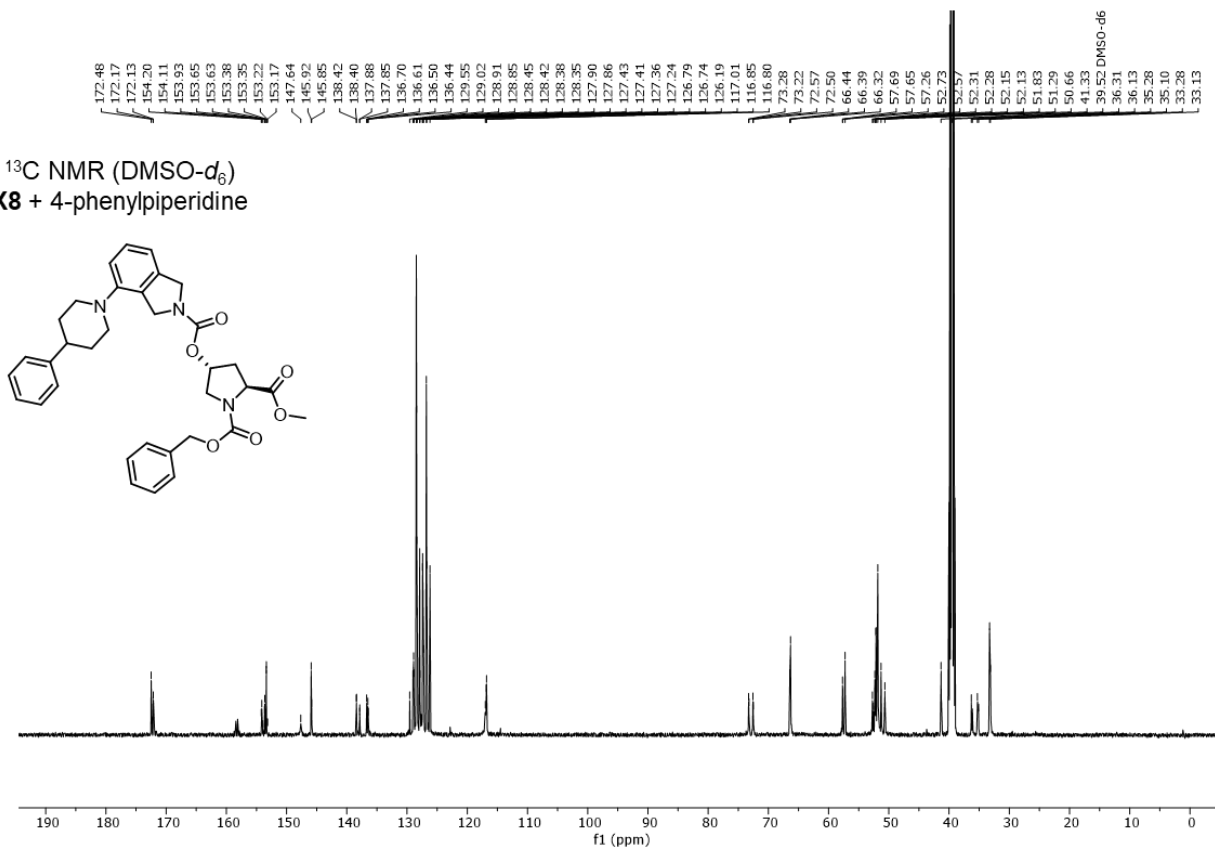
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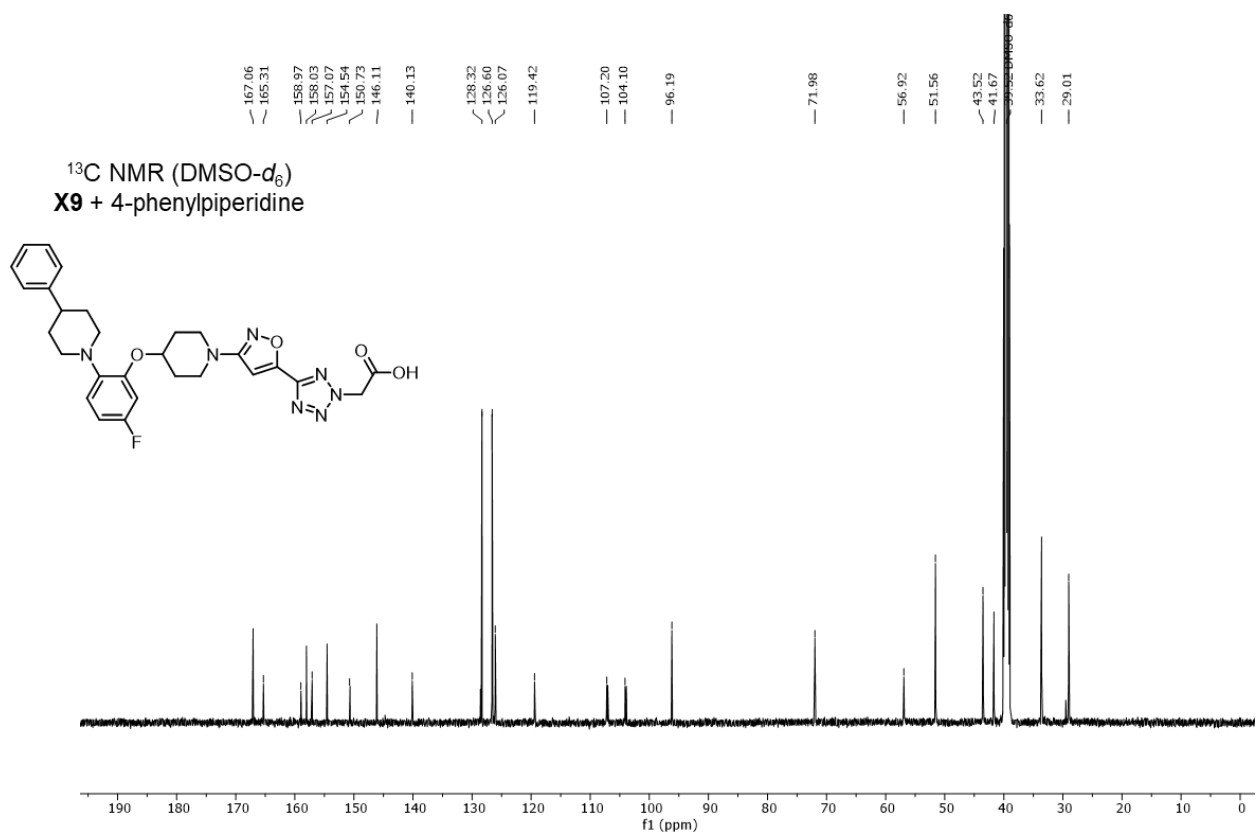
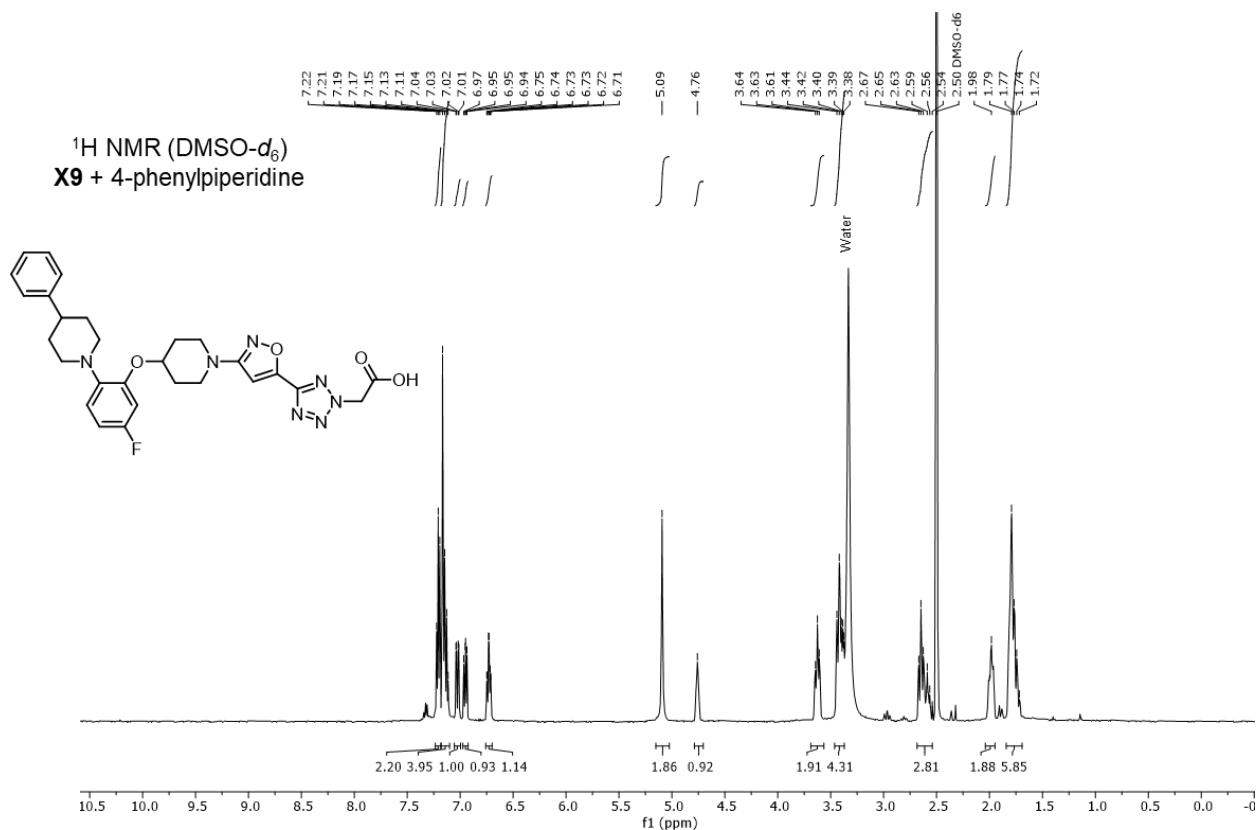


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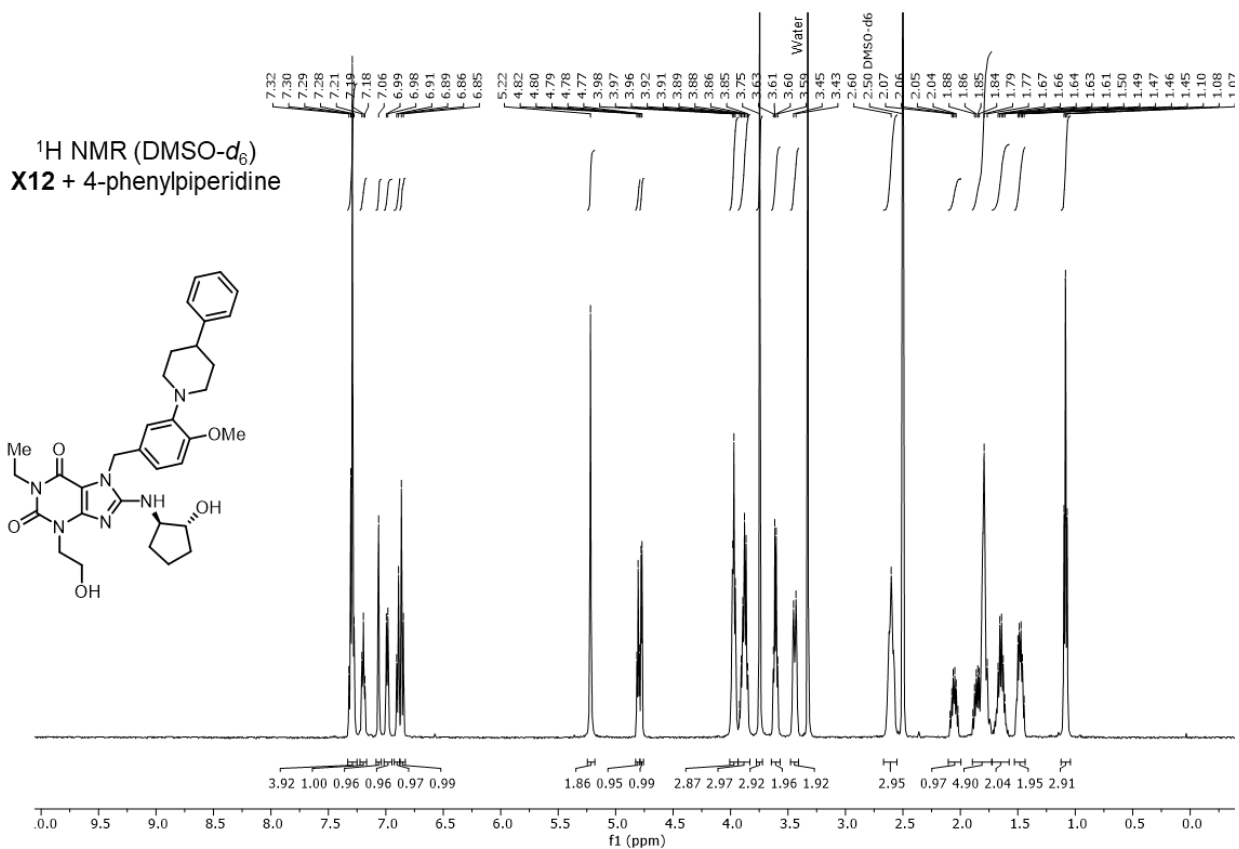


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